

Who Profits from Prediction?

Execution, not Information*

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Abstract

Retail traders pick winners yet lose money. In 222 million resolved Polymarket trades with binary, known-outcome payoffs, we separate two skills, forecasting (the side picked) and execution (the price paid). The two are nearly independent, and execution, not forecasting, determines who profits: bots at coin-flip accuracy (49.9%) earn \$133 million, more accurate retail (51.3%) lose \$79 million. Conventional VWAP benchmarks, built from the trader's own trades, make the two appear to trade off. The independence holds in 14.1 million CBOE option trades; the profit reversal does not. Retail traders lose not because they are wrong, but because they are late.

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1 Introduction

Retail traders pick winners yet lose money. In equities, their directional picks positively predict short-horizon returns (Barber, Lin, and Odean, 2024; Kelley and Tetlock, 2013; Kaniel, Saar, and Titman, 2008; Welch, 2022), while their trading loses money in aggregate (Barber and Odean, 2000). If picking the right side were enough, accuracy and losses could not coexist. A second skill must be at work: *execution*, the ability to enter at a good price. Execution has been studied for particular trader types (Hendershott, Jones, and Menkveld, 2011; Menkveld, 2013), but never separated from *forecasting* within the same group. This paper makes that separation and shows that the price paid, not the side picked, determines who profits.

Canonical informed-trading models and much of the empirical performance attribution literature bundle the two skills. The same latent advantage is treated as producing both better forecasts and better entry prices. In Kyle (1985), an informed trader profits by identifying the right side and trading before prices adjust, so a better forecast obtains a better entry price. Glosten and Milgrom (1985) and the microstructure literature that follows inherit this structure, as does empirical work on trader performance (Anand, Irvine, Puckett, and Venkataraman, 2012; Patton and Weller, 2020; Grinblatt, Keloharju, and Linnainmaa, 2012). We call this the *bundled view*; it predicts that the two skills should rise and fall together across traders.

Testing the bundled view requires two conditions. First, profit must be decomposable: a reference price (fair value) splits profit per contract into a directional part (outcome minus reference) and an execution part (reference minus entry price), which requires a terminal payoff observed independently of the trade sequence. Second, the reference price that splits the two components must not be mechanically affected by the evaluated trader’s own trades. Equities satisfy neither. There is no common terminal payoff, and conventional VWAP benchmarks share trades with the traders they evaluate. Sharing trades biases the measured cross-skill relation toward negative values even when the two skills are independent.

Prediction-market binary contracts meet both conditions. The contract resolves at \$0 or \$1 by an external event, and the price of the trade immediately preceding the trader’s own provides a reference that excludes the evaluated trade by construction.

In this paper, we find that execution, not forecasting, separates profitable from unprofitable traders, and the two skills are nearly independent of each other. The contrast is sharpest between bots and active retail, the two types that trade the same markets in volume. Bots achieve 49.9% directional accuracy, no better than a coin flip, yet earn the only positive aggregate return in the sample: \$133 million. Active retail traders achieve 51.3% accuracy and lose \$79 million. Profitability runs opposite to accuracy because no trader type achieves positive excess accuracy, defined as directional accuracy above the price-implied benchmark. A contract priced at \$0.70 wins about 70% of the time with no skill, so raw accuracy above 50% reflects trading at prices that had already moved toward the outcome, not superior information.

The bundled view fails on both of its predictions. (i) Across trader types, the cross-skill coefficient is zero under a benchmark that does not draw on the evaluated trader’s own prior trades. (ii) Within trader types, higher accuracy buys no entry-price discount. The two-dimensional view holds on both of the predictions.

Consider a retail trader and an automated bot, each trading 100 contracts that resolve to \$1 or \$0 in markets where the average price is \$0.50. The retail trader follows the news closely and picks the winning side in 60 of 100 trades, modestly better than chance, but enters at an average of \$0.65: the market has already moved toward the outcome by the time the retail trader arrives. That premium, paid on winners and losers alike, returns \$60 on the \$65 invested, a loss of \$5.

In contrast, the bot picks the winning side only 50 times, no better than a coin flip, but enters at \$0.40 by taking the other side of incoming demand before prices adjust, earning \$50 on the \$40 invested, a profit of \$10. Decomposed against the market average, the retail trader has a directional edge of \$0.10 per contract (60% beats the 50% baseline) and an execution

penalty of \$0.15 (paying \$0.65 against a \$0.50 average); the bot has no directional edge and a \$0.10 execution gain. The retail trader is the more accurate forecaster, the bot the more profitable trader. The bundled view does not capture the distinction, the two-dimensional view does.

Resource independence is the property behind this pattern: forecasting and execution draw on different inputs.¹ Forecasting draws on cognition and information: reading signals and pricing contracts against terminal payoffs. Execution draws on infrastructure: speed, order priority, and the ability to supply liquidity. An attentive retail trader may form accurate views yet lack the speed to act before prices move; an automated market-maker may quote hundreds of contracts while holding no view at all.

Together with the reference price correction, the two-dimensional view generates four predictions. The bundled view cannot accommodate any of them. First, within a trader type, more accurate traders should not pay lower entry prices once we account for which markets they trade and at what prices.² Second, the relation between the two skills should fall to zero as the benchmark draws progressively less on the trader’s own prior trades. Third, within the same trader over time, execution should look much more persistent under the clean benchmark than under VWAP. Fourth, the two skills should still appear independent in equity options, but the accuracy-profitability inversion should not, because options traders choose strike and expiration as well as direction.

We test these predictions on the complete transaction record of Polymarket, recorded on the Polygon blockchain: 222 million resolved trades by 1.95 million wallets between November 2022 and February 2026, with every contract settling at a known \$0 or \$1. We classify wallets into five behavioral types (bot, sophisticated, active retail, casual, and one-shot) based on trading frequency, volume, diversification, and activity span. The classification is

¹Our execution measure captures the entry price relative to a fair-price benchmark constructed from the trade sequence, not slippage relative to a contemporaneous midpoint as in [Bessembinder \(2003\)](#) and [Dyhrberg, Shkilko, and Werner \(2025\)](#).

²Pure momentum chasing would raise entry prices for all traders in the same market, including those whose own forecasts are uninformative. The within-type test discriminates resource independence from this alternative.

profit-blind, and a separate data-driven k-means clustering replicates the same groups.

Three sources of variation discriminate the two-dimensional view from its alternatives. We compare wallets of the same type, follow the same wallet over time, and vary the reference price while holding the sample and regression fixed.

Within each trader type, higher accuracy does not buy a lower entry price, contrary to what bundling predicts. Bots show this clearly: their more accurate wallets pay higher prices, not lower. For sophisticated and active retail traders, more accurate wallets appear to pay lower prices in the raw data, but the relation disappears once we account for the more accurate wallets trading in different markets and at different price levels than the less accurate ones. The within-type test holds trader type, market, and price range fixed, so a null cannot be attributed to selection on trader characteristics or to entry-time differences across markets.

The cross-trader independence is not a product of the lagged-price benchmark. As the reference price draws less on the trader’s own trades, the cross-skill coefficient moves monotonically toward zero: it is -0.178 under the conventional backward VWAP, -0.013 under a narrower local VWAP, and 0.000 under the lagged-price benchmark. A genuine bundling relation would not move when only the benchmark changes.

Following the same wallet over time confirms the pattern. Each skill predicts itself: first-half directional accuracy predicts second-half accuracy ($\hat{\beta} = 0.270$), and first-half execution predicts second-half execution. Neither predicts the other: first-half accuracy is uninformative about second-half execution. Execution persistence rises from 0.080 under VWAP to 0.434 under the lagged-price benchmark. Pure noise would lower measured persistence, not raise it, so the gap reflects benchmark bias rather than measurement error.

Execution decides profits because the value of a correct forecast decays as prices converge toward their terminal values. A ten-point accuracy gain is worth nearly 12 cents per contract when prices are near \$0.50, but less than 2 cents once prices approach a terminal payoff. Bots enter while prices reveal little of the eventual outcome; retail arrives after prices have already

moved. In [Kyle \(1985\)](#), early entry reflects private information; here it does not, because bots forecast no better than a coin flip. Early entry operates as execution infrastructure rather than as evidence of a signal, and the advantage holds within both roles of the limit-order book (makers and takers), ruling out pure spread capture as the explanation.

The independence between skills travels across markets; the inversion does not. In 14.1 million CBOE equity option contracts with investor-class identifiers, split-sample cross-skill persistence is near zero for every trader type, and first-half forecasting explains under half a percent of the variance in second-half execution. The inversion fails to replicate in options because the two conditions that produce it do not hold. Options traders choose strike and expiration alongside direction, so accuracy compounds across three margins rather than the single binary one. And the upside on a winning option contract is not capped, so accuracy beyond a baseline keeps paying off rather than hitting the binary payoff ceiling that binds the returns to forecasting in prediction markets. More accurate options traders earn 141 cents more per option share from Q1 to Q5 of the accuracy distribution. The boundary ties the inversion to single-margin markets with capped payoffs and rules out the possibility that the inversion is generated by the corrected decomposition itself.

We make three contributions. First, we document that trading skill has two dimensions. Forecasting and execution are nearly independent across traders and persistent within them, with own-skill coefficients of 0.270 and 0.434 and a cross-skill coefficient of 0.000 under a clean benchmark. Models that treat informed trading as a single underlying ability cannot accommodate this separability.

Second, we identify execution timing as the channel behind the inversion and bound the inversion to single-margin markets. As prices converge toward terminal payoffs, the marginal value of a correct forecast collapses, and traders who arrive late forfeit the surplus to traders who arrive early. The inversion replicates only in markets where direction is the single binary choice; in options, where strike and expiration are also free, more accurate traders earn more.

Third, we derive a benchmark-contamination bias that generalizes beyond our setting.

Any decomposition that uses a benchmark built from the same trades it scores inherits the bias; the analytical bound matches the empirical estimate in this paper to within one percent. The magnitude in other settings is an empirical question.

The results carry implications for theory, policy, and practice. Standard informed-trading models should add a second execution skill alongside the forecasting advantage; retail-protection rules should target execution access rather than information asymmetry alone, since retail losses arise mainly from late arrival; and performance-attribution decompositions built on VWAP-style benchmarks should be re-examined for the cross-skill bias documented here.

The most closely related measurement approach is [Bogousslavsky, Fos, and Muravyev \(2024\)](#), who treat informed trading as a single latent variable. We show that trading skill is two-dimensional and that the dimension determining profitability is the one a single-trait measure cannot isolate. The closest methodological precedent is [Hagströmer \(2021\)](#), who shows that the conventional effective bid-ask spread is biased by how its benchmark is constructed; we apply the same logic to the covariance between forecasting and execution, where removing the bias reveals a structural feature of trading skill rather than correcting a level. The HFT literature has long documented that automated traders profit from speed and liquidity provision rather than fundamental information ([Hendershott et al., 2011](#); [Menkveld, 2013](#); [Baron, Brogaard, Hagströmer, and Kirilenko, 2019](#); [Baldauf and Mollner, 2022](#)); our contribution is to separate forecasting from execution within the same trader population.

Within prediction markets, subsequent work by [Akey, Grégoire, Harvie, and Martineau \(2026\)](#) documents that Polymarket profits are highly concentrated and that maker volume share is the strongest cross-sectional predictor of performance. That work characterizes performance descriptively and notes the absence of a benchmark of expected returns to separate skill from luck. We supply that benchmark: the terminal-payoff decomposition identifies forecasting and execution as distinct, orthogonal skills, and shows that the predictor they isolate, liquidity provision, is the execution margin rather than forecasting. [Reichenbach](#)

and Walther (2025), Yang (2026), and Gómez-Cram, Guo, Jensen, and Kung (2026) classify traders by accuracy or profitability as a single latent variable; our decomposition shows this conflates forecasting with execution.

2 Framework and Predictions

Is trading skill one-dimensional or two-dimensional? Standard microstructure theory treats it as one-dimensional. We propose an alternative in which *forecasting* and *execution* are distinct capacities drawing on different underlying resources, neither mechanically generating the other. This section establishes the bundled view as the convention, sets out the two-dimensional alternative, identifies the measurement problem that has prevented prior work from testing between them, describes the prediction-market setting that resolves it, and derives four testable predictions.

2.1 The Bundled View as Maintained Convention

The dominant view of trading skill is one-dimensional. Kyle (1985) and Glosten and Milgrom (1985) formalize a single informational advantage that delivers both correct directional picks and favorable entry prices. The broader microstructure literature inherits this structure.

Empirical work implements the bundled view through single-variable measures of trader skill (Bogousslavsky et al., 2024; Grinblatt et al., 2012). Performance attribution embeds the same structure (Anand et al., 2012; Patton and Weller, 2020; Saglam, Moallemi, and Sotiropoulos, 2019). Where execution has been studied separately (Kaniel et al., 2008; Hendershott et al., 2011; Menkveld, 2013), it has been treated as a feature of specific trader populations rather than as a universal skill margin.

The bundled view requires that traders capture an information rent. Profit comes from signals prices have not yet incorporated, so profitable traders should win more often than their prices imply. A contract’s price is the market’s implied probability that it pays \$1,

so buying at an average price of \$0.70 wins roughly 70% of the time with no skill. Excess accuracy, a trader’s directional accuracy above that price-implied null, is the relevant margin. The bundled view then carries two testable implications. First, if forecasting and execution are the same capacity, more accurate traders within a type should not pay systematically worse entry prices, and may pay better ones if a directional signal also lets them trade before prices adjust. We test the sharpest version of bundling, a negative within-type accuracy-entry-price coefficient. The two-dimensional view predicts no such gradient survives composition controls. Second, because the informational advantage exists independently of the measurement convention, the relation between the two skills should survive clean measurement: a benchmark free of mechanical bias should reveal a positive cross-skill relation, not eliminate it.

The bundled view has not been tested cleanly. Standard equity performance attribution lacks the externally determined terminal payoff a clean test requires, and its conventional benchmarks are built from the evaluated traders’ own trades, which biases any comparison between the two skills regardless of the underlying structure.

2.2 The Two-Dimensional Alternative

The two-dimensional alternative rests on two distinct rents in the microstructure literature, each with its own source and its own requirements for capture. The first is the information rent. A trader who identifies the winning side before prices adjust earns a directional profit. By “information” we mean superior directional forecasting relative to prices; the broader microstructure sense of order-flow or timing information falls under the execution rent in our framework, not the information rent. The informed trader’s profit is funded by uninformed traders: in [Glosten and Milgrom \(1985\)](#) they bear it through the spread the market maker charges to offset adverse selection, and in [Kyle \(1985\)](#) noise traders bear it by trading against an order flow that already embeds the signal. Generating the rent requires observing relevant information (whether private signals or publicly available data), parsing it, and pricing the

contract relative to terminal payoffs.

The second is the execution rent documented in the literature on automated and high-frequency trading (Hendershott et al., 2011; Menkveld, 2013; Budish, Cramton, and Shim, 2015; Baron et al., 2019). A trader who enters markets before demand arrives captures a price discount relative to the fair-price consensus, as does a trader who provides liquidity selectively to uninformed flow. Liquidity demanders (market orders) pay this discount through the spread; late-arriving counterparties pay it through subsequent price impact. Generating this rent requires technological infrastructure (such as latency, queue position, market access) and strategic timing rather than fundamental information about contract outcomes.

The two rents typically accrue to different sides of each transaction. The informed trader demands liquidity to profit on a signal. The fast trader supplies liquidity to capture the spread. A fast market-maker need not possess superior fundamental information. A sharp analyst need not have low-latency execution.

In our framework, forecasting (θ_i) captures the information rent and execution quality (ε_i) the execution rent. The information rent draws on cognition and information; the execution rent draws on technological infrastructure. Because θ and ε draw on different resources, acquiring one does not mechanically deliver the other. The cross-sectional coefficient between the two skills should therefore be zero under clean measurement. Bundling predicts a positive coefficient because a single informational capacity delivers both. Appendix D formalizes the argument.

We do not treat resource independence as a structural primitive. General cognitive ability could, in principle, drive both forecasting and execution, producing a positive cleaned coefficient even when the two resources are nominally distinct (Grinblatt et al., 2012). Our identification targets the contamination that biases the measured relation between the two skills, not the resource structure itself, so the empirical content survives whether or not the resources are strictly independent. Any positive coefficient that survives the correction would reflect such a shared factor, and its magnitude is directly testable here.

The independence prediction concerns variation across traders, not absolute skill levels. Realized profit per contract decomposes into the two components:

$$E[\Pi_i] = \underbrace{(\theta_i - \bar{p})}_{\text{Directional}} + \underbrace{(\bar{p} - p_i^{\text{entry}})}_{\text{Execution}} - \kappa, \quad (1)$$

where $\bar{p}$ is a fair-price benchmark and κ captures transaction costs. Here θ_i is the expected payoff on the side trader i took. Its realized per-trade analog is the terminal payoff on that side (one if the side wins, zero otherwise), and its empirical trader-level counterpart is the fraction of a trader’s resolved trades on the winning side, which we call directional accuracy. The empirical counterpart of ε_i is the wallet-average execution component $(\bar{p} - p_i^{\text{entry}})$, which we call execution edge.

Equation 1 implements the cross-trader test. We call the cross-trader covariance of the directional and execution components the cross-skill covariance. Under a benchmark that does not include the evaluated trade’s own entry price, the two-dimensional view predicts it should be zero. Bundling predicts it to be positive: a single capacity drives both. A structural speed-precision tradeoff predicts it to be negative. Because $\bar{p}$ enters the directional component negatively and the execution component positively, moving the benchmark shifts profit between the components but does not, by itself, induce cross-trader covariance. A clean benchmark adds no term that ties either component to the trader’s own entry, so the covariance they exhibit is an empirical property rather than a mechanical artifact; Proposition 1 characterizes the bias a contaminated benchmark introduces.

2.3 Benchmark Independence as an Identification Requirement

The cross-trader covariance test in Equation 1 requires the benchmark to be independent of the trader. The benchmark that splits profit must not be mechanically affected by the evaluated trader’s own trading, and the directional endpoint must be determined outside the trade sequence. Standard equity performance-attribution settings fail to satisfy both of

these requirements.

The conventional choice in performance-attribution decompositions is a volume-weighted average price (VWAP) constructed from observed trades (Anand et al., 2012; Saglam et al., 2019). Any such benchmark can contain the evaluated wallet’s own prior fills and trades, biasing the cross-skill estimate. Both Hagströmer (2021) and Muravyev and Pearson (2020) make analogous points for measured execution costs: the bid-ask spread is biased upward by midpoint-based benchmark construction in equities, and options effective spreads are similarly overstated. Our contamination falls on the covariance between skill components. The contamination is not specific to VWAP: no widely used equity-market benchmark is constructed independently of the trade flow under analysis.³

Consider a trader who enters a winning contract at \$0.40. The conventional backward VWAP fair price for that contract is \$0.60; the immediately preceding trade is \$0.50. VWAP is a backward average over the trades preceding each evaluated trade, so a trader’s own activity enters it mechanically. An order filled across several trades places its earlier fills in the average that scores the later ones, and a wallet’s earlier trades enter the benchmark of every subsequent one. The lagged price refers only to the single trade immediately preceding the entry, and shares no term with p_i^{entry} .

The VWAP decomposition assigns 40 cents to directional ($1 - 0.60$) and 20 cents to execution ($0.60 - 0.40$). The lagged-price decomposition assigns 50 cents to directional ($1 - 0.50$) and 10 cents to execution ($0.50 - 0.40$). Both attribute the same 60 cents of total profit, but they disagree on the within-trader allocation by $\delta = 10$ cents, shifted from the directional into the execution component under VWAP. Any benchmark gap that varies across wallets biases the measured covariance (Proposition 1). What makes the VWAP gap systematic rather than random noise is its own-trade content: the trader’s prior fills and trades pull the average toward the trader’s own prices, so δ becomes a function of the

³Implementation shortfall (Perold, 1988) uses a pre-trade decision price that reflects prior order flow; closing prices aggregate the day’s trades, including those of the evaluated traders; and factor-model residuals use the asset’s own return as the dependent variable. The contamination channel differs across these benchmarks, but the structural feature is the same.

evaluated trader’s own entries.

The example is a single trade; the bias in cross-trader comparisons comes from the same gap aggregated across a wallet’s trades. Throughout this subsection, θ_w^B and ε_w^B denote the two components of Equation 1 at the wallet level. θ_w^B is the wallet- w average of the directional component $(\theta_i - \bar{p}_B)$. ε_w^B is the wallet- w average of the execution component $(\bar{p}_B - p_i^{\text{entry}})$. Let L be a benchmark that does not share a term with p_i^{entry} and let V denote VWAP. Define $\delta_w = \overline{\text{VWAP}}_w - \bar{L}_w$ as the wallet-level mean gap between the two benchmarks for wallet w , the per-trade δ of the example averaged over w ’s trades. At the wallet level:

$$\theta_w^V = \theta_w^L - \delta_w, \quad \varepsilon_w^V = \varepsilon_w^L + \delta_w. \quad (2)$$

Proposition 1 (Benchmark contamination bias). *If $\text{Cov}(\theta^L, \varepsilon^L) = 0$, δ_w is uncorrelated with θ_w^L and ε_w^L , and $\text{Var}(\delta) > 0$, then*

$$\text{Cov}(\theta^V, \varepsilon^V) = -\text{Var}(\delta) < 0.$$

The VWAP-measured cross-component covariance is biased negative by $-\text{Var}(\delta)$: under orthogonality of the clean-benchmark components, the measured covariance is itself negative.

The result follows from substituting Equation 2 into the covariance: $\text{Cov}(\theta^V, \varepsilon^V) = \text{Cov}(\theta^L - \delta, \varepsilon^L + \delta) = \text{Cov}(\theta^L, \varepsilon^L) + \text{Cov}(\theta^L, \delta) - \text{Cov}(\delta, \varepsilon^L) - \text{Var}(\delta)$. Under the orthogonality and independence conditions, the first three terms vanish; $\text{Var}(\delta) > 0$ makes the covariance strictly negative.

Performance-attribution decompositions in equity markets benchmark execution against the trade record (Anand et al., 2012; Saglam et al., 2019). The proposition implies that such a benchmark contributes a mechanical negative term to the measured cross-skill coefficient, even under orthogonality. A negative measured coefficient therefore cannot, by itself, distinguish true bundling from pure contamination.

The proposition’s second condition, that δ is uncorrelated with θ^L and ε^L , holds only

approximately in our data; we do not claim exact independence. When it fails, the measured covariance is $-\text{Var}(\delta) + \text{Cov}(\theta^L, \delta) - \text{Cov}(\delta, \varepsilon^L)$ rather than $-\text{Var}(\delta)$ exactly. The benchmark-gradient test does not require the residual covariance terms to vanish, only that the contaminated and clean benchmarks order as Equation 2 implies; we verify this empirically using forward-VWAP comparisons and a Monte Carlo benchmark under imposed orthogonality.

Even lagged or pre-trade quotes in equity and bond markets may reflect prior order flow by the traders being evaluated, so the contamination concern is not limited to averaging-based benchmarks. The empirical magnitude scales with $\text{Var}(\delta)$, which depends on the trader’s share of preceding volume: for institutional equity orders that are a small fraction of daily volume the contamination is correspondingly attenuated, while for proprietary HFT order flow that constitutes a meaningful share of trades in a venue the bias can be material.

2.4 The Prediction-Market Test

Prediction-market binary contracts resolve to known terminal payoffs, fixed by the realized outcome and independent of the trade sequence by construction. Moskowitz (2021) exploits the same property in sports-betting markets, treating terminal-payoff resolution as a clean laboratory for asset-pricing tests free of systematic risk; we use it to identify the directional component. The realized analog of θ_i , the terminal payoff on the trader’s side, is directly observable:

$$\underbrace{\text{Total Edge}_t}_{\text{realized profit per contract}} = \underbrace{(\text{Outcome}_m - \text{Fair Price}_{m,t})}_{\text{Directional}} + \underbrace{(\text{Fair Price}_{m,t} - \text{Entry Price}_t)}_{\text{Execution}}, \quad (3)$$

where $\text{Outcome}_m \in \{0, 1\}$ is the asset-level terminal payoff. Equation 3 is written for a buy-side trade; sell-side trades reverse the sign. We drop κ from the empirical decomposition because Polymarket charged no explicit trading fees during the sample period.⁴ The

⁴Section 8 discusses the welfare bound that fees would imply.

endpoint of the directional component is clean by construction: the realized terminal payoff is independent of the entry price.

Both components still depend on the choice of fair-price benchmark. We use the lagged-price benchmark $p_{m,t}^{\text{lag}} = p_{m,t-1}$, where t indexes the trade sequence within market m , as a diagnostic, not as a preferred fair-price estimate. By construction $p_{m,t-1}$ shares no term with $p_{m,t}$, so the lagged-price decomposition carries none of the own-entry contamination that δ measures. The lagged price is a noisy reference for current consensus. Its purpose is to detect residual bias in the execution component, not to recover the level of execution edge.⁵

The bias decomposition in Equation 2 generates a diagnostic that distinguishes mechanical contamination from a genuine cross-skill relation. We call it the *benchmark-gradient test*. If the negative cross-skill coefficient attenuates as the benchmark becomes cleaner and disappears under a benchmark that does not include the evaluated trade’s own entry price, the relation is contamination. A structural tradeoff would not move when only the benchmark changes; bundling predicts the cleaned coefficient remains positive because the underlying skills genuinely co-vary. The identification requirement is not specific to prediction markets: any contract whose terminal payoff is fixed by a realized event and independent of the trade history satisfies it.⁶

2.5 Testable Predictions

The two-dimensional view generates four testable predictions. The empirical tests estimate regression coefficients whose sign and attenuation inherit the covariance predictions in Equations 1 and 2; each test asks whether the data fit bundling or the two-dimensional alternative.

⁵Section 3.5 specifies the three benchmark implementations used in the main tests: backward VWAP as the primary contaminated benchmark, local VWAP as an intermediate, and the lagged price as the diagnostic benchmark. Internet Appendix Section IA.1 adds an independently constructed forward VWAP, skipping the trades nearest entry; Section 4.3 reports it alongside the gradient.

⁶The contamination can also be bounded analytically. Under orthogonality and approximate independence of δ from both skill dimensions, the VWAP regression coefficient reduces to $\hat{\beta}^V \approx -\text{Var}(\delta)/[\text{Var}(\theta^L) + \text{Var}(\delta)]$. Section 4.3 evaluates this bound at the empirical moments of our sample, presents a Monte Carlo benchmark under imposed orthogonality (Internet Appendix Section IA.2), and compares both to the empirical coefficient.

1. **Within-type entry-price gradient.** The sharpest form of bundling implies a negative within-type gradient ($\partial p^{\text{entry}} / \partial \theta_i < 0$): if a directional signal also lets traders enter before prices adjust, more accurate traders pay lower prices. The two-dimensional view predicts no such gradient survives market and price-range composition controls. The test is necessary but not sufficient; Predictions 2 and 3 carry the identification.
2. **Benchmark contamination.** The cross-skill coefficient should attenuate as the benchmark is cleaned: negative under VWAP, smaller under intermediate benchmarks, and zero under benchmarks that do not include the trader’s own entry price. Bundling predicts the cleaned coefficient remains positive. The gradient is the empirical signature of Equation 2.
3. **Execution persistence understated.** Split-sample execution persistence should rise substantially under the lagged-price benchmark relative to VWAP. Pure noise predicts near-zero persistence under both; pure benchmark noise predicts persistence to fall, not rise. Only the contamination mechanism predicts the observed direction.
4. **Cross-setting separability.** If the two skills draw on different resources, orthogonality should replicate in CBOE equity options. The accuracy-profitability inversion should not, because options traders choose strike and expiration as well as direction. Bundling predicts a positive cleaned coefficient in both settings and an inversion in neither. Independence without inversion would tie the inversion to single-margin markets.

Tested together, the four predictions decide between the bundled and two-dimensional views.

Several alternative mechanisms could generate elements of the two-dimensional pattern without requiring independent skill dimensions: momentum chasing, market selection, pure liquidity provision, misclassification, and selection on a single type-level latent capacity. The within-type entry-price test (Prediction 1) discriminates the two-dimensional view from

bundling and from momentum chasing simultaneously, and holding type fixed rules out selection on a type-level latent capacity. Sections 5 and 6 develop further tests that discriminate it from liquidity provision, market selection, and misclassification.

3 Data and Institutional Setting

In this section, we describe the Polymarket trade-event record and analysis sample, classify wallets behaviorally, define the realized-payoff decomposition and its three benchmarks, and specify the empirical design.

3.1 Institutional Setting

Polymarket is a prediction market built on the Polygon blockchain. Users trade binary contracts denominated in USDC (a stablecoin pegged to \$1), with each contract resolving to either \$0 or \$1 based on the outcome of a specified real-world event. Multi-outcome events (such as elections with several candidates) are implemented as bundles of related binary contracts, one per outcome, settled through the NegRisk Conditional Token Framework.

The platform operates a central limit order book through smart contracts. Unlike sports betting platforms where a bookmaker sets prices and extracts a house edge, Polymarket is a peer-to-peer exchange: every dollar won by one trader is lost by another, making the market strictly zero-sum. Every matched trade records an identifiable maker and taker wallet, so order roles are observable at the trade level. The platform charged no explicit trading fees during our sample period; operator revenue comes from sources outside the matching engine and does not affect trader P&L.

Markets span diverse event categories: politics and current affairs (about 45% of volume), sports (25%, across more than 400,000 individual markets), cryptocurrency price markets (15%); the remainder splits among entertainment, economics, science, and other categories. Prior work studies whether prediction-market prices are efficient (Forsythe, Nelson, Neu-

mann, and Wright, 1992; Wolfers and Zitzewitz, 2004; Snowberg, Wolfers, and Zitzewitz, 2013); we study which skills profit, and why.

Four features make Polymarket well-suited to test the two-dimensional framework. The on-chain record covers every trade by every wallet, eliminating the self-selection and survivorship concerns inherent in single-broker samples. The zero-sum central limit order book rules out platform-paid maker rebates and platform-side fees as a source of trader profitability: makers and takers earn or pay only the bid-ask spread. This design distinguishes Polymarket from the automated-market-maker venues studied in much of the decentralized-exchange microstructure literature (Lehar and Parlour, 2025; Capponi and Jia, 2025). Binary terminal payoffs let us measure directional accuracy directly against realized outcomes. Finally, the sequenced trade record makes the price of the trade immediately preceding each entry available as a benchmark that does not include the trader’s own entry price, satisfying the second identification requirement.

Two features support generalization beyond this venue. The sample spans several distinct event regimes (pre-election, election, post-election, and late-sample periods) across which the results hold (Internet Appendix Table IA.1). The Polymarket setting is therefore the cleanest place to identify the decomposition, not the boundary of where it applies.

3.2 Sample Construction

When an order is matched, the smart contract emits an *OrderFilled* event containing the maker address, taker address, asset IDs, fill amounts, and fee. We collect all such events from Polygon, spanning November 2022 through February 2026. The raw dataset contains 394.6 million *OrderFilled* events. Because a single matched trade can generate events on both the maker and taker sides, we deduplicate to the trade level by retaining maker-side events. After mapping tokens to markets, the deduplicated record contains 233 million processed trades by 1,981,408 distinct wallets (Table 1).

We obtain resolution data for 556,632 markets, covering 95.3% of processed trades; the

remaining markets are unresolved or canceled, and we exclude them.⁷ Of the 233 million processed trades, 222 million occur in resolved markets. The 1,945,255 wallets that traded in at least one resolved market form our analysis sample; the remaining 36,153 wallets traded only in unresolved markets.⁸ Because each matched trade involves two wallets (one maker, one taker), the resolved trades yield 444 million wallet-trade observations.

Trade-level analysis is the natural choice given the data structure: each *OrderFilled* event represents a single matched trade, and positions are constructed objects derived from trade sequences. We report wallet-level aggregates computed from trade-level observations. The position-level alternative (aggregating to wallet-market net positions) yields identical type-level totals and reproduces the within-type accuracy-edge relationship in sign and magnitude (Internet Appendix Table IA.2). The trade-level unit is required to separate execution edge at entry from outcome realization, a separation that position-level aggregation collapses by construction.

3.3 Trader Classification

To test whether accuracy and profitability align across trader types, we classify wallets by observable behavioral characteristics: trading frequency, volume, diversification (the Herfindahl-Hirschman index, HHI, of a wallet’s volume across markets), and activity span. Unlike approaches that sort users by ex-post realized P&L, our classification is fixed by behavior observable before outcomes are realized. Our behavioral classification complements data-driven approaches such as [Bogousslavsky et al. \(2024\)](#), who train machine learning clas-

⁷[Sirolly, Ma, Kanoria, and Sethi \(2025\)](#) estimate that approximately 25% of Polymarket share volume reflects wash trading, concentrated at near-zero prices. The volume shares in Table 1 are not wash-adjusted, but our results are dollar-weighted and P&L-based, and wash trades generate near-zero P&L by construction, so the analysis is unaffected; Internet Appendix Section IA.4 provides details.

⁸The unit of observation is the wallet, not the person or firm behind it; a single entity may operate several wallets. Wallet splitting leaves aggregate type-level P&L unchanged, because dollars sum across an entity’s wallets, and for wallet-level regressions it adds measurement noise that attenuates coefficients toward zero. The strong own-skill persistence estimated under the identical wallet unit (Section 4.4) shows that wallet-level measures retain substantial signal, so splitting noise cannot explain a near-zero cross-skill coefficient alongside strong own-skill persistence. Entity-level aggregation through funding-address linkage is left for future work.

sifiers on identified informed trades. The on-chain record enables direct classification from observable activity, without labeled training data or the trade-signing algorithms used to identify retail order flow in equity markets (Boehmer, Jones, Zhang, and Zhang, 2021). The approach also captures the heterogeneity within retail populations documented by Eaton, Green, Roseman, and Wu (2022).

We classify wallets sequentially into five categories: bots, sophisticated traders, active retail, casual traders (the residual), and one-shot wallets.⁹ Bots are defined by trading frequency; sophisticated traders by volume, diversification, and activity span; active retail by trade count.¹⁰ Table 1 reports the exact thresholds, chosen at natural breaks in the empirical distributions of trading frequency and volume; rules are applied in order so that earlier categories take priority. Section 6 verifies that all main results replicate under data-driven k -means clustering, which requires no researcher-chosen thresholds, and under out-of-sample classification based on each wallet’s first 30 days of activity.

Activity concentrates sharply. Bots are 3.8% of wallets but trade 76.8% of volume (\$23.9 billion), while active retail is two-thirds of wallets (66.7%) but an eighth of volume (12.7%). Diversification falls monotonically from bots (median HHI 0.087) to casual and one-shot wallets (0.850 and above), with sophisticated and active retail in between. Table 1, Panel B reports the full breakdown by type.¹¹

[Insert Table 1 here]

⁹We use “bot” as shorthand for wallets whose trading frequency, diversification, and activity span are consistent with automation. The classification is behavioral, not based on platform-assigned identifiers.

¹⁰One-shot traders (single trade) may exhibit selection bias: traders who win on their first trade may exit satisfied, while losers may continue trading and thus be reclassified. One-shot directional accuracy is 53.4%, above the 50% mechanical rate, consistent with selective exit. We report one-shot results separately and exclude them from main specifications.

¹¹Bot profits are concentrated but not dominated by individual wallets; the inversion persists when the top 10 bots by P&L are excluded (Table 2, Panel C).

3.4 P&L Computation

For each trade on a resolved market, we compute realized profit and loss (P&L) at the trade level:

$$\text{P\&L}_t = s_t \times (\text{Outcome}_m \times Q_t - C_t)$$

where t indexes the trade and m the market in which it executes, $\text{Outcome}_m \in \{0, 1\}$ is the binary terminal payoff per contract, Q_t is the contract quantity transacted, C_t is the USDC cost, and $s_t = +1$ for buy-side trades and -1 for sell-side trades. A buy-side trade on a winning contract earns $Q_t - C_t$; on a losing contract, $-C_t$. A sell-side trade reverses the signs: the trader receives C_t at execution and either pays Q_t on a winning contract (earning $C_t - Q_t$) or pays nothing on a losing contract (earning C_t). Trades are matched to terminal payoffs regardless of how long the position is held.

Because sell trades can also reflect inventory reductions or profit-taking rather than a direct opposite-side forecast, our accuracy measure captures whether the trade’s marginal exposure ended on the winning side, not what the trader believed. With no explicit trading fees, the implicit cost of crossing the spread is the only transaction cost, absorbed into the execution component of the decomposition. Variable definitions appear in Appendix A, Table A1.

3.5 Decomposition Implementation

For each trade t in market m , we implement the decomposition in Equation 3. $\text{Outcome}_m \in \{0, 1\}$ is the realized terminal payoff, Entry Price_t is the price the trader paid for the contract, and $\text{Fair Price}_{m,t}$ is the within-market consensus benchmark evaluated at trade t , defined below. Total Edge is the realized profit per contract in dollars (range \$-1 to \$+1), reported throughout the paper in cents. The mechanics are illustrated by the \$0.40 entry example in Section 2.3; the decomposition holds with sign reversal for losing contracts ($\text{Outcome}_m = 0$) and for sell-side trades.

We use three fair-price benchmarks.¹² Let $p_{m,j}$ and $v_{m,j}$ denote the price and USDC volume of the j -th buy-side trade in market m , let t index the trade being decomposed, and let $b(t)$ denote the number of buy-side trades preceding it. The three benchmarks are the *backward VWAP* (our primary), the *local VWAP*, and the *lagged price*:

$$\text{VWAP}_{m,t}^{\text{back}} = \frac{\sum_{j \leq b(t)} p_{m,j} v_{m,j}}{\sum_{j \leq b(t)} v_{m,j}}, \quad \text{VWAP}_{m,t}^{\text{local}} = \frac{\sum_{j=b(t)-49}^{b(t)} p_{m,j} v_{m,j}}{\sum_{j=b(t)-49}^{b(t)} v_{m,j}}, \quad p_{m,t}^{\text{lag}} = q_{m,t-1}, \quad (4)$$

where $q_{m,t-1}$ is the price of the trade immediately preceding t , regardless of side. The backward VWAP averages over all buy-side trades strictly preceding trade t , so it carries no lookahead bias; the local VWAP restricts the average to the 50 most recent of those trades. Both VWAPs can therefore contain the evaluated wallet’s own prior fills and trades in the market; the lagged price, a single adjacent trade, minimizes this own-trade content, and the forward VWAP in the footnote above removes the adjacent trades entirely.¹³ Trades with no preceding history in their market lack a benchmark and are excluded for that benchmark.

The VWAP benchmarks are alternative point estimates of fair price; the lagged price is a diagnostic rather than a fair-price estimate, whose own-entry contamination δ in Equation 2 is zero by construction. Cross-trader variation in the directional component primarily reflects *market selection* (which markets a wallet traded in); variation in the execution component reflects *price selection* (at what prices within those markets).

¹²Internet Appendix Section IA.1 introduces a fourth benchmark, the *forward-VWAP-skip-5*, defined as the volume-weighted average price of trades at positions $[t+5, t+55)$ within market m . This forward-looking construction is independent of both the backward-VWAP construction, which shares the contemporary trade, and the lagged-price construction, which uses only the immediately prior trade. Section 4.3 uses it as additional defense against the concern that the lagged-price diagnostic mechanically captures bid-ask bounce or maker-taker role stickiness.

¹³The lagged price does not fully eliminate own-wallet exposure: in 72% of resolved trades the trade at $t-1$ involves the evaluated wallet as either maker or taker, primarily reflecting order-book chunking where a single posted limit order is filled across multiple trade events as incoming market orders arrive sequentially (self-trades are negligible, 785 of 237.75 million resolved trades). The channel’s effect on the cross-skill regression is economically negligible: restricting to wallets whose lagged observations are predominantly from different counterparties (same-wallet share < 0.10) yields a cross-skill coefficient of $+0.001$, indistinguishable from the full-sample result, and a $\theta \times \text{share}$ interaction in the unrestricted regression has coefficient $+0.003$ ($t = 4.4$). Internet Appendix Section IA.6 reports the diagnostic.

3.6 Summary Statistics

Table 1, Panel C reports the wallet-level summary statistics; five facts stand out. First, mean accuracy is 50.3%, essentially a coin flip. Excess accuracy, which nets out the accuracy already implied by the price paid, averages -20.1 percentage points, and three-quarters of wallets fail to outperform always picking the favorite. Section 3.7 gives the formal $\max(p, 1-p)$ definition. Second, mean entry prices are \$0.39, well below the \$0.50 midpoint, indicating that wallets disproportionately buy cheap (longshot) tokens. This unconditional longshot tilt is distinct from the entry price conditional on the resolved outcome. The elevated prices paid on winning tokens reflect late arrival in markets that resolved favorably, not a preference for high-probability favorites.

Third, the median maker fraction, the share of a wallet’s trading executed as maker, is zero: the median wallet posts no limit orders. Maker activity is not rare, however; it concentrates among the high-volume bots that supply most of the liquidity. Fourth, mean wallet P&L is near zero by zero-sum construction, but mean wallet ROI is -5.6% : for every \$100 traded, the average wallet loses about \$5.60. The wallet-level P&L distribution is heavily skewed: most wallets lose modest amounts, while a small subset of wallets wins large amounts. Fifth, wallet-level total edge averages -1.6 cents per contract, decomposed into -1.3 cents directional and -0.3 cents execution. The typical loss comes mostly from picking the wrong side of the market, with a smaller contribution from paying too much at entry.

3.7 Empirical Design

We test the framework’s predictions using two designs. A within-type design compares wallets in the same trader category, ruling out type-driven differences as the source of any cross-skill correlation. A split-sample persistence design distinguishes stable trader characteristics from transient noise. Within a trader category, the identifying variation is direct: two wallets of the same type achieve different realized accuracy and entry prices on the same

population of resolved markets.¹⁴

The principal specification regresses execution edge on *excess accuracy*. For a trade at price p , the most likely outcome is the one priced above 0.50; an unskilled trader who always picks the favorite is correct $\max(p, 1 - p)$ of the time. For each resolved trade, let *Correct* be a 0/1 variable equal to 1 if the trader bought a token that resolved to \$1 or sold a token that resolved to \$0, and 0 otherwise. Excess accuracy at the trade level is then

$$\text{Excess Accuracy} = \text{Correct} - \max(p, 1 - p), \quad (5)$$

and the wallet-level measure is the average across the trader’s resolved trades. This measure removes the mechanical link between raw accuracy and price levels; Section 4.1 gives the worked example. For each type $\tau \in \{\text{Bot}, \text{Sophisticated}, \text{Active Retail}\}$ we estimate the wallet-level regression¹⁵

$$\text{Execution Edge}_w = \alpha_\tau + \beta_\tau \cdot \text{Excess Accuracy}_w + u_w. \quad (6)$$

As a baseline, we also estimate the entry-price specification on raw accuracy (where Entry Price_w is wallet w ’s volume-weighted average entry price across resolved trades):

$$\text{Entry Price}_w = \alpha_\tau + \beta_\tau \cdot \text{Accuracy}_w + u_w. \quad (7)$$

Section 4.2 reports both specifications by trader type.

For each wallet with at least 50 resolved trades, we split trades chronologically into first and second halves, compute $(\theta_1, \varepsilon_1)$ on the first half and $(\theta_2, \varepsilon_2)$ on the second half, and

¹⁴Standard errors are heteroskedasticity-robust (HC1) at the wallet level for wallet-level specifications and clustered at the wallet level for trade-level specifications. Results are qualitatively unchanged under market-level clustering.

¹⁵Casual wallets (at most nine resolved trades each) and one-shot wallets (exactly one trade) do not support stable wallet-level estimates of both skills; they enter the type-level decomposition and the pooled benchmark tests (Section 4.3).

estimate the within-skill persistence regressions

$$\theta_{w,2} = \rho_{\theta} \cdot \theta_{w,1} + \nu_w, \quad \varepsilon_{w,2} = \rho_{\varepsilon} \cdot \varepsilon_{w,1} + \omega_w, \quad (8)$$

together with the cross-skill regressions $\theta_1 \rightarrow \varepsilon_2$ and $\varepsilon_1 \rightarrow \theta_2$. The cross-skill specifications are estimated under both the VWAP and lagged-price benchmarks; the lagged price is free of own-entry contamination by construction, so any surviving coefficient does not reflect that contamination. To test the benchmark-contamination prediction (Prediction 2), we replicate the cross-skill regression under each benchmark in Section 3.5 and examine whether the coefficient attenuates toward zero. To test the persistence prediction (Prediction 3), we compare execution-persistence estimates across benchmarks and examine whether the clean-benchmark estimate exceeds the VWAP estimate. Section 4.3 reports the benchmark-gradient results; Section 4.4 reports the persistence and within-wallet cross-skill estimates.

4 Empirical Results

This section tests whether execution rather than forecasting determines profit, evaluating the first three predictions of Section 2.5. The starting point is descriptive: profitability inverts across trader types, with the only profitable type picking no better than a coin flip. This alone does not settle the question, as a single underlying skill held in different amounts could produce the same pattern.

The first test compares wallets of the same type and asks whether more accurate traders enter at better prices, as the bundling view requires. They do not, once market and price-range composition are absorbed. The second test holds the sample and regression fixed and varies only the benchmark, asking whether the cross-skill coefficient survives a clean benchmark or collapses toward zero. It collapses to zero. The third test follows individual wallets over time, asking whether a trader’s forecasting predicts their own later execution. It does not. Three independent tests converge on the same answer: the skills are distinct,

and execution rather than forecasting determines who profits. Cross-setting replication in equity options is in Section 7.

4.1 The Inversion and the Return Decomposition

Directional accuracy is the fraction of resolved trades where the trader bought a token that resolved to \$1 or sold a token that resolved to \$0; under no forecasting skill, expected accuracy is 50%. ROI is total realized P&L divided by total USDC invested. If accuracy and profitability were driven by a single underlying skill, the most accurate traders would also be the most profitable. They are not. The clearest contrast is between the two types that trade the same markets in volume. Table 2, Panel A reports accuracy and returns by type, trade-weighted, and Figure 1 plots the five types on the accuracy-ROI plane. Bots achieve 49.9% directional accuracy, a shade below a coin flip, yet earn the only positive aggregate return in the sample, \$133 million. Active retail achieves 51.3% accuracy and loses \$79 million. The more accurate of the two loses. The other three types also lose, but for reasons unrelated to this contrast, so the inversion is properly read as a bot-versus-active-retail comparison.¹⁶

The inversion is not driven by a few large wallets. Bot profits are concentrated, with the top 10 wallets earning roughly 70% of aggregate bot profit and only a third of bot wallets individually profitable, but Table 2, Panel C shows the inversion holds when those 10 are excluded. The within-type and persistence tests that follow are computed at the wallet level, where this concentration does not enter.

[Insert Table 2 here]

[Insert Figure 1 here]

No trader type forecasts better than prices. Raw accuracy above 50% need not reflect forecasting skill: a trader who buys favorites at \$0.80 is correct 80% of the time but earns

¹⁶Casual traders post below-random accuracy because they overweight longshots, the favorite-longshot bias of [Snowberg and Wolfers \(2010\)](#); sophisticated traders' below-random accuracy reflects market selection rather than longshot exposure. Their losses do not bear on the bot-retail inversion.

zero excess accuracy, since the price already implies the outcome.¹⁷ Table 2, Panel B reports excess accuracy against the price-implied benchmark: it is negative for every type, and the negative reading is broad across the wallet distribution rather than driven by a few extremes. The result holds in the competitive price range where prices are least informative and under a chosen-side benchmark that scores accuracy against the price the trader paid.¹⁸ The cross-type differences in raw accuracy are therefore not differences in forecasting ability relative to market prices.

Bots are the only type without a negative value on either component. We decompose each trade’s total edge into a directional component (outcome minus fair price) and an execution component (fair price minus entry price), using the backward cumulative VWAP as the benchmark; Table 2, Panel A and Figure 2 report the decomposition. On a trade-weighted basis, the relevant weighting for who captures the dollar profits, bots are the only type with positive total edge, +\$0.002 per token.¹⁹ Sophisticated and casual traders execute adequately but lose on direction; one-shot wallets lose on both. Neither bots nor active retail beats the price-implied accuracy benchmark (Table 2, Panel B), so what separates the profitable type from the more accurate but unprofitable one is not forecasting but the price at which each enters. Which component gets the credit, directional or execution, depends on the benchmark; what does not depend on the benchmark is that no type profits by forecasting. Bots profit not by excelling on the forecasting margin but by avoiding the adverse entries that sink the other types.

[Insert Figure 2 here]

¹⁷We use prices as tradeable payoff benchmarks, not structural probabilities, which would require assumptions about beliefs and preferences (Manski, 2006; Wolfers and Zitzewitz, 2004).

¹⁸Under the chosen-side benchmark, chosen-side excess accuracy is $\text{Correct}_i - p_i$ for a buyer and $\text{Correct}_i - (1 - p_i)$ for a seller, equivalent to asking whether traders identify mispricing relative to the prices they paid. It is negative for every type (−0.004 for bots, −0.008 for active retail). Bot takers are near zero (−0.001); bot makers are modestly negative (−0.010), reflecting adverse selection in liquidity provision. Sorting traders by realized performance and then measuring accuracy would mechanically select for traders who were both correct and well-timed; the ex-ante behavioral classification (Section 3.3) avoids this. Full breakdowns and the competitive-zone results are in Internet Appendix Tables IA.3 and IA.4.

¹⁹The equal-weighted average across bot wallets is negative; profits concentrate in the high-volume wallets noted above, which carry most of the trading and so dominate the trade-weighted aggregate. Table 2 reports both weightings.

This cross-type pattern cannot by itself distinguish the bundled and two-dimensional views. The directional component depends on which market a trader chose and which side of that market they took, so variation in it across types reflects market and side selection rather than within-market forecasting precision. Type-level differences could reflect different amounts of a single underlying skill rather than independent variation in two distinct skills. The next three subsections supply the discriminating variation: within type, across benchmarks, and within wallet over time.

4.2 The Within-Type Entry-Price Gradient

We compare wallets classified as the same trader type and ask whether more accurate wallets pay higher or lower entry prices. Bundling predicts a robust negative gradient; the two-dimensional view predicts the gradient should not survive once market and price-range composition are absorbed (Prediction 1). Table 3 reports three specifications: the contaminated execution-edge regression on excess accuracy (columns 1–3), the clean wallet-level entry-price regression on accuracy (columns 4–9), and the trade-level entry-price regression with market fixed effects (columns 10–12).

[Insert Table 3 here]

The contaminated benchmark cannot discriminate between the two views. Table 3, columns (1)–(3) regress execution edge on excess accuracy under the backward VWAP benchmark; the coefficients are negative and significant at the 1% level for all three types. Under bundling, the negative sign would confirm that information drives both accuracy and execution; under the two-dimensional view, the benchmark shares a price term with both components (the contamination in Equation 2), which mechanically biases the coefficient negative even when the underlying skills are uncorrelated. Both views predict a negative coefficient; the contaminated specification cannot tell them apart. The clean specifications that follow drop the benchmark from the regression altogether.

The bundled negative gradient does not survive at the wallet level. Table 3, columns (4)–(6) regress mean entry price directly on accuracy. The gradient is positive for bots and sophisticated traders ($\hat{\beta} = 0.190$ and 0.092): more accurate wallets pay *higher*, not lower, entry prices, the opposite of what bundling predicts. Columns (7)–(9) add controls and leave the sign unchanged (0.090 and 0.074), with the bot coefficient attenuating to marginal significance once log volume, log trade count, and maker fraction are included. Active retail returns a negative coefficient (-0.166 , attenuated to -0.068 with controls), but this reflects price-range composition across markets rather than within-market bundling.

Holding the market constant separates within-market timing from cross-market selection. Table 3, columns (10)–(12) estimate the trade-level regression with market fixed effects. The bot coefficient remains positive ($\hat{\beta} = 0.021$): within the same market, more accurate bots still pay higher prices. The sophisticated coefficient turns negative (-0.067), indicating that their wallet-level positive gradient reflects cross-market selection rather than within-market timing. The active retail coefficient attenuates to -0.057 , well below the strong negative bundling requires.²⁰ No type retains the robust negative gradient bundling requires. Cleaned coefficients are economically near zero rather than robustly negative; this is what the two-dimensional view predicts, not evidence for an opposite-signed tradeoff.

Momentum chasing cannot explain the pattern that survives. A pure momentum-chasing story would raise entry prices independently of a trader’s own accuracy, predicting no accuracy-entry-price relation; late arrival predicts a positive one, because traders who are right but late buy after prices have moved toward the outcome.²¹ The data show the positive relation where late arrival predicts it. The maker-side trade-level specification is positive and

²⁰Adding market-by-price-decile fixed effects shrinks the entry-price gradient toward zero for all three types, leaving the strong negative reading of the raw specification as a price-range-composition effect rather than a structural negative. At the wallet-market-cell level the bot coefficient remains positive ($+0.003$ to $+0.007$ across cell-inclusion thresholds), while the sophisticated and active-retail coefficients lie within a few thousandths of zero, economically negligible against the raw retail coefficient of -0.166 ; the sign of these near-zero estimates is sensitive to the cell-inclusion threshold. The composition-absorbed gradient is in every case far from the strong negative the bundled view requires.

²¹Retail losses superficially resemble trend-chasing losses, but the composition-adjusted results indicate that the apparent accuracy-entry-price relation is largely driven by market and price-range composition rather than a stable tendency for more accurate traders to chase prices.

significant for all types, with the bot coefficient at $\hat{\beta} = 0.673$. Accuracy-quintile sorts within type show bots with a positive gradient across the lower four quintiles: \$0.52 at Q1 versus \$0.78 at Q4 for winning tokens, with a slight inflection at Q5 (\$0.67) that reflects the small sample in the top quintile. Internet Appendix Tables [IA.5](#) and [IA.6](#) report the full results.

4.3 The Benchmark Contamination Gradient

Under the contaminated benchmark both views predict the same negative coefficient, so its level cannot discriminate between them. Varying the benchmark can: a genuine tradeoff would not move when only the reference price changes, while contamination would attenuate as the benchmark is cleaned. We hold the regression fixed and vary only the benchmark. Bundling predicts a positive cleaned coefficient; the two-dimensional view predicts convergence to zero (Prediction 2). Table [4](#) reports the cross-skill regression under three benchmarks drawn from the same backward price record, ordered by decreasing content of the evaluated wallet’s own trades (backward cumulative VWAP, local 50-trade VWAP, lagged price), plus an independently constructed forward-VWAP-skip-5 as a further check.

[Insert Table [4](#) here]

The cross-skill coefficient attenuates toward zero as the benchmark is cleaned. Table [4](#), Panel B regresses the execution component on the directional component with trader-type fixed effects: across the three benchmarks of increasing cleanliness, the coefficient falls from $\hat{\beta} = -0.178$ ($R^2 = 0.179$) under backward VWAP to -0.013 under local VWAP and 0.000 ($R^2 = 0.001$) under the lagged-price diagnostic. Panel A confirms the execution hierarchy is preserved throughout: bots sit at or near the top under every benchmark. The independently constructed forward-VWAP-skip-5 delivers the same near-zero cross-skill relation ($+0.019$, $R^2 = 0.007$; Table [4](#), column 4, with the full treatment in Internet Appendix Section [IA.1](#)). The clean benchmarks recover the near-zero coefficient the two-dimensional view predicts.

The bias mechanism reproduces the result quantitatively. Evaluated at the full-sample

moments ($\text{Var}(\theta^L) = 0.0174$, $\text{Var}(\delta) = 0.0037$), the analytical bound of Section 2.4 is -0.176 under strict δ -independence, within one percent of the empirical -0.178 ; the measured covariances between δ and the clean components are negligible, about one percent of $\text{Var}(\delta)$. A Monte Carlo makes the same point from the other side. We simulate traders whose forecasting and execution are independent by construction; the VWAP regression still reproduces 99% of the empirical coefficient on the higher-activity subsample, where the benchmark distortion is largest relative to forecasting (Internet Appendix Section IA.2). The negative number appears even when, by design, no real relation exists. The empirical gradient is not merely consistent with the bias mechanism; it is what the mechanism quantitatively predicts.

The convergence operates within type as well as in the total sample. Under VWAP, every main-type cross-skill coefficient is positive ($+0.089$ for bots, $+0.237$ for sophisticated, $+0.158$ for active retail); the pooled -0.178 is negative only because it also weights casual and one-shot wallets into the regression. Under the lagged-price benchmark, every by-type coefficient is zero (-0.002 for bots, 0.000 for sophisticated, 0.000 for active retail). Cleaning the benchmark drives both to zero from opposite directions: the pooled coefficient rises from -0.178 up to zero, and the within-type coefficients fall from positive values down to zero. The convergence is not driven by pooling; it operates within each type.²²

The data match the two-dimensional prediction. The cleaned cross-skill coefficient converges to zero rather than to the positive value bundling requires. The gradient from -0.178 under VWAP to 0.000 under the clean benchmark is the empirical signature of the bias mechanism in Equation 2. If a single capacity delivered both forecasting and execution, the cleaned correlation would be positive regardless of the benchmark. Applied to our data, the conventional VWAP decomposition (Saglam et al., 2019; Anand et al., 2012) produces this

²²Within-half regressions, $n_{\text{resolved}} \geq 50$; sample as in Table 5. The pooled -0.178 is the full-sample estimate of Table 4; the by-type coefficients are computed on the restricted sample, where each wallet has enough trades to estimate both components in each half. The within-type VWAP coefficients are positive rather than the negative the analytical bound predicts because δ_w is not approximately independent of θ^L within type; the bound holds in the pooled specification, where the cross-covariances are small relative to $\text{Var}(\delta)$. Prediction 2 requires the gradient toward zero under the cleaner benchmark, which holds at both levels of aggregation.

negative coefficient and would read it as a genuine tradeoff, as if sharper forecasting came at the cost of worse execution. It does not. The negative sign appears only because the VWAP benchmark is built partly from the trader’s own trades, so it enters both skill measures and pushes them apart; remove that overlap from the benchmark and the coefficient goes to zero. The pattern is robust to specification: the raw Pearson correlation, with no type controls at all, falls from 0.40 under VWAP to 0.03 under the lagged-price benchmark, the same attenuation the type-FE regression captures.²³ Figure 3 visualizes the gradient: Panel A plots execution against accuracy under both benchmarks; Panel B confirms cross-skill independence out-of-sample via the split-sample rank-rank scatter. Internet Appendix Section IA.11 reconciles every cross-skill estimate in the paper.

[Insert Figure 3 here]

4.4 Execution Persistence Under Clean Benchmarks

The benchmark gradient of the preceding subsection establishes independence across wallets at a single point in time, but a cross-section leaves two openings. First, a near-zero cross-sectional correlation can be measurement error: noisy wallet-level estimates of forecasting and execution attenuate their correlation toward zero, so the zero could mask a genuine positive relation. Second, a cross-section cannot show that either skill is a durable trait of the individual trader rather than a one-period draw; if execution quality does not repeat, it cannot be what determines who profits. Following the same wallet across its first and second halves closes both. Whether each skill predicts itself reveals whether it is a stable trait and whether the wallet-level measures carry signal at all; whether first-half forecasting predicts second-half execution tests independence within the trader, free of cross-wallet noise. Bundling predicts that first-half forecasting should predict second-half execution; the two-

²³The Pearson is positive while the regression with type fixed effects is negative because the two measures isolate different variation: the Pearson captures cross-type variation in θ and ε alongside within-type variation, while the regression with type fixed effects isolates the within-type relationship. Both measures collapse to near-zero under the clean benchmark, which is what the two-dimensional view requires.

dimensional view predicts the cross-skill coefficient should be zero (Prediction 3).

The design also tests whether VWAP understates execution persistence. The contamination in Equation 2 mechanically depresses execution persistence measured under VWAP, so the clean-benchmark estimate should be substantially higher; if execution were pure noise, persistence would be near zero under both benchmarks, and if the lagged price merely added noise, the clean-benchmark estimate should be lower, not higher. We split each wallet’s resolved trades chronologically at the midpoint and estimate own-skill and cross-skill regressions under both benchmarks (Equation 8); the same wallet appears on both sides of every regression, so any time-invariant wallet trait is absorbed by the design.

The conventional benchmark paints execution as close to noise. Table 5, columns 1–4 report the backward VWAP results: directional accuracy is strongly persistent ($\hat{\beta}(\theta_1 \rightarrow \theta_2) = 0.270$), execution is only weakly persistent ($\hat{\beta}(\varepsilon_1 \rightarrow \varepsilon_2) = 0.080$), and first-half accuracy is nearly uninformative about second-half execution. Forecasting reads as a stable trait, execution as noise, and the two as unrelated.

The clean benchmark inverts the picture for execution while leaving cross-skill independence intact. Under the lagged-price benchmark, Table 5, columns 5–8 show execution persistence strengthens to $\hat{\beta}(\varepsilon_1^{\text{lag}} \rightarrow \varepsilon_2^{\text{lag}}) = 0.434$ ($R^2 = 0.217$), while the cross-skill coefficient $\hat{\beta}(\theta_1 \rightarrow \varepsilon_2^{\text{lag}}) = 0.000$ remains indistinguishable from zero; Figure 4 plots the persistence coefficients under both benchmarks. The reverse direction is the same: first-half execution enters with a raw coefficient of 0.302 but explains none of the variance in second-half accuracy ($R^2 < 0.001$), and the standardized coefficient is 0.008. The large raw coefficient reflects scale (lagged-price execution is measured in cents per contract while accuracy is bounded in $[0, 1]$).²⁴

²⁴Two concerns about the lagged-price benchmark as a persistence measure are worth addressing directly. Low variance: lagged-price execution persistence is 5.4 times the VWAP estimate (0.434 vs. 0.080), the opposite of what a low-variance benchmark would produce. Role stickiness: the within-role analysis in Section 5.2 shows the bot-retail execution gap holds within both maker and taker roles separately. The benchmark gradient and the independently constructed forward-VWAP-skip-5 check established in Section 4.3 provide additional defense by showing that the independence finding does not depend on any one benchmark construction. Adding wallet-level maker fraction as a control, or splitting the sample at the median maker fraction, leaves execution persistence and the near-zero cross-skill relation intact (Internet

Resource independence holds within the trader. Within the same wallet, knowing first-half directional accuracy is almost uninformative about second-half execution quality once benchmark contamination is removed: the cross-skill coefficient is 0.000 under the clean benchmark, and the independently constructed forward-VWAP-skip-5 benchmark delivers the same near-zero result (Section 4.3). Under a one-dimensional alternative in which a single capacity drives both skills, first-half forecasting and second-half execution should covary positively because both load on the same underlying trait. They do not. The capacity that produces a wallet’s first-half accuracy is uninformative about the capacity that produces its second-half execution quality, consistent with the two skills drawing on distinct underlying resources.

The conventional benchmark substantially understates how stable execution skill is. The $0.080 \rightarrow 0.434$ gap is a 5.4-fold ratio that prior measurement would have read as transient noise. The bias mechanism reproduces it from independent quantities. Under Equation 2, the persistence you measure under VWAP is a blend of two things: how stable true execution is, and how stable the benchmark distortion is. (Formally, a variance-weighted average of the two, when the distortion is uncorrelated with clean execution across halves.) Feeding that formula the measured quantities (0.434 clean persistence, 0.080 gap persistence, half-sample variances) predicts a VWAP persistence of 0.079; the observed value is 0.080 (Table 5 notes). The match is not a calibration: each input is an independently measured moment, and the variance-weighted attenuation formula was derived analytically in Section 2.4 before any matching.

The cross-skill slope reads the same contamination against the other component: at the measured full-sample moments, the strict-independence bound is -0.176 , within one percent of the empirical -0.178 . One contamination magnitude, read twice from the same mechanism, reproduces the persistence collapse to the third decimal and the cross-skill slope

Appendix Table IA.8). Thin-market staleness: the forward-VWAP-skip-5 is defined only for trades followed by at least 55 further trades in the same market, so the independent-construction check operates precisely where order flow is active and a stale lagged price is least likely.

to within one percent. Execution quality is a durable wallet-level characteristic, and prior measurement has been hiding it.²⁵ A pure-noise account of execution predicts no persistence gap across benchmarks; the empirical $5.4\times$ ratio rejects it.

[Insert Figure 4 here]

The type-level patterns reveal meaningful heterogeneity. Forecasting accuracy is moderately to strongly persistent across all trader types, with bots showing the strongest persistence, consistent with algorithmic strategies producing stable directional patterns. What persists is each wallet’s accuracy *level*, not a forecasting advantage over prices: some wallets are persistently near 52%, others near 48%, but no type achieves accuracy above the price-implied benchmark. Execution quality is more persistent for bots than for human traders, consistent with automated execution advantages, built on stable timing and liquidity provision, being more durable than human ones. Cross-skill regression R^2 remains low even within type: under 1% for human traders’ $\theta_1 \rightarrow \varepsilon_2$, and under 1% in either direction for bots (0.97% for $\theta_1 \rightarrow \varepsilon_2$ and 0.23% for $\varepsilon_1 \rightarrow \theta_2$). At the wallet level, the Pearson correlation between accuracy and ROI is $\hat{\rho} = 0.17$ (under 3% shared variance): knowing how accurate a wallet is says little about whether it profits.

Our first three predictions survive a sweep of the alternatives. Superior forecasting cannot account for the inversion: excess accuracy is negative for every type against both price-implied and chosen-side benchmarks. A single underlying trading skill held in varying amounts cannot either: within-wallet cross-skill regressions show zero relation between first-half forecasting and second-half execution under the clean benchmark. Bundling fails on the same evidence, since the cleaned cross-skill coefficient is zero rather than the positive value bundling requires. Momentum chasing and price-range composition fare no better: chasing

²⁵Subsequent work by [Akey et al. \(2026\)](#) reports that monthly PnL on Polymarket is only weakly persistent at the user-month level. Our split-sample design measures persistence in skill components (θ and ε) at the wallet level under the clean benchmark; the two findings are complementary because monthly dollar PnL aggregates skill components with realized-outcome variance, while our split-sample components separate signal from realization noise.

moves entry prices without regard to accuracy, so it cannot produce the positive accuracy-price relation that survives within market and on the maker side; the raw negative gradient reflects price-range composition, with no type retaining the strong negative bundling requires. What remains is the two-dimensional view: at the margin between the profitable and unprofitable types, execution rather than directional accuracy is the discriminating component, and the bias mechanism in Equation 2 explains why prior decompositions reported a negative tradeoff where none exists. Section 5 identifies the mechanism behind that execution advantage; Section 6 rules out the microstructure and classification alternatives that remain.

5 Mechanism: Timing and Execution

Section 4 establishes that execution, not forecasting, drives profits but does not identify the channel. The surplus that determines profitability is not the gap between any trader’s accuracy and the resolved outcome (no type beats prices); it is the gap between bot entry prices and retail entry prices, captured at the moment of trade. We propose lifecycle timing: as prices converge toward terminal values, the marginal value of a correct directional forecast declines, and the binary terminal payoff bounds the surplus available to a late entrant.

We build the case in four steps. First, profitable and unprofitable types differ in when they enter: bots trade while prices still reveal little of the outcome, retail after prices have moved toward it. Second, the gap is not pure spread capture: bots beat retail within both the maker and taker roles, not merely by making while retail takes. Third, the role decomposition traces bots’ net edge to operating disproportionately as makers, the role where their early entry pays off most. The discriminating test is the accuracy-price-distance interaction, which asks whether a correct forecast loses value as the price approaches its terminal value, and it does.

5.1 Trade Timing and Entry Prices

If the execution gap is a timing phenomenon, profitable and unprofitable types should differ in *when* they enter relative to resolution: late arrivals trade once prices already reveal the outcome, while early entrants trade when prices still reveal little of it. The mean entry price on trades that resolve to \$1 versus \$0 reveals how much the price has already moved at the moment of trade. The winning-minus-losing price gap is 9.8 cents wider for casual traders than for bots (0.455 versus 0.357; Internet Appendix Table [IA.9](#)). Casual traders thus enter at prices that already partially reveal the outcome, while bots enter before prices have moved toward the eventual outcome. This timing pattern is consistent with the attention-driven trading that [Barber, Huang, Odean, and Schwarz \(2022\)](#) document in equity markets.

Bot edge is a lifecycle phenomenon. Edge-at-entry, the realized edge per contract on trades placed at a given distance from resolution, decays monotonically for bots over the market lifecycle, from large and positive more than a year out to approximately zero at 7–14 days and negative at the close ([Figure 5](#)). The monotonic decay is primarily a feature of automated trading, where early entry translates into execution advantage.²⁶ The descriptive pattern is also consistent with an information-arrival channel, but the excess-accuracy results rule it out: excess accuracy is negative for every type and broadly across the wallet distribution, so late arrival does not reflect superior information.

[Insert [Figure 5](#) here]

²⁶For other types the lifecycle pattern is not monotonic ([Internet Appendix Figure IA.1](#) plots all types): early trades by retail and casual traders show large negative edge (adverse selection against the small subset entering earliest), while casual traders show positive edge at 0–7 days reflecting near-resolution trades on contracts whose prices have already moved close to \$0 or \$1, with thin dollar margins that do not offset losses elsewhere in the lifecycle ([Table 2](#)). Within-type accuracy-arrival correlations corroborate the late-arrival reading for retail: the Pearson correlation between wallet accuracy and days-to-resolution at entry is -0.028 for active retail, -0.003 for sophisticated traders, and $+0.005$ for bots (who trade continuously across the lifecycle, so this cross-sectional measure does not capture their timing). These correlations are computed across maker-side trade observations on an early-sample extract of the trade record (4.28M, 257K, and 372K trades for bots, sophisticated, and active retail, respectively), each trade carrying its wallet’s overall accuracy and wallets required to have at least ten resolved maker-side trades; they are descriptive context rather than identifying evidence. The timing reading for bots is supported by the within-type entry-price gradient ([Section 4.2](#)) and the price-distance interaction ([Section 5.4](#)).

5.2 Maker/Taker Decomposition

If bots profit through liquidity provision rather than information, they should earn the spread as makers while retail pays it as takers. Appendix Table A2 reports effective spreads by trader type, and bots do earn it: they are net liquidity suppliers, with a negative effective spread of -4.25 basis points on average, consistent with Menkveld (2013) and Hendershott et al. (2011). Unlike the aggregate retail contrarian flow of Barrot, Kaniel, and Sraer (2016), individual retail traders in our sample pay the spread rather than earning it: every non-bot type other than one-shot wallets posts a positive effective spread, the mirror image of the bots' negative one.²⁷

The sharpest evidence against pure liquidity provision is that the profitability hierarchy survives within each role separately. If bots simply earned the spread, the gap would close once we compare bots to retail within the taker role, where neither side provides liquidity. It does not: Table 6, Panel B shows active retail trails bots on both the maker and taker sides (by 0.68 and 0.73 cents), as do casual traders (by 2.26 and 3.33 cents), with every non-bot type significant at the 1% level in both roles (Internet Appendix Table IA.10). The within-role regressions in Internet Appendix Table IA.5 regress entry price on wallet-level accuracy separately for maker and taker trades within each type. The price-accuracy gradient within the maker role is large and significant ($\hat{\beta} = 0.67$ for bots), meaning that more accurate bots pay higher prices on their maker-side trades. This is the late-arrival pattern operating within the maker role: makers who trade at prices that have already moved toward the outcome appear more accurate but capture less execution surplus. The taker-side gradient is small ($\hat{\beta} = -0.05$), and adding maker fraction as a continuous control in the cross-sectional ROI regression does not alter the accuracy coefficient (Internet Appendix Table IA.11).

Corroborating evidence comes from a different platform: Becker (2026) documents that

²⁷Bot profits concentrate in market-making bots: those with maker fraction at or above the median (39%) earn their surplus on the maker side, while directional bots (below 39%) trade more often as takers and collectively lose money. The highest maker-fraction quartile accounts for the majority of aggregate bot dollar profits despite the lowest mean per-trade ROI, reflecting concentration of dollar profits in high-volume wallets.

Kalshi makers buying YES and makers buying NO earn nearly identical returns (Cohen’s $d \approx 0.02$), consistent with liquidity providers profiting through execution structure rather than forecasting. Subsequent work by [Akey et al. \(2026\)](#) reaches related conclusions: maker volume share is the strongest cross-sectional predictor of profitability, and a meaningful share of the most profitable users’ gains comes from liquidity provision rather than directional bets.

5.3 Directional and Execution Edge by Order Role

The role-level decomposition shows that bot profits arise from role composition, not from either role being unconditionally profitable. Two pieces of evidence point to lifecycle timing as the central channel: edge-at-entry decays monotonically for bots over the market lifecycle, and the accuracy-price-distance interaction shows that the marginal value of directional accuracy collapses as prices approach terminal payoffs. We report here the role decomposition the data identify cleanly.²⁸

[Insert Table 6 here]

Table 6 shows the two roles mirror each other. Bot makers earn their edge through execution and pay it back in directional costs, absorbing the adverse selection that liquidity provision entails; bot takers do the reverse, paying the spread they cross but gaining directional exposure. Each of these bot roles nets near zero on its own, though the maker role ends slightly positive once its offsetting directional and execution components are combined. The execution components are near mirror images, makers earning approximately what takers pay, which is what a neutral book produces. The positive net edge for bots therefore arises not from either role being structurally favorable but from bots operating disproportionately as makers (57% of trades), the role where favorable timing pays off most directly. This pattern is consistent with the theoretical equilibrium in [Baldauf and Mollner \(2022\)](#),

²⁸We do not assign a precise share to the timing channel: partitioning execution into timing and within-period components is sensitive to the time-bucket definition.

in which fast traders profit not from the spread itself but from the timing that lets them supply liquidity while avoiding adverse selection, rather than from directional positioning.

5.4 Interaction Tests: Accuracy and Price Distance

Lifecycle timing predicts that the marginal value of accuracy should decline as prices approach terminal payoffs. We test this by examining how accuracy’s value varies with distance from the uninformative prior. The binary payoff ceiling binds the value of accuracy under either timing or an information channel: a trader who buys at $p = 0.95$ can earn at most 5 cents regardless of accuracy. The interaction gradient alone, therefore, does not separate the two mechanisms. The separation comes from the excess-accuracy null: no type beats the price-implied benchmark in any price range, leaving no residual mispricing for superior information to capture.

We define Price Distance as $|p-0.5|$, the distance of the trade price from the uninformative prior. High price distance indicates that the market has already moved toward a terminal payoff, so that even a correct directional bet earns a thin margin. The unit of observation is the wallet-market pair, weighted by the pair’s trade count; we estimate:

$$\begin{aligned} \text{Edge}_{w,m} = & \alpha + \beta_1 \text{Accuracy}_{w,m} + \beta_2 \text{PriceDist}_{w,m} \\ & + \beta_3 (\text{Accuracy}_{w,m} \times \text{PriceDist}_{w,m}) + u_{w,m} \end{aligned}$$

where $\text{Edge}_{w,m}$ is wallet w ’s total edge in dollars per contract in market m , $\text{Accuracy}_{w,m}$ is its fraction of resolved trades in that market on the winning side, and $\text{PriceDist}_{w,m}$ measures how far its trades in that market lie from the uninformative prior. Standard errors are clustered at the wallet level.

[Insert Table 7 here]

Table 7 reports the interaction regression. Accuracy and price distance work against each other, and the interaction is what the timing channel predicts: a correct forecast is worth

progressively less the further the price has already moved toward its terminal value. The effect is large. A ten-point gain in accuracy is worth nearly 12 cents per contract at the indifference price but about a cent and a half once the market has converged near a payoff, a nearly eightfold collapse across the price range. The interaction coefficient is strongly negative and reorganizes the fit, lifting the R^2 from 0.45 to 0.78, and it is unchanged under market fixed effects and added controls.²⁹

Accuracy is valuable only while prices remain uninformative; once the market has moved, being right confers almost no edge, consistent with timing rather than information. This pattern mirrors the adverse-selection logic of Kyle (1985) and Glosten and Milgrom (1985): as prices converge toward the outcome, the informational edge that pays a directional trader disappears, and execution becomes the only remaining way to earn a surplus. Liquidity-provision rents do not scale with a trader’s forecasting accuracy, so the accuracy-price-distance interaction isolates the decay in forecasting value rather than spread capture.

Taken together, the evidence points to lifecycle timing as the central channel: directly, through the collapse of accuracy’s marginal value near terminal payoffs, and by exclusion, since the bot-retail gap survives within both roles and so cannot be spread capture alone. The timing channel reinforces the resource-independence conclusion of the results section: the infrastructure behind early entry is distinct from the cognitive resources behind directional accuracy. The robustness section evaluates the remaining alternative mechanisms, addresses sample and measurement concerns, and uses the 2024 U.S. presidential election as a natural attention shock.

²⁹The interaction coefficient is $\hat{\beta}_3 = -2.244$ (Table 7, column 3), -2.250 under market fixed effects (column 4) and -2.221 with additional controls (column 5). The implied marginal value of a ten-point accuracy gain, computed as $0.10(\hat{\beta}_1 + \hat{\beta}_3 \text{PriceDist})$ with $\hat{\beta}_1 = 1.162$, is 11.6 cents per contract at $p = 0.50$, 4.9 cents at $p = 0.80$, and 1.5 cents at $p = 0.95$. Internet Appendix Table IA.12 replicates the interaction with excess accuracy in place of raw accuracy, recovering a nearly identical coefficient (-2.145). The level-specification R^2 is high because wallet-market edge is mechanically linked to realized correctness; the wallet-level ROI specifications in Internet Appendix Section IA.18 show accuracy explains less than 5% of cross-sectional profitability.

6 Robustness

The execution-dominance finding rests on three elements: the behavioral wallet classification, the profit decomposition, and the neutrality of the matching engine. We subject it to four sets of checks. First, we address the four alternative explanations for the inversion: private information, market selection, liquidity provision, and misclassification. Second, we address sample and measurement concerns: survivorship, multi-wallet entities, and wash trading. Third, we examine matching-engine neutrality, the one assumption on-chain data cannot fully verify. Finally, we use the 2024 election as an attention shock to test the timing mechanism directly.

6.1 Alternative Explanations

Four channels could in principle generate the patterns we document. Three fail specific tests; the fourth, liquidity provision, is real but cannot account for the gap. Execution timing is the channel consistent with the full evidence chain.

Private information predicts that accurate traders beat the price-implied benchmark $\max(p, 1 - p)$. None do: every type posts negative excess accuracy in every price bin, including the competitive zone where the benchmark is least informative. Raw accuracy above 50% reflects entry into already-informative prices, not forecasting skill.

Market selection predicts the execution gap vanishes once markets are held fixed. It does not. Restricting to the 229,457 markets where both trade leaves the bot-retail execution gap at 0.80 cents per contract, identical to the full sample, and the bot within-type gradient stays positive within market (Table 3, columns 10–12). Market and market-by-week fixed effects with trade-size, maker, and price controls, and a panel regression controlling for wallet-level accuracy, leave the type gaps intact (Internet Appendix Tables IA.14 and IA.15).

Pure liquidity provision predicts the profitability hierarchy disappears within the taker role. It persists in both roles (Internet Appendix Table IA.10), and the lifecycle decay

of edge-at-entry points to timing as a central component of the bot advantage. Liquidity provision matters but is not the channel.

Misclassification predicts the inversion is sensitive to where we draw type boundaries. It is not. The sharpest test discards our threshold rules entirely. An unsupervised k -means partition ($k = 5$) sorts all 1.95 million wallets on five standardized behavioral features (log volume, log trades, HHI, buy ratio, log active span), using no profit, return, or accuracy information. It recovers the inversion intact. Table A3 and Figure A1 in Appendix C report the cluster results: exactly one of the five clusters is profitable, the high-volume, bot-dense cluster sitting at coin-flip accuracy (0.498, +0.27% ROI), while the most accurate clusters lose money (0.526 accuracy, -1.39% ROI). A profit-blind algorithm draws the same line our thresholds do, so the inversion does not depend on our specific classification cutoffs. The within-wallet persistence test and the price-distance interaction hold within each type, ruling out the broader concern that any algorithm identifying early-trading behavior would mechanically recover the surplus.

The inversion survives both threshold-sensitivity and out-of-sample reclassification. Varying the bot and sophisticated thresholds by $\pm 50\%$ (nine specifications; Internet Appendix Table IA.16) and classifying wallets out-of-sample on their first 30 days of activity both preserve the inversion. Bot edge is positive and significant across classification windows from 7 to 180 days. Among wallets with stable type, the apparent non-bot profitability under out-of-sample classification disappears (Internet Appendix Table IA.18). The inversion is not driven by endogenous reclassification.

No alternative explains the full chain; execution timing is consistent with every pattern documented in the mechanism section.

6.2 Sample and Measurement

Survivorship cannot drive the inversion in the direction we document. Retail wallets that lose money are if anything more likely to exit, so conditioning on observed activity selects toward

higher-performing retail. The full-sample finding is therefore conservative: if we observed the trades of wallets that exited early, retail aggregate P&L would be lower, widening the bot-retail gap rather than narrowing it. The wallet-level metrics also include one-shot and casual wallets, the populations most prone to immediate exit, which exhibit the largest per-trade losses and weakest accuracy in Table 2.

Multi-wallet entities only attenuate the inversion. The wallet, not the entity behind it, is the unit of observation. A single person or firm operating multiple wallets would split its activity across rows, adding measurement noise that attenuates cross-wallet coefficients toward zero. The strong own-skill persistence under the clean benchmark in Table 5 ($\hat{\beta}(\varepsilon_1^{\text{lag}} \rightarrow \varepsilon_2^{\text{lag}}) = 0.434$) bounds this attenuation: wallet-level measures retain substantial signal, so wallet-splitting noise cannot explain a near-zero cross-skill coefficient alongside the observed own-execution persistence.

Wash trading cannot account for the bot-retail dollar gap. [Siroly et al. \(2025\)](#) estimate that approximately 25% of Polymarket share volume reflects wash trading, concentrated at near-zero prices, but such trades generate near-zero P&L by construction. The bot-retail gap is denominated in realized dollar profits (+\$133 million versus −\$79 million; Table 2), not in trade counts, so the wash-trading channel cannot be its source. Internet Appendix Section [IA.4](#) provides additional detail.

6.3 Matching-Engine Neutrality

The available evidence is consistent with matching-engine neutrality, though we cannot bound the residual with on-chain data alone. Our analysis treats the matching engine as a neutral limit order book, and the off-chain infrastructure is not publicly documented; latency asymmetries between API and web access could in principle contribute to the bot advantage. The maker-taker decomposition is internally consistent with a neutral book: bot maker earnings (+3.17 cents) are close to bot taker payments (3.97 cents; Table 6). As a comparison among bot trades rather than between bots and retail, it is not decisive on

its own. On Kalshi, with the same order-book design, [Becker \(2026\)](#) finds maker returns indistinguishable across sides (Cohen’s $d \approx 0.02$).

To the extent a latency asymmetry exists, it reinforces the timing mechanism rather than substituting for it: both operate through the speed advantage of automated traders, and both imply policy at the execution margin, not the informational one. The options analysis that follows takes the framework to a regulated exchange, where none of these engine-specific concerns apply.

6.4 The 2024 Election as an Attention Shock

If bots capture surplus from uninformed retail flow, bot execution edge should rise when retail attention surges and recede when it does. The 2024 U.S. presidential election provides a natural attention shock: election-related contracts, identified by keyword matching on market titles, drew unprecedented retail participation during the June through November 2024 window. Bot edge rose with retail attention: it was roughly half again as large on election markets as on the rest of the sample (+0.47 versus +0.31 cents per contract), consistent with a wider pool of retail flow generating more surplus to capture (Internet Appendix Table [IA.19](#)). The casual-trader $\times$ election and one-shot $\times$ election interactions are positive (+2.2 cents and +8.7 cents respectively, both significant at the 1% level), though election markets have higher unit prices, which compress the cents-per-contract scale of retail losses. The comparison cannot separate timing-mechanism effects from bot compositional shifts that may have responded to election volume; we report it as suggestive evidence rather than identification of the timing channel.

7 Cross-Setting Evidence: Equity Options

The two-dimensional decomposition is a structural claim about trading skill, not a Polymarket-specific finding. Prediction 4 requires that the independence of forecasting and execution

replicate across market designs, while the accuracy-profitability inversion should not, because it requires the single-margin binary payoff structure options markets lack. We test both in 14.1 million CBOE equity option contracts from an independent window (options end July 2022; Polymarket begins November 2022). The two settings differ on the design features that matter most for the framework (binary versus continuous payoffs, blockchain versus regulated exchange, one choice margin versus three), so a finding that replicates across both is not a product of either setting. Our cross-setting test complements recent evidence on heterogeneity in option-trader performance ([Hu, Kirilova, Park, and Ryu, 2024](#)).

The data impose one constraint. The benchmark correction needs a trade-by-trade lagged-price benchmark, which requires intraday quotes. The CBOE Open-Close Volume Summary files are daily, so we test separability through split-sample persistence under the contract-level VWAP rather than through the benchmark gradient. [Muravyev and Pearson \(2020\)](#) show conventional options benchmarks misstate execution costs, so this benchmark is known to be contaminated. The contamination works against us: by Proposition 1, a contaminated benchmark pushes measured cross-skill relations toward the negative even when the skills are independent. A near-zero measured relation therefore rules out a substantial negative tradeoff, though it cannot by itself distinguish exact independence from mild positive dependence. Internet Appendix Section [IA.24](#) provides the full analysis.

7.1 Data and Setting

The CBOE Open-Close Volume Summary data aggregate every equity option transaction into daily, contract-level buy and sell volume by exchange-assigned origin code. The code classifies the trading account as one of eight types: Firm (Proprietary), Broker-Dealer, three retail size categories (Small, Medium, Large), and three professional categories (under 100 contracts, 100–199, and 200 or more). The code is assigned from account registration, not inferred from behavior, so it carries none of the endogeneity of a behavioral classification. Internet Appendix Table [IA.20](#) describes the sample, which spans January 2011 through

July 2022 and covers 14.1 million contracts.

We apply the design of Section 3.7 with the contract-level VWAP as the benchmark. The return measure is edge-at-entry, the difference between terminal payoff and entry price, which isolates the entry decision from interim price moves and early exercise.

One adjustment is needed. Option positions carry exposure to the underlying, so types that are net long the market earn the equity risk premium (ERP) mechanically in the directional component. Prediction markets have no analog: terminal payoffs are determined by event outcomes, not by systematic market risk. We remove it by contract demeaning, subtracting the contract-level mean payoff from each trader’s edge. Demeaning strips every common contract-level component (genuine contract-selection skill along with ERP), so the demeaned type comparisons are conservative. A short-dated, near-the-money restriction in Table 8, Panel B independently strips out ERP exposure.

Designated market makers are excluded because their hedged, delta-neutral positions cannot be linked to individual contracts. The separability result holds across all eight non-market-maker types, so the exclusion would matter only if market makers reversed sign, which is unlikely given they trade on infrastructure rather than information.

7.2 Decomposition Results

[Insert Table 8 here]

[Insert Table 9 here]

Table 8 reports the decomposition of edge into directional and execution components by trader type; Table 9 reports the contract-demeaned version that nets out the equity risk premium. Small-scale professionals (Prof. <100) are the structural analog to prediction-market active retail: taker-side traders who pay for execution without the infrastructure to supply liquidity. If independence generalizes, it should appear most clearly here, and it does. Table 9 shows that after contract demeaning, this group’s excess execution is -22.5 cents

and its demeaned directional component is +50.3 cents, both significant at the 1% level and opposite in sign: roughly half the directional advantage is given back through adverse execution. The bundled view predicts the two move together; instead, they are offsetting rather than bundled, in a market structurally unlike prediction markets.

Broker-dealers are an apparent counterexample. Table 8, Panel A shows them at below-average accuracy (31.2%) with the largest positive raw directional edge in the sample (+66.8 cents). This edge is ERP exposure, not forecasting, as the short-ATM subsample confirms: with ERP stripped out, broker-dealer directional edge reverses to -15.9 cents while execution rises to +20.4 cents (Panel B), the bot pattern. The main analysis is conservative because demeaning removes ERP from direction without crediting it back to execution.

We do not construct an excess-accuracy test for options. The naive benchmark $\max(p, 1-p)$ is model-free in prediction markets, but its options analog requires a model of the implied probability surface and assumptions about beliefs and preferences (Manski, 2006; Wolfers and Zitzewitz, 2004). The independence test instead uses the relation between the two components directly.

7.3 Within-Type and Cross-Skill Tests

A clean within-type prediction exists only for the type whose advantage operates on the single entry-price margin against which execution is measured. For multi-margin types (direction, strike, expiration), the sign is ambiguous: accuracy in direction may go with higher or lower entry prices depending on whether it comes from direction-picking or from strike-and-expiration selection. For Prof. <100, the type that maps most directly onto the prediction-market mechanism, the within-type coefficient is $\beta = -0.31$, consistent with the contamination Proposition 1 predicts. The options benchmark, unlike the prediction-market data, cannot be cleaned to confirm that reading (Table 9). Across the eight types the within-type β ranges from -0.31 (Prof. <100) to $+0.31$ (one of the multi-margin types), and the cross-type $\rho(\theta, \varepsilon)$ from -0.16 to $+0.16$, centered near zero despite a contamination

that pushes the within-type readings negative. The within-type and cross-type readings are consistent, not in tension: the within-type regression shares the demeaning reference and inherits the contamination, while the cross-type comparison is less exposed to it.

The cross-skill test in this setting uses split-sample persistence: we split each trader’s window at the midpoint and ask whether first-half forecasting predicts second-half execution. Within-period contamination cannot drive this comparison unless the benchmark gap itself persists across halves (its split-half persistence is 0.08 in the Polymarket data; we cannot measure it in the options data). Cross-skill persistence is small for all eight types, the largest coefficient explaining under half a percent of variance (Internet Appendix Table [IA.21](#)), consistent with the below-one-percent shared variance in prediction markets (Table [5](#)). The test is weaker here than in the resolved-trade panel, since daily data give θ less signal and own- θ persistence is itself low (-0.08 for Prof. <100), so the near-zero coefficient is corroborative rather than decisive.

7.4 Mechanisms and Boundary Conditions

The channel differs across settings, but the two-rent structure holds. In prediction markets it is lifecycle timing: bots enter while prices still reveal little of the eventual outcome and capture discounts that vanish as prices converge. In options it is adverse selection ([Kyle, 1985](#)): market makers price against informed flow at the contract level (Internet Appendix Table [IA.22](#)). The Prof. <100 excess execution reflects this pricing, and its negative within-type coefficient under the contaminated benchmark is the pattern Proposition [1](#) predicts, as in prediction markets. Both channels are instantiations of the execution rent: consistent with distinct resources behind the information and execution rents, the two-rent structure survives even as the delivering channel changes.

Prediction 4 resolves as anticipated. Separability holds in both settings. The inversion does not: sorting by accuracy quintile, the Q5–Q1 spread in total edge is +140.8 cents per share (Internet Appendix Section [IA.27](#)), with more accurate traders more profitable.

The non-replication is structural, because the inversion needs a single binary payoff margin with diminishing returns to accuracy, whereas options span three margins (direction, strike, expiration) so accuracy compounds rather than hitting a ceiling.

The boundary conditions clarify what is general and what is structure-specific. Individual θ persistence is weaker in options, but the independence result is cross-sectional and needs only that the two skills be distinct at a point in time, which the cross-sectional decomposition establishes. Type-level execution rankings are stable, with premium-normalized Spearman correlations of 0.90 to 0.95 across five regime splits (Internet Appendix Table [IA.24](#)). The inversion appears specific to single-margin markets; the independence of the two skills travels, holding in both settings despite different market structures, information environments, and economic mechanisms.

8 Conclusion

Two readings of the data do not survive a clean benchmark. The negative cross-skill correlation that conventional decompositions report vanishes, and the bias we derive accounts for it to within one percent. Execution persistence, which VWAP reads as transient noise, rises more than fivefold once the contamination is removed, and the same bias reproduces the contaminated estimate from independent moments. Execution quality is a durable, wallet-level trait that prior measurement has been hiding, with the bias scaling in each trader's share of preceding volume.

The evidence points to lifecycle timing as the central mechanism. As prices converge toward their terminal values, the value of a correct forecast declines, and a trader who arrives once the information is in the price gives up the execution edge that drives profit. Early entry runs on infrastructure, accurate forecasting on cognition, two inputs acquired separately, which fits the independence we measure. Whether the split extends to richer settings, such as multi-leg options or dynamic portfolio management, remains open; a formal

model with two independent skills is the natural next step.

The result reframes retail protection. Retail loses from late arrival, not from weaker information: its accuracy exceeds bots' on average, and its losses come from the price paid, not the side chosen. Giving active retail the bot-median execution edge, holding sides fixed, cuts the \$79 million loss to about \$23 million, a partial-equilibrium number the independence of the two skills makes well-defined. Rules aimed at the price paid target the margin that matters here, while rules aimed at information asymmetry target a secondary channel; in our fee-free sample the loss is a zero-sum transfer, and fees would make it a lower bound.

The negative cross-skill relations in the performance-attribution literature are measured against own-trade benchmarks that carry this bias. In the one setting where the benchmark can be corrected, the tradeoff disappears, and two independent skills appear where the contaminated reading showed one. What determines who profits is execution, not the directional accuracy that performance evaluation has long rewarded.

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Table 1: Sample Classification and Descriptive Statistics

This table reports the classification of 1,945,255 Polymarket wallets into five behavioral types and their wallet-level descriptive statistics. Panel A reports the classification rules, applied sequentially with earlier rules taking priority. Panel B reports type-level composition: wallet counts, volume shares, the within-type median of total resolved trades per wallet, and the within-type median wallet-level Herfindahl-Hirschman index (HHI) of a wallet’s volume distribution across markets, where $\text{HHI} = 1$ indicates concentration in a single market and $\text{HHI} \rightarrow 0$ indicates uniform diversification. Panel C reports wallet-level descriptive statistics for the 1,944,050 wallets with non-missing values for all variables, equal-weighted across wallets. Accuracy is the fraction of a wallet’s resolved trades on the winning side; excess accuracy is observed accuracy minus $\max(p, 1-p)$ averaged across the wallet’s resolved trades. Directional edge is outcome minus backward VWAP; execution edge is backward VWAP minus entry price; total edge is their sum. Entry price is the volume-weighted mean execution price across the wallet’s trades. Maker fraction is the share of a wallet’s trades placed as limit orders. ROI is realized profit and loss divided by total volume, and P&L is realized profit and loss in U.S. dollars. Volume is total U.S. dollar notional traded. Active days is the count of distinct calendar days on which a wallet placed at least one trade. Wallet HHI is computed on the analysis sample of resolved-market trades; classification at the data-build stage used full-sample HHI, which differs slightly from the values reported here. p25, Median, and p75 are the 25th, 50th, and 75th percentiles. The sample is 222 million resolved trades across 556,632 markets on Polymarket, November 2022 through February 2026, restricted to resolved markets (95.3% of 233 million processed trades).

Panel A: Classification Rules

Type	Definition
Bot	> 50 trades/day OR > 1,000 total trades
Sophisticated	Volume > \$10,000 AND $\text{HHI} < 0.5$ AND > 30 active days
Active Retail	10 to 1,000 total trades
One-Shot	Exactly 1 trade
Casual	All remaining wallets

Panel B: Wallet Classification

Type	Wallets	% Wallets	Volume (\$B)	% Volume	Med. Trades	Med. HHI
Bot	73,615	3.8	23.93	76.8	511	0.087
Sophisticated	64,896	3.3	2.81	9.0	153	0.141
Active Retail	1,298,085	66.7	3.96	12.7	33	0.278
Casual	401,669	20.6	0.40	1.3	4	0.850
One-Shot	106,990	5.5	0.04	0.1	1	1.000

Table 1: Continued

<i>Panel C: Wallet-Level Descriptive Statistics ($N = 1,944,050$)</i>					
	Mean	SD	p25	Median	p75
Resolved trades	228.6	11,190.5	8.0	23.0	57.0
Volume (\$)	16,018.9	531,833.9	48.1	330.3	1,914.8
Accuracy	0.503	0.243	0.419	0.500	0.600
Excess accuracy	-0.201	0.285	-0.441	-0.159	0.000
Directional edge (\$)	-0.013	0.146	-0.036	0.000	0.026
Execution edge (\$)	-0.003	0.062	-0.016	-0.001	0.004
Total edge (\$)	-0.016	0.132	-0.029	-0.001	0.009
Entry price (\$)	0.388	0.246	0.184	0.434	0.525
Maker fraction	0.168	0.256	0.000	0.000	0.333
ROI (%)	-5.557	142.134	-8.893	-0.683	0.529
P&L (\$)	-0.1	33,555.3	-19.3	-1.0	0.6

Table 2: Performance Summary by Trader Type

This table reports the return decomposition, excess accuracy, and concentration robustness for the five trader types, using the backward VWAP benchmark. Panel A decomposes profit into directional (θ , outcome minus fair price) and execution (ε , fair price minus entry price) components under two weighting blocks. The wallet-level block weights every wallet equally; the aggregate block weights by resolved trades. The two blocks differ in every column because volume is concentrated in a few high-volume wallets (bots) or spread across many small wallets (active retail, one-shot); within-type accuracy and entry price are therefore not expected to match across blocks. In the wallet-level block, N is the number of wallets and ROI is the equal-weighted mean of individual wallet ROIs. In the aggregate block, N is the number of resolved wallet-trade observations in millions; Accuracy, Directional, Execution, and Total are trade-weighted means, and Entry Price is volume-weighted; ROI is aggregate P&L divided by aggregate volume. Entry Price, the edge components, and Total are in U.S. dollars per token. P&L is aggregate realized profit in millions of U.S. dollars, identical across weightings, and the five types sum to zero by the zero-sum construction of the market (no explicit fees in sample period). Panel B reports excess accuracy, defined as observed accuracy minus the naive price-implied benchmark $\max(p, 1-p)$ averaged across a wallet's resolved trades and equal-weighted across wallets; the t -statistic tests the null of zero excess accuracy with wallet-level standard errors. Panel C reports the aggregate decomposition after excluding the 10 bot wallets with the highest aggregate P&L; non-bot types are identical to the aggregate block of Panel A. Types appear in the same order in every panel.

<i>Panel A: Return Decomposition</i>										
Weighting	Type	N	Accuracy	Entry (\$)	Direct. (\$)	Exec. (\$)	Total (\$)	Volume (\$B)	ROI (%)	P&L (\$M)
Wallet-level	Bot	73,615	0.479	0.475	-0.007	+0.001	-0.006	23.93	-2.28	+133
	Sophisticated	64,896	0.489	0.495	-0.005	-0.001	-0.006	2.81	-1.32	-38
	Active Retail	1,298,085	0.516	0.358	-0.005	-0.004	-0.008	3.96	-3.79	-79
	Casual	401,669	0.462	0.424	-0.035	0.000	-0.034	0.40	-11.55	-14
	One-Shot	106,990	0.534	0.487	-0.042	-0.016	-0.058	0.04	-10.38	-2
Aggregate	Bot	356.6M	0.499	0.522	+0.001	+0.001	+0.002	23.93	+0.56	+133
	Sophisticated	15.5M	0.468	0.492	-0.008	+0.001	-0.007	2.81	-1.37	-38
	Active Retail	70.4M	0.513	0.520	-0.003	-0.004	-0.006	3.96	-1.99	-79
	Casual	1.8M	0.463	0.612	-0.032	+0.001	-0.031	0.40	-3.45	-14
	One-Shot	0.1M	0.534	0.834	-0.042	-0.016	-0.058	0.04	-5.94	-2

Table 2: Continued

<i>Panel B: Excess Accuracy (Observed – Naive Benchmark)</i>		
Type	Excess Accuracy	<i>t</i> -statistic
Bot	−0.242	−262.71
Sophisticated	−0.111	−141.77
Active Retail	−0.184	−862.88
Casual	−0.244	−448.13
One-Shot	−0.285	−195.48

<i>Panel C: Concentration Robustness (Aggregate, Excluding Top 10 Bots)</i>						
Type	<i>N</i> Wallets	Accuracy	ROI (%)	P&L (\$M)	Direct. (\$)	Exec. (\$)
Bot (full)	73,615	0.499	+0.56	+133	+0.001	+0.001
Bot (ex top 10)	73,605	0.499	+0.17	+40	+0.001	+0.001
Sophisticated	64,896	0.468	−1.37	−38	−0.008	+0.001
Active Retail	1,298,085	0.513	−1.99	−79	−0.003	−0.004
Casual	401,669	0.463	−3.45	−14	−0.032	+0.001
One-Shot	106,990	0.534	−5.94	−2	−0.042	−0.016

Table 3: Within-Type Evidence: Accuracy, Entry Price, and Execution Quality

This table reports within-type regressions for three trader categories. Columns (1)–(3) regress execution edge (backward VWAP minus entry price, in dollars per token) on excess accuracy, defined as observed accuracy minus the price-implied benchmark $\max(p, 1-p)$, at the wallet level. Columns (4)–(6) regress mean entry price on raw directional accuracy at the wallet level. Columns (7)–(9) augment columns (4)–(6) with controls (log volume, log trade count, and maker fraction). Columns (10)–(12) regress entry price on directional accuracy at the trade level with market fixed effects; the unit of observation is one wallet-side (maker or taker) per trade, and R^2 is the within-market R^2 . The wallet-level sample (columns 1–9) restricts to wallets with at least 50 resolved trades; the column (1) sample drops one wallet with missing execution-edge data. The trade-level sample (columns 10–12) restricts to wallets with at least 10 resolved trades trading in markets with at least 5 trades. Columns 1–9 are WLS with resolved-trade weights; columns 10–12 are unweighted OLS. Standard errors are heteroskedasticity-robust (columns 1–9) or clustered by wallet (columns 10–12); t -statistics in parentheses. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

	Execution Edge on Excess Accuracy			Entry Price on Accuracy			Entry Price + Controls			Entry Price + Market FE		
	(1) Bot	(2) Soph	(3) Retail	(4) Bot	(5) Soph	(6) Retail	(7) Bot	(8) Soph	(9) Retail	(10) Bot	(11) Soph	(12) Retail
Excess Accuracy	−0.150*** (−19.90)	−0.152*** (−77.10)	−0.104*** (−135.44)									
Accuracy				0.190*** (3.51)	0.092*** (14.92)	−0.166*** (−56.03)	0.090** (1.96)	0.074*** (12.19)	−0.068*** (−23.62)	0.021*** (4.07)	−0.067*** (−78.89)	−0.057*** (−242.47)
Log(Volume)							0.034*** (12.02)	0.017*** (19.85)	0.057*** (256.59)			
Log(Trades)							−0.036*** (−8.35)	−0.028*** (−30.24)	−0.043*** (−66.77)			
Maker Fraction							−0.064*** (−4.53)	0.024*** (7.16)	0.054*** (31.00)			
Unit	Wallet	Wallet	Wallet	Wallet	Wallet	Wallet	Wallet	Wallet	Wallet	Trade	Trade	Trade
Controls	No	No	No	No	No	No	Yes	Yes	Yes	No	No	No
Market FE	No	No	No	No	No	No	No	No	No	Yes	Yes	Yes
Observations	73,092	57,050	422,024	73,093	57,050	422,024	73,093	57,050	422,024	170.6M	10.5M	51.2M
R^2	0.259	0.285	0.188	0.026	0.012	0.017	0.137	0.041	0.244	0.001	0.011	0.008

Table 4: Benchmark Sensitivity: Execution Edge and Cross-Skill Correlation

This table reports how the choice of fair-price benchmark affects measured execution edge and the cross-skill coefficient between the directional and execution components. Columns (1) through (3) are ordered by increasing benchmark cleanliness: the backward VWAP averages all preceding trades, including the evaluated wallet’s own; the local VWAP restricts to a 50-trade window; the lagged price uses only the single immediately preceding trade and does not include the evaluated trade’s own entry price. Panel A regresses wallet-level mean execution edge on trader-type dummies (bot is the omitted base category; the constant reports bot’s mean execution edge and each type coefficient reports the difference from bot). Panel B regresses the wallet-level execution component on the directional component with type fixed effects. [†]Column (4) reports the independently constructed forward-VWAP-skip-5 benchmark and runs a different regression in Panel B: it regresses the forward-VWAP execution edge on benchmark-independent excess accuracy rather than on the directional component, with type fixed effects; Panel A is not reported for this benchmark because its decomposition is not comparable to columns (1)–(3). Standard errors are heteroskedasticity-robust throughout. Internet Appendix Section [IA.1](#) reports the full treatment. Wallet-level OLS regressions on the full sample of Polymarket wallets with resolved trades, November 2022 through February 2026; execution and directional components are in U.S. dollars per token; t -statistics in parentheses. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

<i>Panel A: Execution Edge by Benchmark</i>				
	(1)	(2)	(3)	(4)
	Backward VWAP	Local VWAP	Lagged Price	Forward VWAP [†]
Sophisticated	−0.002*** (−7.03)	−0.003*** (−33.75)	−0.001*** (−33.63)	—
Active Retail	−0.005*** (−18.68)	−0.002*** (−30.11)	−0.001*** (−49.71)	—
Casual	0.000 (−1.62)	−0.003*** (−38.15)	−0.002*** (−57.41)	—
One-Shot	−0.017*** (−38.00)	−0.003*** (−27.78)	−0.001*** (−18.83)	—
Bot (constant)	0.001*** (3.62)	0.000 (1.40)	−0.001*** (−34.11)	—
Observations	1.94M	1.93M	1.94M	—
R^2	0.003	0.001	0.001	—
<i>Panel B: ε on θ (Cross-Skill Regression)</i>				
	(1)	(2)	(3)	(4)
	Backward VWAP	Local VWAP	Lagged Price	Forward VWAP [†]
Directional (θ)	−0.178*** (−280.83)	−0.013*** (−51.11)	0.000 (1.28)	
Excess accuracy [†]				+0.019*** (29.14)
Observations	1.94M	1.93M	1.94M	1.39M
R^2	0.179	0.008	0.001	0.007

Table 5: Skill Persistence: Split-Sample Regressions

This table reports split-sample persistence regressions for directional accuracy (θ) and execution quality (ε , fair price minus entry price in dollars per token). Each wallet's resolved trades are split chronologically at the midpoint; subscripts denote first ($\cdot_1$) and second ($\cdot_2$) halves. Columns (1)–(4) use the backward VWAP benchmark. Columns (5)–(8) use the lagged-price benchmark $p_{m,i-1}$, which by construction does not include the evaluated trade's own entry price (Section 2.4); the execution component is $\varepsilon_{m,i}^{\text{lag}} = p_{m,i-1} - p_{m,i}$. Within each benchmark block, columns (1)/(5) and (2)/(6) report own-skill persistence; columns (3)/(7) and (4)/(8) report cross-skill regressions. Columns (1) and (5) are identical because θ does not depend on the execution benchmark. Sample restricted to wallets with at least 50 resolved trades, the same inclusion criterion as Table 3. Wallet-level moments computed on the 550,860 wallets with both halves of all components present, a subsample of the 552,167 wallets in the persistence regressions: $\text{Var}(\theta^L) = 0.0027$, $\text{Var}(\varepsilon^L) = 0.00002$, $\text{Var}(\delta_w) = 0.0022$. These moments differ from the full-sample moments reported in Section 4.3 because the persistence subsample restricts to highly active wallets, where measured θ has converged closer to each wallet's true accuracy. The benchmark gap is partially persistent across sample halves (correlation 0.08); the variance-weighted attenuation formula evaluated at the measured persistences and half-sample variances predicts a VWAP execution persistence of 0.079, against 0.080 observed. OLS regressions with heteroskedasticity-robust standard errors; t -statistics in parentheses. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

	VWAP Benchmark				Lagged-Price Benchmark			
	Own-Skill		Cross-Skill		Own-Skill		Cross-Skill	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	θ_2	ε_2	ε_2	θ_2	θ_2	$\varepsilon_2^{\text{lag}}$	$\varepsilon_2^{\text{lag}}$	θ_2
θ_1	0.270*** (116.26)		0.028*** (44.46)		0.270*** (116.26)		0.000 (−0.61)	
$\varepsilon_1 / \varepsilon_1^{\text{lag}}$		0.080*** (19.67)		0.124*** (27.03)		0.434*** (30.72)		0.302*** (5.55)
Observations	552,167	552,167	552,167	552,167	552,167	552,167	552,167	552,167
R^2	0.068	0.007	0.006	0.002	0.068	0.217	0.000	0.000

Table 6: Directional and Execution Edge by Trader Type and Order Role

Panel A reports the directional (θ) and execution (ε) decomposition of total edge by trader type and order role (maker versus taker), using equal-weighted trade-level means under the backward cumulative VWAP benchmark. Total edge is the sum of its directional and execution components. The maker (passive) side posts quotes; the taker (active) side crosses the spread. Panel B reports each non-bot type’s gap in total edge relative to bots, computed from Panel A and expressed in cents per token. The sample is all resolved Polymarket trades, November 2022 through February 2026. One-shot wallets (single trade, negligible volume) are omitted. All values are in cents per token.

<i>Panel A: Decomposition by Type and Role</i>					
Type	Role	Total ($\theta + \varepsilon$)	Directional (θ)	Execution (ε)	Trades
Bot	Maker	+0.292	−2.879	+3.171	202.0M
Bot	Taker	+0.018	+3.986	−3.967	154.1M
Sophisticated	Maker	−0.107	−4.941	+4.834	4.9M
Sophisticated	Taker	−0.929	−2.555	+1.626	10.5M
Active Retail	Maker	−0.383	−3.113	+2.730	14.7M
Active Retail	Taker	−0.710	+1.296	−2.005	55.6M
Casual	Maker	−1.972	−2.257	+0.284	254,651
Casual	Taker	−3.307	−4.668	+1.361	1.6M

<i>Panel B: Within-Role Total Edge Gap Relative to Bots</i>		
Type	Maker Gap	Taker Gap
Sophisticated	−0.40	−0.95
Active Retail	−0.68	−0.73
Casual	−2.26	−3.33

Table 7: Mechanism Identification: Edge on Accuracy $\times$ Price Distance

This table regresses total edge (dollars per token) on accuracy, price distance, and their interaction. Accuracy is the wallet-market cell's fraction of resolved trades on the winning side. Price Distance is the wallet-market mean of $|p_i - 0.5|$, measuring proximity to resolution: higher values indicate prices closer to terminal payoffs. Columns (1)–(3) progressively add covariates; columns (4)–(5) include market fixed effects, and column (5) additionally controls for log volume, log trade count, and maker fraction. The unit of observation is the wallet-market cell, aggregating all trades by wallet i in market m . WLS with trade-count weights; standard errors are clustered at the wallet level; t -statistics in parentheses. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

	(1)	(2)	(3)	(4)	(5)
Accuracy	0.451*** (64.33)	0.451*** (66.23)	1.162*** (404.52)	1.163*** (406.99)	1.160*** (439.08)
Price Distance		−0.018* (−1.68)	1.109*** (303.87)	1.108*** (280.06)	1.095*** (321.31)
Accuracy $\times$ Price Distance			−2.244*** (−331.64)	−2.250*** (−333.33)	−2.221*** (−429.18)
Controls	No	No	No	No	Yes
Market FE	No	No	No	Yes	Yes
Observations	55.4M	55.4M	55.4M	55.4M	55.4M
R^2	0.454	0.454	0.775	0.776	0.779

Table 8: Options Skill Decomposition: Full Sample and Short-ATM Subsample

This table decomposes option returns into directional and execution components using contract-level VWAP benchmarks. The sample is CBOE equity options from the CBOE Open-Close Volume Summary data, January 2011 through July 2022; market makers are excluded because their positions are typically hedged. Accuracy is the fraction of contracts that expire in-the-money. Total, directional, and execution edges are in cents per option share (100 shares per contract); ROI is total P&L divided by total premium invested, expressed as a percent. Panel A uses the full sample. Panel B restricts to short-dated ($DTE \leq 30$), near-the-money (moneyness 0.95–1.05) contracts, where the equity risk premium has minimal influence on payoffs. t -statistics in parentheses, computed from standard errors of the volume-weighted means across type-by-underlying cells. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

<i>Panel A: Full Sample</i>					
Type	Accuracy (%)	Total Edge	Directional	Execution	ROI (%)
Firm (Prop.)	32.8	0.845 (1.15)	−0.888 (−1.27)	1.734*** (11.46)	0.5
Broker-Dealer	31.2	74.252*** (7.80)	66.760*** (7.33)	7.492*** (7.43)	33.5
Small Retail	33.2	17.998*** (32.57)	8.903*** (16.72)	9.096*** (122.45)	6.6
Medium Retail	30.9	13.508*** (8.99)	1.648 (1.16)	11.860*** (39.85)	8.2
Large Retail	29.4	6.355*** (4.46)	0.866 (0.65)	5.489*** (17.83)	3.5
Prof. (<100)	39.1	27.796*** (16.38)	41.884*** (25.59)	−14.088*** (−32.56)	8.4
Prof. (100–199)	33.9	6.832* (1.69)	−2.298 (−0.60)	9.131*** (8.29)	4.9
Prof. (>199)	29.5	5.081 (0.89)	−4.240 (−0.78)	9.321*** (7.48)	3.1
<i>Panel B: Short-ATM ($DTE \leq 30$, Moneyness 0.95–1.05)</i>					
Type	Accuracy (%)	Total Edge	Directional	Execution	ROI (%)
Firm (Prop.)	34.5	−7.557*** (−12.24)	−17.919*** (−29.65)	10.362*** (69.60)	−6.6
Broker-Dealer	29.2	4.505** (2.22)	−15.917*** (−7.71)	20.421*** (33.31)	4.8
Small Retail	35.0	−21.814*** (−30.58)	−27.785*** (−40.14)	5.972*** (68.72)	−10.1
Medium Retail	32.4	−12.278*** (−10.72)	−26.953*** (−24.83)	14.674*** (51.40)	−10.8
Large Retail	29.4	−13.983*** (−11.87)	−23.034*** (−21.74)	9.051*** (27.45)	−13.5
Prof. (<100)	42.1	0.830 (0.60)	−4.087*** (−3.02)	4.917*** (13.28)	0.4
Prof. (100–199)	37.5	2.176 (0.38)	−16.985*** (−3.08)	19.161*** (14.12)	2.4
Prof. (>199)	31.6	27.152*** (3.27)	14.502* (1.79)	12.650*** (6.72)	27.3

Table 9: Options: Contract-Demeaned Execution and the Within-Type θ - ε Relation

This table reports contract-demeaned execution quality and within-type correlations between forecasting skill (θ) and execution quality (ε) for CBOE equity options, January 2011 through July 2022 (market makers excluded). Excess Execution removes the contract-level mean, isolating whether a type pays more or less than the average trader for the same contract. Demeaned Directional analogously removes the contract-level mean of the directional component. Within-type β regresses execution edge on the directional component θ across underlyings within each type: $\varepsilon_{i,u} = \alpha + \beta \cdot \theta_{i,u} + e$, where u indexes underlyings. $\rho(\theta, \varepsilon)$ is the Pearson correlation across underlyings within type. Excess Execution and Demeaned Directional are in cents per option share (100 shares per contract); t -statistics in parentheses, computed from standard errors of the volume-weighted means across type-by-underlying cells (classical OLS standard errors for the within-type β). Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

Type	Excess Execution	Demeaned Directional	Within-type β	$\rho(\theta, \varepsilon)$
Firm (Prop.)	−0.985*** (−8.16)	1.854** (2.58)	−0.027 (−1.14)	−0.015
Broker-Dealer	−0.830 (−1.19)	75.043*** (7.99)	0.308*** (3.68)	0.078
Small Retail	−0.576*** (−18.74)	18.593*** (33.68)	0.173*** (13.21)	0.155
Medium Retail	6.365*** (28.37)	7.166*** (4.86)	0.153*** (5.74)	0.074
Large Retail	1.077*** (5.13)	5.304*** (3.82)	0.194*** (9.25)	0.128
Prof. (<100)	−22.457*** (−55.29)	50.316*** (30.10)	−0.314*** (−10.81)	−0.161
Prof. (100–199)	5.421*** (5.53)	1.431 (0.36)	0.038 (0.92)	0.020
Prof. (>199)	6.233*** (5.94)	−1.104 (−0.20)	0.142*** (3.32)	0.077

Figures

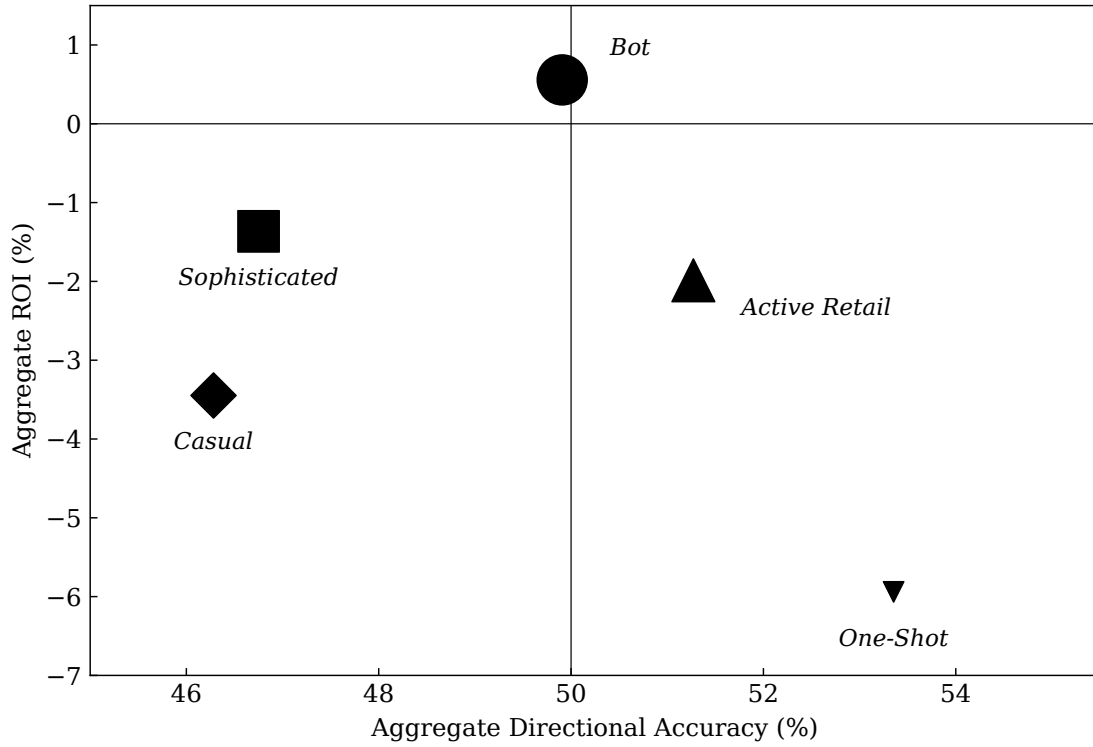


Figure 1: Accuracy and Profitability by Trader Type

This figure plots aggregate directional accuracy (x-axis) against aggregate return on investment (y-axis) for each of the five trader types. Marker shapes identify trader types (circle = bot, square = sophisticated, triangle-up = active retail, triangle-down = one-shot, diamond = casual), and marker size is proportional to log aggregate volume. Reference lines mark 50% accuracy (no forecasting skill) and 0% ROI (breakeven). Values from Table 2, Panel A aggregate block.

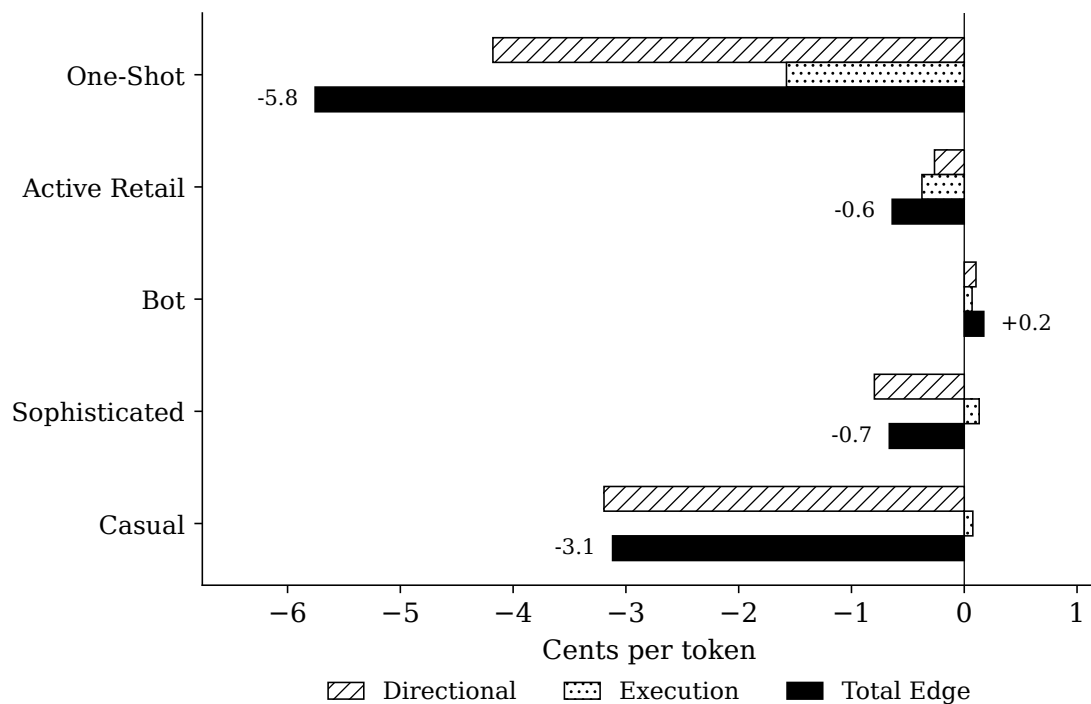


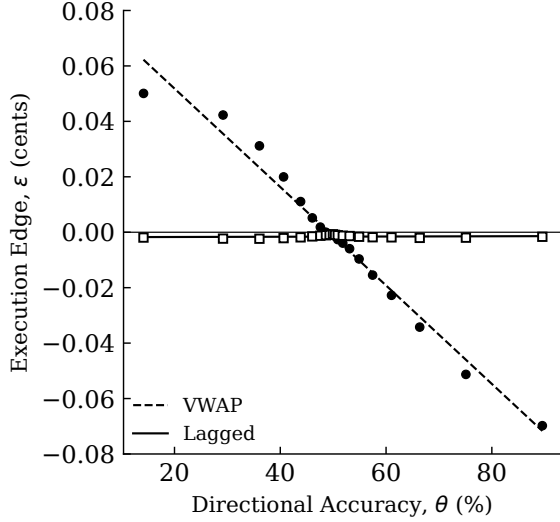
Figure 2: Return Decomposition by Trader Type

This figure plots average edge-at-entry by trader type, in cents per token, decomposed into the directional component (slashed bars), the execution component (dotted bars), and net total edge (solid black bars). Types are ordered by aggregate directional accuracy, ascending from bottom to top. Value labels report total edge only. The vertical line marks zero. Values from Table 2, Panel A aggregate block, converted from dollars to cents per token.

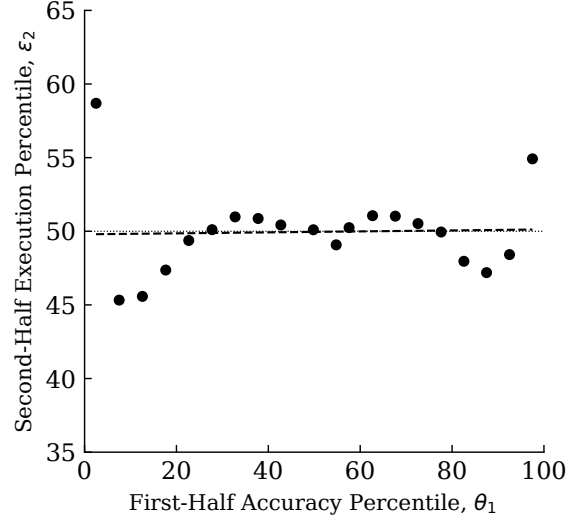
Figure 3: Benchmark Contamination and Cross-Skill Independence

This figure plots execution edge against directional accuracy under two benchmarks. Panel (a) plots binned means (20 ventile bins) of execution edge (ε , cents per token) against directional accuracy (θ , percent) for 552,167 wallets with at least 50 resolved trades; filled circles and the dashed line use the backward VWAP benchmark, open squares and the solid line use the lagged-price benchmark. Panel (b) plots the split-sample rank-rank relation, the percentile rank of second-half lagged-price execution (ε_2) against the percentile rank of first-half accuracy (θ_1), using the same wallets as Table 5; the dotted line marks the 50th-percentile independence null and the dashed line is the fitted regression.

(a) Benchmark contamination



(b) Out-of-sample cross-skill persistence



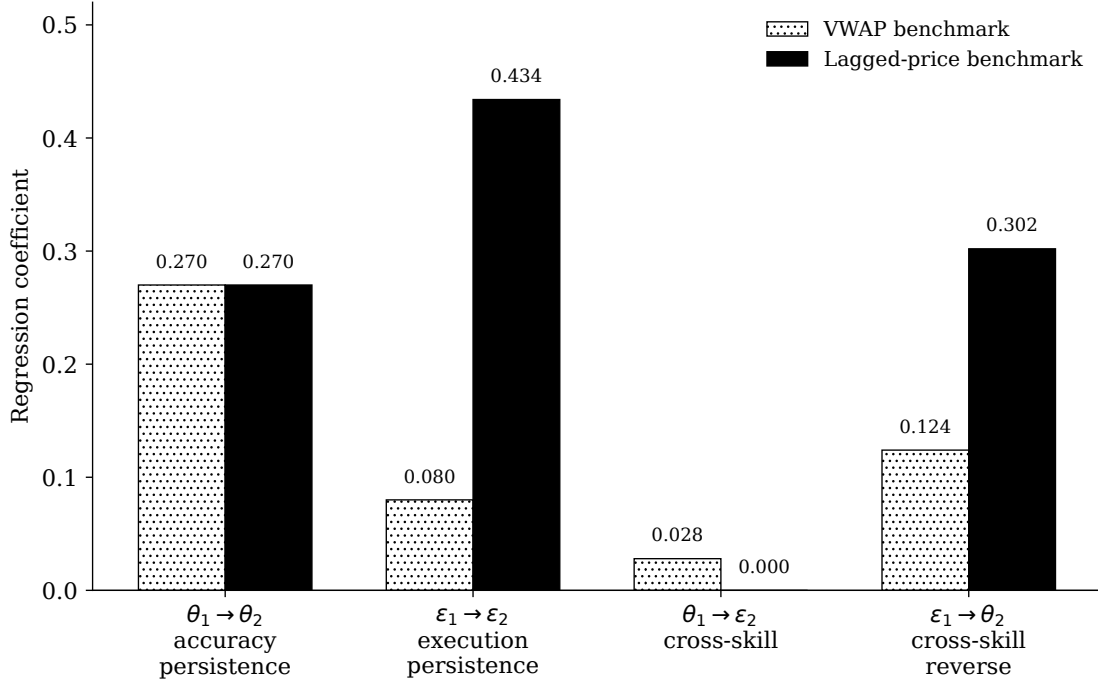


Figure 4: Skill Persistence: Split-Sample Regressions Under Two Benchmarks

This figure plots split-sample regression coefficients for directional accuracy (θ) and execution quality (ϵ) under the backward VWAP benchmark (light bars) and the lagged-price benchmark (dark bars). For each wallet with at least 50 resolved trades, trades are split chronologically at the midpoint into first and second halves. The first two bar pairs show own-skill persistence: first-half skill predicting second-half skill in the same dimension. The second two bar pairs show cross-skill regressions: first-half skill in one dimension predicting second-half skill in the other. Table 5 reports the full regression results.

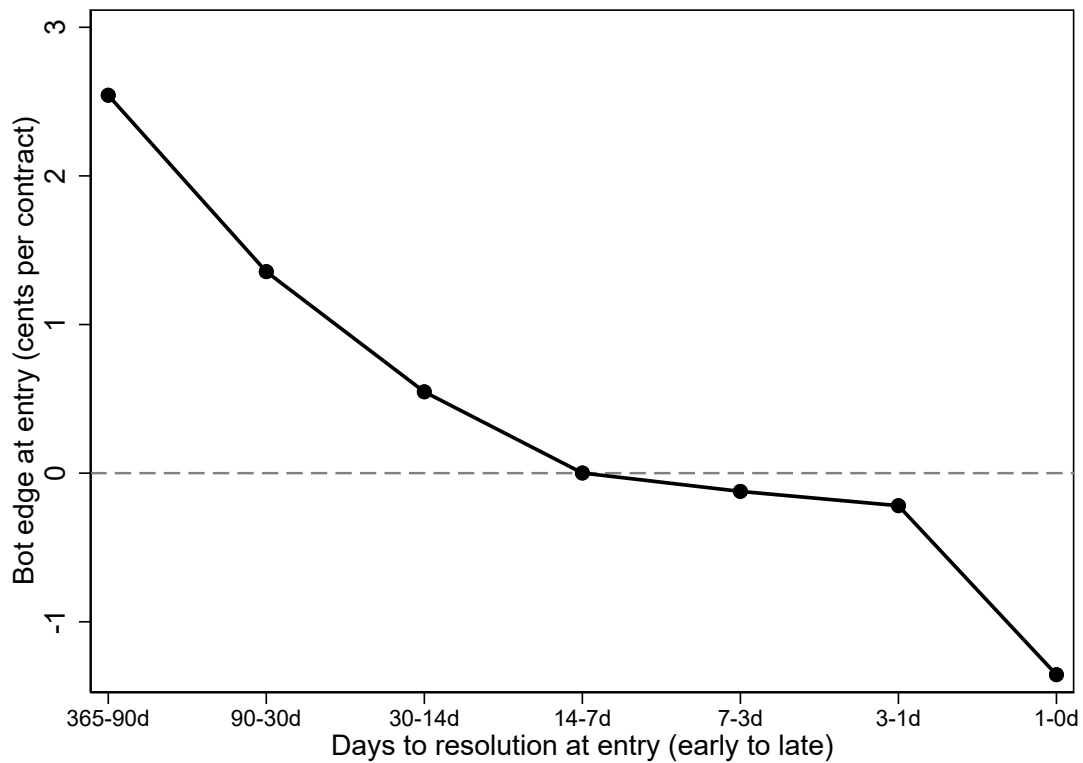


Figure 5: Bot Edge-at-Entry over the Market Lifecycle

This figure plots bots' average realized edge per contract (terminal payoff minus entry price for buys, sign-reversed for sells, in cents) against days to resolution at the time of entry, equal-weighted within each bucket. Buckets run from earliest entry (left) to latest (right); each label gives the day range, so "90-30d" is 30 to 90 days before resolution. The data includes the full resolved-trade record, November 2022 through February 2026.

A Variable Definitions

Table A1: Variable Definitions

Variable	Definition
Directional Accuracy	Fraction of resolved trades that are correct: buying tokens that resolve to \$1, or selling tokens that resolve to \$0. Under no skill, expected accuracy is 50%.
ROI	Return on investment: total P&L divided by total USDC invested.
Entry Price	Price paid per contract at trade execution, computed as USDC amount / token quantity for buy trades.
Days to Resolution	Calendar days between the trade date and the market’s resolution date.
Maker	Indicator equal to 1 if the wallet was the liquidity provider (passive side) in the trade, 0 if liquidity taker (active side).
Price Distance	Absolute distance of the trade price from 0.50: $ p - 0.5 $.
HHI	Herfindahl-Hirschman Index of a wallet’s volume across markets. Lower values indicate more diversification.
Forecasting Skill (θ)	Wallet-level directional accuracy: fraction of resolved trades where the trader was on the correct side.
Execution Quality (ε)	Fair price minus entry price, averaged across a wallet’s buy-side resolved trades. Positive means favorable execution.
Directional Edge	Outcome minus fair price: the component of edge attributable to entering a market where the consensus underpriced the outcome.
Execution Edge	Fair price minus entry price: the component of edge attributable to paying a better (or worse) price than the market average.
Fair Price	Volume-weighted average transaction price across buy-side trades preceding the evaluated trade (the backward VWAP).

B Effective Spreads

Table A2 reports volume-weighted effective spreads by trader type. Bots earn the spread (-4.25 bps), while active retail pays it ($+11.68$ bps), the mirror-image pattern underlying the maker-taker decomposition.

Table A2: Effective Spreads by Wallet Type

This table reports the volume-weighted effective spread by wallet type, computed as $2 \times \text{direction} \times (\text{price} - \text{midpoint})$, where the midpoint is estimated from surrounding trades. The sample is Polymarket trades, November 2022 through February 2026. Negative values indicate the wallet type is on average earning the spread (as maker); positive values indicate paying the spread (as taker). Volume is total U.S. dollar notional traded by the type.

Type	N Trades	Vol-Weighted Spread	Total Volume
Bot	356,490,376	-4.25 bps	\$23.9B
Sophisticated	15,462,333	19.37 bps	\$2.8B
Active Retail	70,469,200	11.68 bps	\$4.0B
Casual	1,843,505	4.28 bps	\$401.3M
One-Shot	107,102	-17.93 bps	\$38.9M

C Data-Driven Classification

This appendix reports the data-driven classification cited in the robustness section as the sharpest test of classification endogeneity, with the full per-cluster aggregates in Table A3 and Figure A1.

The profitable cluster’s ROI (+0.27%) is lower than the baseline bot ROI (+0.56%) because k -means assigns wallets by behavioral similarity rather than strict threshold rules, pulling in non-bot wallets that meet the frequency criterion but lack the execution advantage of dedicated automated traders. This dilution is the mirror image of the out-of-sample sign-flip mechanism documented in Internet Appendix Section IA.22: when classification boundaries shift, reclassified wallets carry their characteristic edge into the receiving category.

Table A3: P&L by Data-Driven Cluster (k-means, $k = 5$)

This table reports P&L by cluster from k-means clustering ($k = 5$) on log volume, log trades, HHI, buy ratio, and log active span. Features are standardized before clustering; clustering uses no profit, return, or accuracy information. Clusters are estimated on the 1,945,255 analysis-sample wallets (November 2022 through February 2026). “Dominant Type” is the most common threshold-based wallet type within each cluster. “Bot %” is the fraction of wallets in each cluster classified as bots under the threshold rules. ROI is aggregate gross profit divided by aggregate volume; accuracy is the aggregate fraction of resolved trades on the winning side.

Cluster	N Wallets	Trades	Volume	ROI	Accuracy	Dominant Type	Bot %
1	426,086	407.0M	\$27.2B	+0.27%	0.498	Active Retail	13.2
2	336,602	2.6M	\$198.4M	−5.75%	0.444	Casual	0.6
3	629,730	18.7M	\$300.8M	−3.04%	0.523	Active Retail	1.6
4	221,510	1.0M	\$229.1M	−3.51%	0.478	Casual	0.1
5	331,327	15.0M	\$3.2B	−1.39%	0.526	Active Retail	1.5

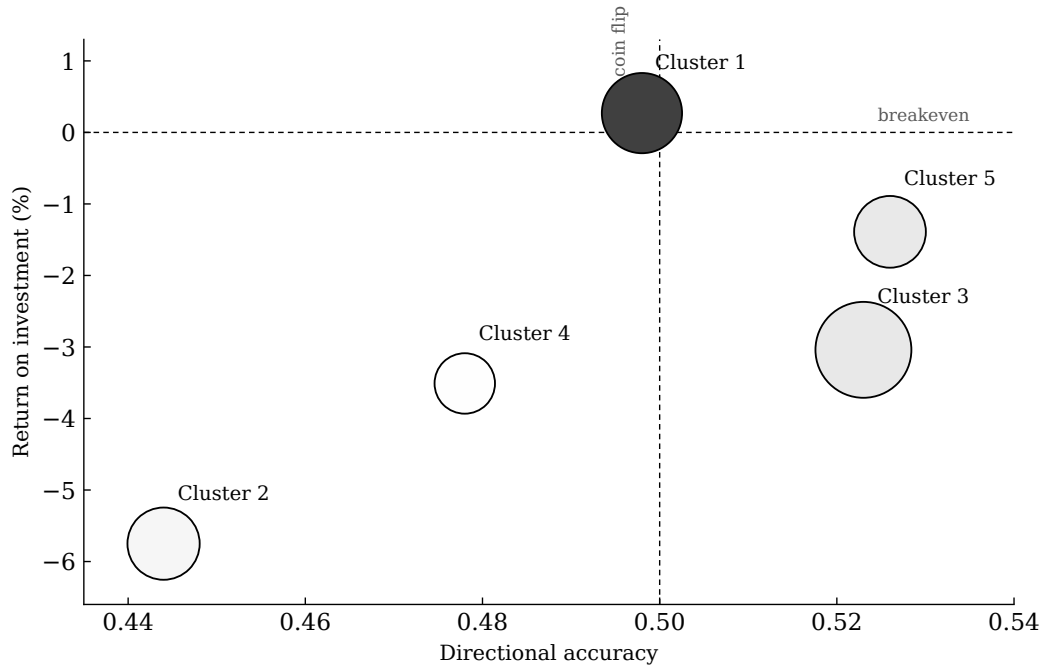


Figure A1: Profitability by Data-Driven (k-means) Cluster

This figure plots the five k -means clusters on the directional-accuracy (horizontal) and return-on-investment (vertical) plane. Bubble area scales with the number of wallets in the cluster, and bubble shade encodes bot density (darker is more bot-dense). The dashed vertical line marks coin-flip accuracy (0.50) and the dashed horizontal line marks breakeven (0% ROI). Cluster aggregates are reported in Table A3.

D Resource Independence and Orthogonality

This appendix formalizes the resource-independence argument behind the orthogonality prediction in Section 2.2. Let each trader i have two capacities: an information-processing capacity C_i^I , which captures signal observation, interpretation, and pricing relative to terminal payoffs, and an execution capacity C_i^E , which captures latency, queue position, market access, and order-placement technology. Forecasting accuracy and execution quality are increasing functions of these capacities,

$$\theta_i = f(C_i^I), \quad \varepsilon_i = g(C_i^E).$$

If C_i^I and C_i^E are independent across traders, then $f(C_i^I)$ and $g(C_i^E)$ are independent, so $\text{Cov}(\theta_i, \varepsilon_i) = 0$. Orthogonality is therefore the baseline prediction under resource independence. This baseline does not require traders to lack skill; it requires only that the capacity generating directional accuracy is distinct from the capacity generating execution quality.

The bundled view is the alternative case in which a single latent capacity C_i generates both components, $\theta_i = f(C_i)$ and $\varepsilon_i = g(C_i)$. If both mappings are increasing, the cleaned cross-trader covariance should be positive under standard regularity conditions. More generally, a positive cleaned covariance can arise either from a single latent capacity or from correlated acquisition of the two resources, for example a common sophistication factor Z_i driving both C_i^I and C_i^E . The empirical implication is therefore not that every nonzero covariance uniquely identifies bundling, but that the bundled view predicts a positive cleaned relation, whereas the resource-independence benchmark predicts that the relation collapses toward zero once benchmark contamination is removed.

Proposition 1 shows why this distinction cannot be tested with a shared-price benchmark. Its empirical counterpart is the covariance of the directional and execution components measured against a benchmark L , $\text{Cov}(\theta^L, \varepsilon^L)$, which coincides with the skill covariance only up to the shared-benchmark term, so a cleaned covariance of zero is an empirical property

rather than a mechanical consequence of resource independence. When that cleaned covariance is zero and the benchmark gap δ is approximately independent of the two components, a benchmark containing the trader's own prior entry prices maps it to a negative measured covariance, $\text{Cov}(\theta^V, \varepsilon^V) \approx -\text{Var}(\delta)$. The signature that distinguishes contamination from bundling is therefore the attenuation of the measured relation toward zero as the benchmark is cleaned, not its sign under the contaminated benchmark; where the independence condition fails, as within trader types or in the raw cross-sectional correlation, the contaminated relation can itself be positive. The benchmark-gradient and split-sample tests in Sections 4.3 and 4.4 accordingly test whether the cleaned relation remains economically positive, as bundling predicts, or approaches zero, as resource independence predicts.

IA Internet Appendix

“Who Profits from Prediction? Execution, not Information”

This Internet Appendix reports additional results referenced in the main text.

IA.1 Forward-VWAP-Skip-5: Independent-Construction Check

A natural concern with the lagged-price benchmark $p_{m,i-1}$ used in Section 4.3 is that it may mechanically reflect maker-versus-taker role rather than execution skill, since makers systematically transact at favorable prices relative to the prior trade. To address this, we compute an additional benchmark, the forward-VWAP-skip-5, which is independent of both the backward-VWAP construction and the lagged-price construction.

Definition. For each trade i in market m , the forward-VWAP-skip-5 benchmark is the volume-weighted average price of trades at positions $[i + 5, i + 55)$ within the same market:

$$p_{m,i}^{\text{fwd}} = \frac{\sum_{j=i+5}^{i+54} p_{m,j} \cdot v_{m,j}}{\sum_{j=i+5}^{i+54} v_{m,j}}.$$

The skip-5 gap reduces own-trade contamination since wallets rarely contribute 5 consecutive trades in a 50-trade window. The forward-looking window does not share a term with the trader’s entry price $p_{m,i}$ (excludes the trade itself), does not share construction with $p_{m,i-1}$ (forward-looking, averages many subsequent trades), and has substantially more variance than the lagged-price benchmark (50-trade VWAP vs single trade).

We compute the forward-VWAP-skip-5 for trades in resolved markets, matching the resolved-trade sample used throughout the main text (222.7 million trade-level observations, 95.2% of resolved-market trades; the remainder fall in the last 55 positions of each market, where the forward window is incomplete). Applying the same minimum-activity filter as in Section 4.3 yields a regression sample of 1.39 million wallets.

We regress the forward-VWAP-skip-5 execution edge on benchmark-independent excess

accuracy with type fixed effects, on the full resolved sample of 1.39 million wallets. This specification differs from the main-text contamination gradient, which regresses the execution component on the directional component measured against the same benchmark; here the forecasting measure (excess accuracy) is independent of the benchmark by construction, so the test isolates whether execution skill co-varies with forecasting skill under a benchmark unrelated to both the VWAP and the lagged price. The pooled coefficient is $\hat{\beta}^{\text{fwd}} = 0.019$ (t -statistic 29.14 on heteroskedasticity-robust (HC1) standard errors, $R^2 = 0.007$), near zero and far from the contaminated backward-VWAP cross-skill estimate of -0.178 . Table 4, column (4) reports the same coefficient with standard errors clustered by trader type ($t = 3.98$); the coefficient is identical and remains significant under either basis. By trader type, the bot coefficient is $+0.068$, sophisticated is -0.064 , and active retail is $+0.019$; the bot and sophisticated coefficients have opposite signs of comparable magnitude, averaging out in the pooled estimate and providing no by-type evidence of either bundling or a speed-precision tradeoff.

The orthogonality finding survives a benchmark that is independent of both VWAP-construction and lagged-price-construction. The persistence and cross-skill results in the main text are not driven by bid-ask bounce or maker-taker role stickiness in the lagged-price benchmark. Both the contamination gradient under the shared-benchmark specification (-0.178 backward VWAP, -0.013 local VWAP, 0.000 lagged price) and the near-zero forward-VWAP coefficient under the independent-construction specification ($+0.019$) point to the same conclusion: as the benchmark draws progressively less on the trader’s own prior trades, the measured cross-skill relation collapses toward zero.

IA.2 Monte Carlo Diagnosis of Benchmark Contamination

This section specifies the Monte Carlo simulation referenced in Section 4.3. The simulation tests whether the empirical pooled type-FE coefficient under the backward VWAP benchmark is consistent with true orthogonality between forecasting and execution skill,

contaminated by a shared VWAP term.

For each wallet i in the sample with at least 25 resolved trades under both the backward VWAP and lagged-price benchmarks, we observe forecasting and execution components $(\theta_i^V, \varepsilon_i^V)$ under VWAP and $(\theta_i^L, \varepsilon_i^L)$ under the lagged-price benchmark. The wallet-level benchmark gap is $\delta_i = \varepsilon_i^V - \varepsilon_i^L$, equivalently $\theta_i^L - \theta_i^V$ from the algebraic identity. The empirical sample has $N = 943,782$ wallets. On this ≥ 25 -trade sample, the pooled type-FE VWAP coefficient is -0.370 ; the corresponding full-sample coefficient is the -0.178 reported in Table 4, the difference reflecting the larger benchmark-gap variance relative to directional variance among higher-activity wallets.

For each of $S = 1,000$ draws, we generate a synthetic sample of size N as follows. (i) Draw $\theta_i^L \sim \mathcal{N}(\bar{\theta}^L, s_{\theta^L}^2)$ and independently $\varepsilon_i^L \sim \mathcal{N}(\bar{\varepsilon}^L, s_{\varepsilon^L}^2)$ where bars denote sample means and s^2 denotes sample variance, both computed in the empirical wallet sample. The two skill draws are independent by construction, imposing the null of true orthogonality. (ii) Bootstrap δ_i from its empirical distribution, preserving any heteroskedasticity in δ across the wallet population. (iii) Construct the VWAP-based components from the algebraic identity: $\theta_i^V = \theta_i^L - \delta_i$, $\varepsilon_i^V = \varepsilon_i^L + \delta_i$. (iv) Run the same pooled type-FE regression as in Section 4.3: $\varepsilon_i^V = \alpha_t + \beta^V \theta_i^V + u_i$, with type fixed effects.

The simulated VWAP coefficient under true orthogonality has mean -0.365 across the 1,000 draws (SD 0.001, 5th–95th percentile range $[-0.367, -0.364]$), reproducing 99% of the empirical VWAP coefficient on the same ≥ 25 -trade sample (-0.370). The simulated mean coincides with the analytical first-order bound of -0.366 by construction, because the simulation builds the VWAP components from the same identity the bound differentiates; the Monte Carlo therefore illustrates Proposition 1 numerically rather than testing it independently. The simulated lagged-price coefficient under the same draws is statistically and economically zero (mean 0.000, SD 0.0001). The two simulated coefficients reproduce the empirical gradient from contaminated to clean benchmarks under a data-generating process in which the true cross-skill correlation is exactly zero.

The empirical VWAP-based negative correlation is consistent with no underlying structural relationship between forecasting and execution, reflecting instead the mechanical sharing of the VWAP benchmark term across the two components. The residual gap of -0.005 (-0.370 empirical versus -0.365 simulated) is attributable to the small measured covariances between δ and the clean components that the simulation’s independence assumption excludes, and is about one percent of the empirical estimate.

IA.3 Subperiod Stability

Table [IA.1](#) splits the sample into four periods: pre-election (through May 2024), election season (June–November 2024), post-election (December 2024–June 2025), and late sample (July 2025 onward). The inversion is present in all four periods.

IA.4 Wash Trading

[Siroly et al. \(2025\)](#) estimate that approximately 25% of Polymarket share volume reflects wash trading, concentrated at near-zero prices and peaking at roughly 60% of share volume during December 2024. Wash trades are economically immaterial for our analysis for three reasons. First, our analysis is dollar-weighted: wash trades cluster at extreme prices ($p < 0.04$ or $p > 0.96$), so their dollar weight in P&L is small. Second, wash trades generate near-zero P&L by construction, because both sides of a self-matched trade settle to the same wallet (or affiliated wallets) at the same price, leaving realized P&L approximately zero net of platform-level matching frictions. Third, our headline within-wallet results (the split-sample persistence in [Section 4.4](#) and the within-type entry-price regressions in [Section 4.2](#)) hold within each wallet across time, and wash trades affect neither the persistence coefficient nor the entry-price gradient because they enter both halves of the split sample symmetrically. The threshold sensitivity ([Internet Appendix Section IA.21](#)) and the data-driven clustering analysis confirm robustness to wash-trade exclusion: both classify wash-heavy wallets into a low-edge group whose exclusion does not alter the inversion.

Table IA.1: P&L by Wallet Type Across Subperiods

This table reports trades, volume, P&L, ROI, and accuracy by wallet type across four subperiods: pre-election (2022–May 2024), election season (June–November 2024), post-election (December 2024–June 2025), and late sample (July 2025 onward). Volume is total USDC traded; P&L is aggregate realized profit; ROI is aggregate P&L divided by aggregate volume; accuracy is the fraction of resolved trades on the winning outcome.

Period	Type	Trades	Volume	P&L	ROI	Accuracy
Pre-Election (2022–May 2024)	Bot	1.0M	\$266.7M	\$1.7M	+0.64%	0.521
	Sophisticated	136,828	\$39.5M	\$734K	+1.86%	0.466
	Active Retail	266,738	\$38.9M	−\$1.8M	−4.66%	0.449
	Casual	30,973	\$5.1M	−\$504K	−9.83%	0.416
	One-Shot	3,207	\$469K	−\$111K	−23.67%	0.455
Election Season (Jun–Nov 2024)	Bot	25.4M	\$4.7B	\$58.2M	+1.24%	0.508
	Sophisticated	2.8M	\$528.3M	−\$14.8M	−2.80%	0.462
	Active Retail	14.7M	\$1.2B	−\$36.0M	−3.00%	0.495
	Casual	310,149	\$60.5M	−\$5.9M	−9.83%	0.431
	One-Shot	35,138	\$6.2M	−\$1.5M	−23.69%	0.621
Post-Election (Dec 2024–Jun 2025)	Bot	54.7M	\$5.2B	\$24.0M	+0.46%	0.493
	Sophisticated	3.6M	\$814.0M	−\$7.9M	−0.97%	0.447
	Active Retail	24.4M	\$1.0B	−\$14.7M	−1.42%	0.524
	Casual	523,107	\$117.7M	−\$1.6M	−1.38%	0.474
	One-Shot	27,571	\$14.5M	\$191K	+1.31%	0.592
Late 2025+ (Jul 2025–)	Bot	275.5M	\$13.8B	\$50.2M	+0.36%	0.500
	Sophisticated	8.9M	\$1.4B	−\$16.5M	−1.16%	0.478
	Active Retail	31.0M	\$1.7B	−\$27.0M	−1.59%	0.512
	Casual	978,242	\$218.3M	−\$5.8M	−2.65%	0.469
	One-Shot	41,074	\$17.7M	−\$929K	−5.23%	0.426

IA.5 Position-Level vs Trade-Level Aggregation

The main text uses trade-volume-weighted wallet aggregates. As a robustness check on the unit of analysis, we re-estimate the within-type accuracy-edge regression after aggregating trades to wallet-market positions (one observation per wallet per market) and computing each wallet’s mean as the equal-weighted average across its positions. This alternative aggregation downweights wallets that trade many times within a small number of markets and gives equal weight to wallets active across many markets.

Table IA.2 reports the position-level and trade-level coefficients side-by-side. Across all three trader types, the coefficient signs are identical and the magnitudes are within an order of magnitude of one another; the position-level t -statistics remain well above conventional significance thresholds in every cell. The within-type accuracy-edge relationship in the main text is therefore not driven by trade-count concentration in high-volume markets and is robust to the choice between trade-weighted and position-weighted aggregation.

Table IA.2: Position-Level vs Trade-Level Aggregation: Within-Type Robustness

This table reports the within-type regression of wallet-level per-contract realized edge on excess accuracy under two alternative aggregation rules. The *Position* columns aggregate trades to wallet-market positions and compute the wallet-level mean as the equal-weighted average across positions. The *Trade* columns compute the wallet-level mean as the trade-volume-weighted average across all the wallet’s trades (the baseline used in the main text). Per-contract realized edge is the wallet’s average P&L per contract; excess accuracy is the wallet-level hit rate minus the price-implied null $\max(p, 1-p)$. Sample restricted to wallets with at least 10 resolved trades. t -statistics in parentheses based on heteroskedasticity-robust (HC1) standard errors. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

	Bot		Sophisticated		Active Retail	
	Position	Trade	Position	Trade	Position	Trade
Excess Accuracy	0.032*** (40.70)	0.044*** (54.43)	0.104*** (89.67)	0.155*** (114.05)	0.077*** (295.79)	0.090*** (349.83)
Observations	73,518	73,518	64,525	64,525	1,259,919	1,259,919
R^2	0.022	0.039	0.111	0.168	0.065	0.089

IA.6 Lagged Price: Same-Wallet Exposure Diagnostic

The lagged price $p_{m,i-1}$ shares no algebraic term with the trader’s entry $p_{m,i}$, but the immediately preceding trade can still involve the wallet being evaluated as either maker or taker. This subsection quantifies the resulting exposure and tests whether it biases the cross-skill regression of Table 4, Panel B, column (3).

Share. Streaming the 237.75 million resolved trades with a within-market lag, we flag each trade for whether the immediately preceding trade involves either the maker or the taker of the current trade. The same-wallet share is 72.0%: in 72% of resolved trades the wallet at t is also a party at $t - 1$. The maker-side share is 41.6%; the taker-side share is 43.3%. Self-trades (the maker address equal to the taker address on the same trade event) are negligible (785 of 237.75 million, or 0.0003%), so the share reflects *order-book chunking*, where a single posted limit order is filled across multiple trade events as incoming market orders arrive sequentially, not wash trading.

Per-wallet exposure and the cross-skill coefficient. For each wallet, we compute the share of its lagged-benchmark observations for which the same wallet appears at $t - 1$, then re-estimate the cross-skill regression on progressively cleaner subsets of wallets. Among 944,448 wallets with at least 25 resolved trades, the share’s median is approximately 0.7. Restricting to wallets with same-wallet share below 0.50, 0.30, and 0.10 yields cross-skill coefficients of +0.0011, +0.0005, and +0.0013 respectively, against the full-subsample coefficient of +0.0022. All four are within 0.003 of zero, less than 2% of the magnitude of the backward-VWAP coefficient (−0.178). Adding a $\theta \times \text{share}$ interaction to the full-subsample regression yields an interaction coefficient of +0.0031 ($R^2 = 0.014$): the contamination loads on the cross-skill coefficient at +0.003 per unit of same-wallet share, a real but economically negligible channel.

Why a walked-back benchmark is not the right alternative. A natural stricter benchmark walks back from $t - 1$ to find the most recent prior trade in market m where the evaluated wallet was not a party. We do not adopt this construction because it introduces its own

bias: in prediction markets whose prices move monotonically toward resolution, the walked-back lag for an active wallet can be from much earlier in the market lifecycle, conflating own-wallet contamination with stale-time effects. The standard lagged-price benchmark $p_{m,i-1}$ is temporally adjacent to the trade being evaluated and is the appropriate diagnostic instrument; the per-wallet exposure regressions above show that its residual contamination does not bias the cross-skill estimate.

IA.7 Chosen-Side Excess Accuracy

Table [IA.3](#) reports chosen-side excess accuracy, which measures whether each trader type systematically purchases tokens at prices below their realized payoff probability. The chosen-side benchmark assigns each trade the reference p_i for a buyer and $1 - p_i$ for a seller, asking whether the trader identifies mispricing relative to the prices they actually paid. Under no skill, the expected chosen-side excess is zero; negative values indicate that traders paid prices above their realized winning frequency on the side they chose.

Chosen-side excess accuracy is negative for every type across all three panels. Bot takers are nearest zero (-0.001), consistent with bots profiting through lifecycle timing and spread capture rather than systematic mispricing identification (Section [5](#)). Bot makers are modestly negative (-0.010), reflecting the adverse selection inherent in liquidity provision. One-shot wallets exhibit the most negative chosen-side excess overall (-0.053) and, among makers, -0.192 , consistent with their single-trade behavior placing them at late lifecycle stages. Panel C shows that the most negative active retail chosen-side excess occurs in the $[0.6, 0.8)$ price bin (-0.022), where prices are close to resolution and late arrivals pay near-terminal prices.

IA.8 Competitive Price Zone

The accuracy-profitability inversion could reflect mechanical effects at extreme prices. Table [IA.4](#) restricts to wallets whose average entry price falls in $[0.20, 0.80]$, excluding wal-

Table IA.3: Chosen-Side Excess Accuracy

This table reports chosen-side excess accuracy: for a buyer of token i , $\text{Correct}_i - p_i$; for a seller, $\text{Correct}_i - (1 - p_i)$. This measures whether a trader systematically purchases tokens at prices below their realized payoff probability on the side they chose. Panel A reports type-level summary statistics. Favorite excess accuracy ($\text{Correct}_i - \max(p_i, 1 - p_i)$) is included for comparison. Panel B decomposes by maker and taker role. Panel C reports wallet-equal-weighted chosen-side excess by entry-price bin. Sample: 1,600,204 wallets with at least one resolved trade at a strictly interior price ($0 < p < 1$), November 2022 through February 2026; trades executed at exactly \$0 or \$1 are excluded because the chosen-side benchmark is degenerate there, which is why the wallet count is below the full-sample count of wallets with resolved trades.

<i>Panel A: Type-Level Summary</i>					
Type	N Wallets	Correct	Mean p	Fav. Excess	Chosen Excess
Bot	53,042	0.481	0.476	−0.391	−0.004
Sophisticated	60,043	0.488	0.489	−0.298	−0.007
Active Retail	1,128,158	0.522	0.338	−0.345	−0.008
Casual	279,249	0.465	0.401	−0.358	−0.035
One-Shot	79,712	0.559	0.492	−0.256	−0.053
<i>Panel B: By Maker/Taker Role</i>					
Type	Maker		Taker		
	N Wallets	Chosen Excess	N Wallets	Chosen Excess	
Bot	40,957	−0.010	52,733	−0.001	
Sophisticated	48,343	−0.006	59,033	−0.007	
Active Retail	530,814	−0.006	1,120,539	−0.007	
Casual	61,278	−0.031	270,344	−0.034	
One-Shot	3,287	−0.192	76,425	−0.047	
<i>Panel C: By Entry-Price Bin</i>					
Type	Entry Price Range				
	[0.0, 0.2)	[0.2, 0.4)	[0.4, 0.6)	[0.6, 0.8)	[0.8, 1.0)
Bot	−0.003	−0.015	−0.012	−0.008	−0.004
Sophisticated	−0.006	−0.014	−0.012	−0.013	−0.006
Active Retail	−0.004	−0.015	−0.017	−0.022	−0.005
Casual	−0.014	−0.077	−0.042	−0.080	−0.022
One-Shot	−0.005	−0.170	−0.077	−0.150	+0.002

lets that trade predominantly at extreme prices where the $\max(p, 1-p)$ benchmark is near-uninformative. In the competitive zone, directional accuracy is virtually identical across bots and active retail (both 51.1%); the inversion persists in returns: bot ROI is +0.59% while active retail loses -2.17% , a 2.8 percentage point gap attributable entirely to execution quality.

Table IA.4: Performance Decomposition in the Competitive Price Zone

This table replicates the performance decomposition from Table 2 restricted to wallets whose average entry price falls in $[0.20, 0.80]$, excluding wallets that trade predominantly at extreme prices where the $\max(p, 1-p)$ benchmark is near-uninformative. Accuracy, directional edge, execution edge, total edge, and excess accuracy are trade-weighted means within each type (weight = n_{resolved} per wallet); ROI is aggregate P&L divided by aggregate volume; P&L is the aggregate sum. Directional, execution, and total edge are in dollars per token. Excess accuracy is observed accuracy minus the price-implied null $\max(p, 1-p)$, averaged across the wallet’s resolved trades.

Type	N Wallets	Accuracy	Excess Acc.	ROI (%)	P&L (\$M)	Dir. Edge	Exec. Edge	Total Edge
Bot	46,316	0.511	-0.099	+0.59	+126	+0.004	-0.002	+0.002
Sophisticated	60,389	0.468	-0.092	-1.45	-38	-0.009	+0.002	-0.007
Active Retail	898,109	0.511	-0.084	-2.17	-73	-0.004	-0.004	-0.008
Casual	281,303	0.443	-0.161	-4.10	-12	-0.042	+0.001	-0.041
One-Shot	43,658	0.349	-0.278	-12.87	-2	-0.123	-0.007	-0.131

IA.9 Within-Role Execution Quality

A natural concern is that the negative price-accuracy correlation documented in the main text simply reflects the distinction between market makers (who set prices) and market takers (who accept them). Table IA.5 addresses this by estimating separate trade-level regressions of entry price on wallet accuracy within each trader type, split by execution role. Panel A shows that among maker-side trades, the price-accuracy gradient is large and highly significant for all types: bot makers who are more accurate pay higher prices ($\hat{\beta} = 0.673$), a 6.7-cent penalty per 10 percentage points of accuracy, consistent with the execution-accuracy tradeoff operating primarily through limit-order pricing. Panel B shows that taker-side coefficients are near zero, consistent with takers accepting posted prices. The within-type gradient is present even *within* the maker role, inconsistent with the aggregate

correlation being driven entirely by a maker/taker composition effect.

Table IA.5: Within-Type Price–Accuracy Correlation by Execution Role

Each row reports a trade-level OLS regression of entry price on wallet accuracy within a given trader type, estimated separately for maker-side trades (Panel A) and taker-side trades (Panel B). A positive $\hat{\beta}$ indicates that more accurate wallets pay higher prices. Casual and one-shot wallets are excluded due to insufficient maker-side observations. t -statistics in parentheses based on heteroskedasticity-robust (HC1) standard errors. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

Trader Type	N Trades	$\hat{\beta}$ (t -stat)	R^2
<i>Panel A: Maker-side trades</i>			
Bot	63.8M	0.673*** (2878.90)	0.115
Sophisticated	1.6M	0.439*** (494.50)	0.134
Active Retail	4.0M	0.282*** (514.50)	0.063
<i>Panel B: Taker-side trades</i>			
Bot	16.0M	−0.052*** (−198.00)	0.002
Sophisticated	1.3M	0.004*** (4.30)	0.000
Active Retail	4.4M	0.002*** (5.50)	0.000

IA.10 Accuracy Quintile Entry Prices

Table IA.6 reports average entry prices by accuracy quintile within each trader type, reinforcing the within-type evidence of the “right but late” pattern.

Table IA.6: Entry Prices by Accuracy Quintile and Trader Type

This table reports average entry prices by accuracy quintile within each trader type, for buy-side trades on winning tokens. Accuracy quintiles are defined within each trader type. Quintile assignment is based on wallet-level realized accuracy among wallets with at least 50 resolved trades.

Accuracy Quintile	Average Entry Price		
	Bot	Sophisticated	Active Retail
Q1 (Lowest)	0.521	0.557	0.748
Q2	0.624	0.613	0.687
Q3	0.679	0.654	0.659
Q4	0.780	0.688	0.703
Q5 (Highest)	0.673	0.793	0.751

IA.11 Reconciliation of Cross-Skill Estimates

The paper reports cross-skill estimates under several regressors, benchmarks, and levels of variation. Table [IA.7](#) catalogs every member of the family in one place, with each entry copied from its source exhibit; nothing is re-estimated. The estimates' signs differ across rows because the regressor and the conditioning differ: regressions of the execution component on the directional component inherit the contamination in Equation [2](#) with a negative sign, regressions on benchmark-independent excess accuracy do not, and pooled estimates mix cross-type composition with the within-type relation. Every estimate of the execution-forecasting relation attenuates toward zero as the benchmark draws progressively less on the evaluated trader's own prior trades. The one negative coefficient under a clean construction, the sophisticated -0.064 under the forward VWAP, is opposite-signed to the bot $+0.068$ and economically small relative to the contaminated -0.178 ; the family provides no consistent by-type evidence of either bundling or a tradeoff.

IA.12 Persistence with Maker-Fraction Control

We test whether the lagged-price persistence and cross-skill results of Table [5](#) reflect maker-role composition rather than wallet-level execution skill. Table [IA.8](#) adds wallet-level maker fraction as a control and splits the sample at the median maker fraction. Execution persistence remains strong in every specification (0.358 to 0.446 against the baseline 0.434). The cross-skill relation remains economically negligible: the accuracy-to-execution coefficient never exceeds 0.002 in absolute value, and no specification attributes more than 0.6% of variance to the skill regressor. The persistence of execution quality is not a maker-role effect.

Table IA.7: Reconciliation of Cross-Skill Estimates

This table catalogs every cross-skill estimate in the paper. Each entry is copied from the source exhibit listed in the first column; nothing is re-estimated. Panel A reports pooled estimates; Panel B reports within-type estimates for the three main trader types. The directional and execution components are defined in Equation 3 and depend on the benchmark; excess accuracy, observed accuracy minus the price-implied benchmark $\max(p, 1-p)$, is benchmark-independent. Realized edge per contract (terminal payoff minus entry price) includes the directional component, which co-moves with excess accuracy by construction; those rows are therefore positive and do not measure the execution-forecasting relation in isolation. Samples, weighting, and standard errors follow the source exhibits: Table 3 columns (1)–(3) are wallet-level WLS with heteroskedasticity-robust standard errors on wallets with at least 50 resolved trades; Table 4 Panel B includes type fixed effects; the forward-VWAP-skip-5 estimates are described in Internet Appendix Section IA.1; the realized-edge estimates are from Table IA.2. The Section 4.3 within-type coefficients are reported in the text without t -statistics and appear here without significance stars. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively, where the source reports it.

Source	Dependent variable	Regressor	Benchmark	Type FE	Estimate			
<i>Panel A: Pooled estimates</i>								
Table 4, Panel B	Execution comp.	Directional comp.	Backward VWAP	Yes		−0.178***		
Table 4, Panel B	Execution comp.	Directional comp.	Local VWAP	Yes		−0.013***		
Table 4, Panel B	Execution comp.	Directional comp.	Lagged price	Yes		0.000		
Section 4.3	$\rho(\theta, \varepsilon)$, Pearson	–	Backward VWAP	No		0.40		
Section 4.3	$\rho(\theta, \varepsilon)$, Pearson	–	Lagged price	No		0.03		
Table 4, col. (4)	Execution edge	Excess accuracy	Fwd-VWAP-skip-5	Yes		+0.019***		
<i>Panel B: Within-type estimates</i>								
						Bot	Soph.	Retail
Table 3, (1)–(3)	Execution edge	Excess accuracy	Backward VWAP	–	−0.150***	−0.152***	−0.104***	
Section 4.3	Execution comp.	Directional comp.	Backward VWAP	–	+0.089	+0.237	+0.158	
Section 4.3	Execution comp.	Directional comp.	Lagged price	–	−0.002	0.000	0.000	
IA.1	Execution edge	Excess accuracy	Fwd-VWAP-skip-5	–	+0.068***	−0.064***	+0.019***	
Table IA.2 (position)	Realized edge	Excess accuracy	Terminal payoff	–	+0.032***	+0.104***	+0.077***	
Table IA.2 (trade)	Realized edge	Excess accuracy	Terminal payoff	–	+0.044***	+0.155***	+0.090***	

Table IA.8: Lagged-Price Persistence and Cross-Skill Regressions: Maker-Fraction Robustness

This table reports the lagged-price persistence and cross-skill regressions of Table 5 with wallet-level maker fraction (the share of a wallet's trades executed as the limit-order maker, from the wallet panel) as a control and within maker-fraction halves. Column (1) is the baseline from Table 5. Column (2) adds maker fraction as a continuous control. Columns (3) and (4) split the sample at the median maker fraction (0.124). The reverse cross-skill regression's raw coefficient varies in scale across specifications because lagged-price execution is measured in dollars per token while accuracy is bounded in $[0, 1]$; its explanatory power is below 0.7% of variance in every specification. Sample: wallets with at least 50 resolved trades. OLS regressions; t -statistics in parentheses based on heteroskedasticity-robust (HC1) standard errors. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

	(1) Baseline	(2) + Maker Frac.	(3) Low Maker	(4) High Maker
<i>Own-skill persistence: ε_2^{lag} on ε_1^{lag}</i>				
ε_1^{lag}	0.434*** (30.72)	0.389*** (24.00)	0.446*** (19.30)	0.358*** (18.94)
Maker Fraction		0.003*** (29.51)		
R^2	0.217	0.238	0.206	0.159
<i>Cross-skill: ε_2^{lag} on θ_1</i>				
θ_1	0.000 (-0.61)	0.001*** (18.45)	0.002*** (34.67)	-0.001*** (-22.82)
R^2	0.000	0.082	0.005	0.002
<i>Cross-skill: θ_2 on ε_1^{lag}</i>				
ε_1^{lag}	0.302*** (5.55)	1.074*** (15.73)	2.781*** (19.80)	-2.325*** (-16.36)
R^2	0.000	0.004	0.006	0.003
Observations	552,167	552,167	276,084	276,083

IA.13 Entry Prices by Outcome

Table [IA.9](#) reports the average entry price paid by each trader type, conditioning separately on whether the token resolved to \$1 (winning) or \$0 (losing). The price gap (winning minus losing) is wider for casual traders (0.455) than for bots (0.357), indicating that retail traders face more informative prices by the time they arrive.

Table IA.9: Entry Price Conditional on Outcome by Wallet Type

This table reports the average entry price by wallet type, conditional on buying the eventual winning or losing outcome token. “Avg Price|Win” is the volume-weighted average price paid for tokens that ultimately resolve to \$1. A higher price indicates the trader arrived later in the price discovery process. The Gap column is the difference between winning and losing entry prices. Volume is the total USDC traded on the relevant side. The sample is all resolved Polymarket trades, November 2022 through February 2026.

Type	N (Winner)	Avg Price Win	N (Loser)	Avg Price Lose	Gap	Volume
Bot	81.1M	0.835	103.9M	0.478	0.357	\$23.9B
Sophisticated	3.0M	0.810	3.8M	0.472	0.338	\$2.8B
Active Retail	11.3M	0.844	18.2M	0.414	0.430	\$4.0B
Casual	273,791	0.856	488,890	0.401	0.455	\$401.6M
One-Shot	25,472	0.634	20,690	0.403	0.231	\$38.9M

IA.14 Edge-at-Entry over the Market Lifecycle

Figure [IA.1](#) plots realized edge per contract by days-to-resolution at entry. Bot edge-at-entry decays monotonically across the lifecycle, from +14.1 cents per contract more than a year from resolution to approximately zero at 7–14 days and negative inside three days. No other type shows a monotonic profile: early trades by active retail and casual wallets carry large negative edge, and casual wallets show positive but thin-margin edge near resolution. Section [5.1](#) discusses the pattern.

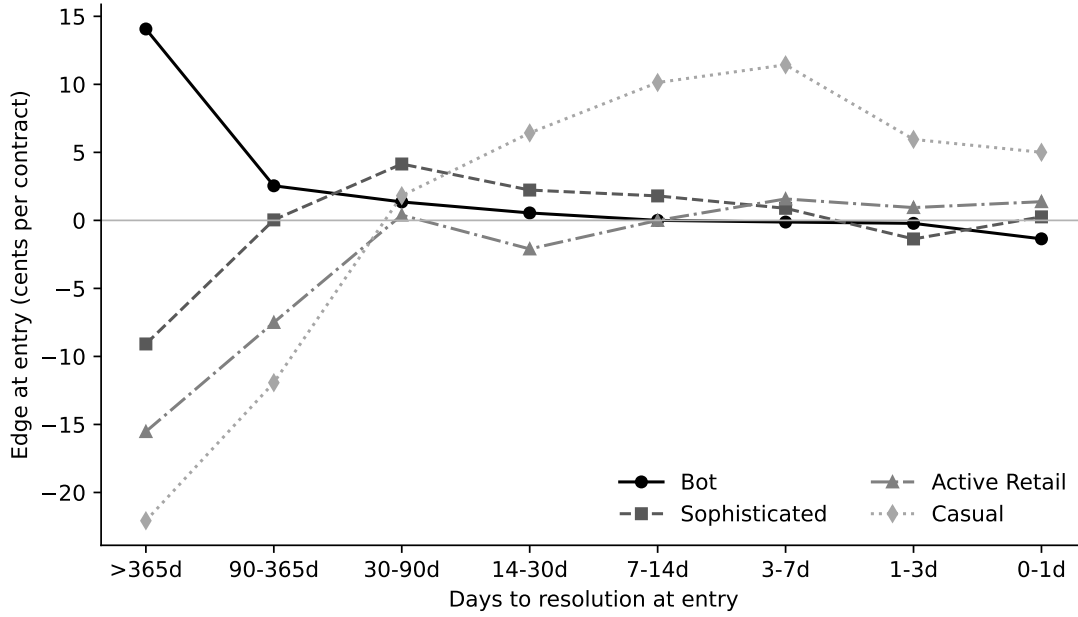


Figure IA.1: Edge-at-Entry by Days to Resolution

Average realized edge per contract (terminal payoff minus entry price for buys, sign-reversed for sells, in cents) by trader type and days-to-resolution bucket at entry, equal-weighted across trades within each type-bucket cell. One-shot wallets are omitted because their per-bucket trade counts are negligible. Source: full resolved-trade record, November 2022 through February 2026.

IA.15 Maker/Taker Decomposition

Table IA.10 regresses trade-level edge on trader-type indicators separately within the maker and taker roles. The profitability hierarchy survives in both roles: active retail trails bots by 0.7 cents per token on each side, and casual and one-shot wallets trail by more. The bot constant is positive on the maker side and indistinguishable from zero on the taker side; the role-level decomposition of bot edge into directional and execution components is in Table 6.

IA.16 Maker Fraction as Continuous Control

We add each wallet's maker fraction as a continuous control in the cross-sectional ROI regression (Table IA.11). The accuracy coefficient is virtually unchanged when maker fraction is added (+1.6%, from 0.080 to 0.081); the interaction of accuracy with maker fraction is

Table IA.10: Edge by Trader Type Within Maker and Taker Roles

This table regresses edge (dollars per token) on trader-type indicators separately for maker-side and taker-side trades. Bot is the omitted category. WLS with trade-count weights; t -statistics in parentheses based on heteroskedasticity-robust (HC1) standard errors. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

	(1) Makers	(2) Takers
Sophisticated	−0.004*** (−6.57)	−0.010*** (−23.88)
Active Retail	−0.007*** (−22.53)	−0.007*** (−24.50)
Casual	−0.023*** (−25.44)	−0.033*** (−83.25)
One-Shot	−0.159*** (−26.07)	−0.054*** (−53.60)
Bot (constant)	0.003*** (9.83)	0.000 (0.59)
Observations	863,984	1,919,176
R-squared	0.003	0.012

positive ($\hat{\beta} = 0.041$), indicating that the accuracy–ROI relationship is steeper for wallets with higher maker fractions. The documented patterns are not driven by differential market-making activity.

Table IA.11: Cross-Sectional ROI Regression with Maker Fraction Control

The dependent variable is wallet-level ROI, expressed as a decimal fraction, for wallets with 50+ resolved trades. ROI and accuracy in this table are recomputed from the trade record by the maker-fraction pipeline, which accumulates P&L and volume over resolved trades only; the accuracy coefficient is therefore not numerically comparable to Table IA.13, where ROI divides gross P&L by total volume including unresolved trades. Accuracy is the fraction of trades where the wallet chose the winning outcome. Maker Fraction is the share of all trades (resolved and unresolved) where the wallet appears as the limit-order maker. Column (1) is the baseline specification. Column (2) adds maker fraction as a continuous control. Column (3) adds the interaction of accuracy with maker fraction. All specifications include wallet-type fixed effects (Bot omitted). t -statistics in parentheses based on heteroskedasticity-robust (HC1) standard errors. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

	(1) Baseline	(2) + Maker Frac.	(3) + Interaction
Accuracy	0.080*** (105.96)	0.081*** (107.82)	0.072*** (80.07)
Log(Volume)	0.000* (−1.65)	−0.001*** (−10.65)	−0.001*** (−13.28)
Log(Trades)	0.001*** (14.10)	0.001*** (13.78)	0.002*** (15.97)
Maker Fraction		0.014*** (44.38)	−0.005*** (−3.96)
Accuracy × Maker Frac.			0.041*** (14.66)
Type FE	Yes	Yes	Yes
N (wallets)	552,167	552,167	552,167
R^2	0.038	0.042	0.043
Adj. R^2	0.038	0.042	0.043

IA.17 Excess Accuracy Interaction

A concern with the mechanism identification test (Table 7) is that the interaction between raw accuracy and price distance may have mechanically high R^2 because raw accuracy is itself a function of price levels. Table IA.12 replicates the interaction specification using excess accuracy, defined as accuracy net of the naive benchmark $\max(p, 1-p)$, instead of raw

accuracy. The interaction coefficient is -2.145 with market fixed effects and full controls, highly significant, compared to -2.221 using raw accuracy. The R_w^2 drops from 0.779 to 0.744, confirming that some explanatory power was mechanical, but the bulk of the interaction is genuine: the timing mechanism operates on excess accuracy as well. The same specification within each type yields a stable interaction coefficient across bots (-2.131), sophisticated (-2.129), and active retail (-2.144).

Table IA.12: Mechanism Identification: Edge on Excess Accuracy $\times$ Price Distance

This table regresses total edge (dollars per token) on excess accuracy, price distance, and their interaction. Excess accuracy is $\text{accuracy} - \max(p, 1 - p)$. Price distance is $|p - 0.5|$. Columns (1)–(3) progressively add covariates; columns (4)–(5) include market fixed effects. Sample and unit of observation follow Table 7. WLS with trade-count weights at the wallet-market level; t -statistics in parentheses based on wallet-clustered standard errors. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

	(1)	(2)	(3)	(4)	(5)
Excess Accuracy	0.321*** (55.34)	0.385*** (64.17)	1.095*** (304.17)	1.115*** (359.68)	1.112*** (383.45)
Price Distance		0.342*** (27.36)	-0.022 *** (-4.67)	0.003 (0.77)	0.008** (2.35)
Excess Acc. $\times$ Price Dist.			-2.116 *** (-218.14)	-2.183 *** (-311.86)	-2.145 *** (-420.59)
Log(Volume)					0.001** (2.46)
Log(Trades)					-0.005 *** (-10.70)
Maker Fraction					0.034*** (18.05)
Constant	0.057*** (35.44)	-0.029 *** (-16.22)	0.051*** (29.94)	0.043*** (32.85)	0.042*** (24.53)
Market FE	No	No	No	Yes	Yes
Controls	No	No	No	No	Yes
Observations	55.4M	55.4M	55.4M	55.4M	55.4M
R-squared	0.322	0.389	0.731	0.739	0.744

IA.18 Cross-Sectional Regressions

The univariate results establish that accuracy and profitability are inversely related across trader types, but this pattern could reflect confounding by trading intensity or other wallet characteristics. To isolate the partial effect of accuracy on profitability, we implement cross-sectional regressions of wallet-level ROI on directional accuracy with controls for log volume, log number of trades, and trader type fixed effects. Each observation is a wallet; standard errors are heteroskedasticity-robust (HC1).

Table [IA.13](#) reports the estimates. The accuracy coefficient is positive and significant across all three specifications, rising from 0.170 univariate to 0.181 with log volume, log trades, and trader-type fixed effects. The economic magnitude is modest: a 10 percentage point increase in accuracy is associated with a 1.8 percentage point increase in ROI, and the full-specification R^2 of 4.6% indicates that accuracy, trading intensity, and trader type collectively explain less than 5% of cross-sectional profitability variation.

IA.19 Market Fixed Effects and Trade-Level Controls

Different wallet types may select into different markets or differ systematically in trade characteristics. Table [IA.14](#) reports regressions of trade-level ROI, in percentage points, on wallet type indicators with progressively richer controls. The wallet type coefficients attenuate as controls are added but remain economically and statistically significant throughout. In the full specification, casual traders still underperform bots by 2.35 percentage points. The maker coefficient confirms that liquidity provision is profitable, but the persistence of large type gaps after controlling for maker status indicates the inversion is not fully explained by differential liquidity provision.

Table IA.13: Accuracy-Profitability Inversion: ROI on Accuracy

This table reports OLS regressions of wallet-level ROI on directional accuracy, with controls for trading intensity. ROI is gross P&L divided by total volume, including unresolved trades. The sample includes wallets with 50+ resolved trades. Column (1) is the bivariate regression. Column (2) adds log volume and log trade count. Column (3) adds wallet-type indicators for bots and sophisticated traders; active retail is the omitted category (casual and one-shot wallets do not meet the 50-trade threshold). t -statistics in parentheses are based on heteroskedasticity-robust (HC1) standard errors. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

	(1) ROI	(2) ROI	(3) ROI
Accuracy	0.170*** (83.76)	0.182*** (89.73)	0.181*** (88.92)
log(Volume)		0.007*** (59.17)	0.008*** (58.66)
log(N_{trades})		-0.001*** (-6.31)	0.001* (1.90)
Bot			-0.010*** (-11.91)
Sophisticated			-0.010*** (-14.39)
N	552,167	552,167	552,167
R^2	0.035	0.045	0.046

Table IA.14: P&L Regressions with Market Fixed Effects and Trade-Level Controls

This table reports OLS regressions of trade-level ROI (percentage points) on wallet-type indicators with progressively richer controls. The unit of observation is a trade-side: each of the 222 million resolved trades contributes a maker-side and a taker-side observation, giving 444 million observations. Bot is the omitted category. Standard errors clustered by wallet are in parentheses. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

	(1) No FE	(2) Mkt FE	(3) Mkt×Wk	(4) +Size	(5) +Maker	(6) Full
Bot (reference)	—	—	—	—	—	—
Sophisticated	−1.93*** (0.12)	−1.78*** (0.11)	−1.52*** (0.14)	−1.41*** (0.13)	−1.18*** (0.12)	−1.08*** (0.12)
Active Retail	−1.12*** (0.04)	−1.04*** (0.04)	−0.89*** (0.05)	−0.82*** (0.05)	−0.71*** (0.05)	−0.64*** (0.05)
Casual	−3.30*** (0.08)	−3.12*** (0.08)	−2.87*** (0.10)	−2.71*** (0.09)	−2.48*** (0.09)	−2.35*** (0.09)
One-Shot	−3.85*** (0.15)	−3.64*** (0.14)	−3.41*** (0.17)	−3.18*** (0.16)	−2.92*** (0.15)	−2.76*** (0.15)
Log(Trade Size)				−0.42*** (0.02)	−0.38*** (0.02)	−0.35*** (0.02)
Maker					1.84*** (0.03)	1.76*** (0.03)
Price Dist. from 0.5						−2.15*** (0.08)
Market FE	No	Yes	Yes	Yes	Yes	Yes
Week FE	No	No	Yes	Yes	Yes	Yes
Observations	444M	444M	444M	444M	444M	444M
R^2	0.008	0.142	0.168	0.181	0.203	0.215

IA.20 Panel Regression and Variance Decomposition

Table IA.15 regresses wallet-market edge on trader-type indicators with progressively richer specifications. Adding market fixed effects in column (2) raises within R^2 from 0.001 (pooled) to 0.005, indicating that market selection accounts for a small share of variation in edge. The full specification in column (4) adds wallet-level accuracy and shows that the cross-type gap absorbed by accuracy (+0.469 on the accuracy coefficient, -0.034 residual one-shot coefficient) reflects the late-arrival pattern documented in Section 4.2.

Table IA.15: Panel Regression: Edge on Trader Type with Market Fixed Effects

This table regresses edge (dollars per token) on trader-type indicators with progressively richer controls. The unit of observation is the wallet-market cell, aggregating all trades by a wallet in a market (55.4 million cells). Column (1) pools all cells. Column (2) adds market fixed effects. Column (3) adds wallet-level controls. Column (4) adds wallet accuracy. Bot is the omitted category. WLS with trade-count weights; t -statistics in parentheses based on wallet-clustered standard errors. The R^2 row reports within-market R^2 for the fixed-effects specifications and the unadjusted R^2 for column (1). Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

	(1) Pooled	(2) Market FE	(3) + Controls	(4) + Accuracy
Sophisticated	-0.008*** (-21.10)	-0.010*** (-25.50)	-0.010*** (-19.86)	0.026*** (13.68)
Active Retail	-0.008*** (-41.10)	-0.011*** (-35.67)	-0.007*** (-14.42)	0.015*** (5.10)
Casual	-0.033*** (-82.75)	-0.036*** (-72.80)	-0.033*** (-47.43)	0.018*** (6.81)
One-Shot	-0.060*** (-59.90)	-0.064*** (-53.33)	-0.058*** (-44.31)	-0.034*** (-11.03)
Log(Volume)			0.002*** (23.20)	0.001 (1.50)
Log(Trades)			-0.003*** (-12.80)	0.000 (-0.23)
Maker Fraction			0.005*** (7.78)	0.066*** (13.10)
Price Distance			0.041*** (21.79)	-0.053*** (-5.99)
Accuracy				0.469*** (83.75)
Constant	0.002*** (8.75)	0.002*** (7.40)	-0.017*** (-20.75)	-0.260*** (-66.67)
Market FE	No	Yes	Yes	Yes
Observations	55.4M	55.4M	55.4M	55.4M
Within R^2	0.001	0.005	0.004	0.471
Wallet clusters	1.95M	1.95M	1.95M	1.95M

IA.21 Alternative Classification Thresholds

Our results could depend on specific classification cutoffs. We vary the classification thresholds by $\pm 50\%$, creating nine specifications: the baseline, jointly loosened or tightened thresholds, and one-at-a-time variations of the bot, sophisticated, and retail boundaries. Table [IA.16](#) reports the corresponding edge regressions. Aggregate bot ROI is positive across all nine specifications, $+0.33\%$ under the loosest threshold, $+0.56\%$ at baseline, and $+0.80\%$ under the strictest, while active retail ROI is negative throughout (-1.59% to -2.18%). The qualitative finding holds regardless of classification boundaries.

Table IA.16: Edge by Trader Type Under Alternative Classification Thresholds

This table reports edge regressions on trader-type indicators under nine alternative wallet classification thresholds. Column (1) is the baseline. Columns (2)–(3) simultaneously loosen or tighten all thresholds. Columns (4)–(9) vary one type boundary at a time. Bot is the omitted category. Edge coefficients in dollars per token (four decimal places for precision at the sub-cent level). WLS with t -statistics in parentheses based on heteroskedasticity-robust (HC1) standard errors. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

	(1) Baseline	(2) Loose	(3) Strict	(4) Bot Loose	(5) Bot Strict	(6) Soph Loose	(7) Soph Strict	(8) Ret Loose	(9) Ret Strict
Sophisticated	−0.0004 (−1.00)	0.0012*** (4.00)	−0.0015*** (−3.00)	0.0014*** (3.50)	−0.0017*** (−4.25)	−0.0005* (−1.67)	−0.0002 (−0.40)	−0.0004 (−1.00)	−0.0004 (−1.00)
Active Retail	−0.0029*** (−14.50)	−0.0043*** (−21.50)	−0.0026*** (−8.67)	−0.0013*** (−6.50)	−0.0042*** (−14.00)	−0.0030*** (−15.00)	−0.0028*** (−14.00)	−0.0054*** (−27.00)	−0.0013*** (−6.50)
Casual	−0.0289*** (−72.25)	−0.0345*** (−69.00)	−0.0202*** (−67.33)	−0.0273*** (−68.25)	−0.0302*** (−75.50)	−0.0290*** (−72.50)	−0.0289*** (−72.25)	−0.0360*** (−72.00)	−0.0189*** (−63.00)
One-Shot	−0.0524*** (−58.22)	−0.0509*** (−56.56)	−0.0538*** (−53.80)	−0.0509*** (−56.56)	−0.0538*** (−53.80)	−0.0524*** (−58.22)	−0.0524*** (−58.22)	−0.0524*** (−58.22)	−0.0524*** (−58.22)
Bot (constant)	−0.0057*** (−28.50)	−0.0072*** (−36.00)	−0.0044*** (−14.67)	−0.0072*** (−36.00)	−0.0044*** (−14.67)	−0.0057*** (−28.50)	−0.0057*** (−28.50)	−0.0057*** (−28.50)	−0.0057*** (−28.50)
Observations	1.95M	1.95M	1.95M	1.95M	1.95M	1.95M	1.95M	1.95M	1.95M
R-squared	0.012	0.011	0.010	0.012	0.012	0.012	0.012	0.011	0.010

IA.22 Out-of-Sample Classification

We classify wallets based only on their first N days of activity, then measure P&L on subsequent trades. Under the baseline 30-day window (Table IA.17), the bot-versus-casual ranking is preserved: bot-like wallets earn +0.36 cents, while casual wallets lose -0.21 cents. Sign flips for intermediate categories reflect contamination from reclassified bots, as shown by the retention decomposition. Panel B of Table IA.18 shows that these patterns are robust across classification windows ranging from 7 to 180 days: bot OOS edge is positive and significant at every horizon.

Table IA.17: Out-of-Sample Classification Test

This table reports out-of-sample performance for wallets classified using only their first 30 days of activity and then evaluated on all subsequent trades. One-shot wallets are excluded. The OOS Performance columns report edge (cents per token) and the associated t -statistic measured on post-classification trades; the Baseline Performance columns report the same quantities under the full-sample classification. t -statistics, reported in their own columns, are based on heteroskedasticity-robust (HC1) standard errors. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

Type (First 30 Days)	N Wallets	OOS Performance		Baseline Performance	
		Edge (cents)	t -stat	Edge (cents)	t -stat
Bot	32,396	+0.36***	93.1	+0.29***	112.9
Sophisticated	24,366	+0.41***	38.7	-0.10^{***}	-6.2
Active retail	413,309	+0.38***	52.0	-0.38^{***}	-48.9
Casual	176,207	-0.21^{***}	-11.7	-2.00^{***}	-30.1

IA.23 The 2024 Election as an Attention Shock

The timing mechanism predicts that when retail attention surges, creating more late-arriving traders, the execution gap should widen and bot profitability should increase. The 2024 U.S. presidential election provides a quasi-experimental attention shock that allows a diagnostic test of this prediction. Election markets were the highest-volume event category in Polymarket history, generating \$2.9 billion in bot volume and \$994 million in retail volume during the June–November 2024 election season alone. The media salience of the election drew a large influx of retail participants, many arriving late in the price discovery process.

Table IA.18: Out-of-Sample Performance by Retention Status and Classification Window

This table reports out-of-sample edge for wallets classified using their first N days of trading and then tested on all subsequent trades. Panel A decomposes the 30-day classification by retention status: “Retain” wallets receive the same type under both the first-window scheme and the full-sample baseline; “Change” wallets are reclassified. Panel B reports pooled out-of-sample edge (combining retain and change wallets) across classification windows of 7, 14, 30, 60, 90, and 180 days. One-shot wallets are excluded throughout. Edge in cents per token; t -statistics in parentheses, computed from standard errors of mean edge across trades. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

<i>Panel A: 30-Day Classification Window by Retention Status</i>				
Wallet Type	Status	Trades	Edge (t -stat)	Accuracy
Bot	Change	166K	−0.610*** (−7.59)	43.4%
	Retain	88.9M	+0.360*** (93.48)	44.9%
Sophisticated	Change	9.2M	+0.450*** (42.13)	46.1%
	Retain	508K	−0.450*** (−9.19)	47.1%
Active Retail	Change	22.6M	+0.440*** (56.30)	44.6%
	Retain	2.7M	−0.170*** (−10.15)	46.7%
Casual	Change	3.3M	−0.210*** (−11.62)	44.8%
	Retain	7K	−0.820*** (−2.73)	69.3%
<i>Panel B: Pooled OOS Edge by Classification Window Length</i>				
Wallet Type	Window	Trades	Edge (t -stat)	Accuracy
Bot	7d	92.8M	+0.340*** (86.90)	45.0%
	14d	97.2M	+0.370*** (96.20)	45.2%
	30d	89.1M	+0.360*** (93.10)	44.9%
	60d	75.0M	+0.370*** (91.00)	44.7%
	90d	57.7M	+0.400*** (88.30)	44.1%
	180d	32.8M	+0.440*** (75.80)	43.7%
Sophisticated	7d	7.9M	+0.210*** (17.50)	46.3%
	14d	9.8M	+0.320*** (30.00)	46.1%
	30d	9.7M	+0.410*** (38.70)	46.1%
	60d	7.9M	+0.490*** (43.20)	44.6%
	90d	5.1M	+0.460*** (31.70)	45.9%
	180d	3.9M	+0.440*** (26.90)	44.1%
Active Retail	7d	44.9M	+0.360*** (67.60)	44.3%
	14d	35.7M	+0.310*** (51.90)	44.0%
	30d	25.3M	+0.380*** (52.00)	44.8%
	60d	12.7M	+0.440*** (44.60)	44.8%
	90d	10.5M	+0.360*** (33.40)	45.1%
	180d	4.7M	+0.340*** (19.70)	44.3%
Casual	7d	7.3M	+0.090*** (6.70)	45.2%
	14d	6.8M	+0.100*** (7.00)	45.9%
	30d	3.3M	−0.210*** (−11.70)	44.8%
	60d	3.0M	−0.100*** (−6.00)	45.4%
	90d	1.5M	−0.270*** (−10.60)	42.7%
	180d	974K	−0.050 (−1.50)	43.1%

We compare election and non-election markets trading contemporaneously on the same platform during the June–November 2024 window, applying the same decomposition and the same trader classifications. Election markets are identified by category (“Politics,” “US-current-affairs”) and question keywords (“president,” “election,” “Trump,” “Harris,” etc.). The estimand is descriptive: we ask whether the execution gap is larger on high-attention markets, not whether the election *caused* the gap in a narrow causal sense, since election markets differ from non-election markets along dimensions beyond attention (e.g., information structure, contract duration). The results are consistent with the mechanism’s prediction. Bot ROI on election markets is +0.97%, nearly four times the +0.27% earned on non-election markets during the same period. Bot edge is larger on the highest-attention markets in the sample: +0.47 cents on election markets versus +0.31 cents on non-election markets. The inversion holds on both market types, but the magnitudes are amplified on election markets, consistent with a larger pool of late-arriving retail traders generating more surplus for bots to capture.

Active retail accuracy on election markets (49.0%) is slightly *lower* than on non-election markets (50.2%), despite election outcomes being the most publicly discussed and widely analyzed events in the sample. If abundant public information were sufficient to generate profitable forecasts, election markets are where this advantage should appear most clearly. Its absence strengthens the claim that raw accuracy above 50% in the main sample reflects price-level mechanics rather than forecasting skill.

Bot accuracy on election markets (51.2%) is modestly above 50% and above the full-sample bot average of 49.9%. This likely reflects the strong price discovery in election markets: with high information arrival, prices converge quickly toward terminal values, so that even non-directional traders who enter at prevailing prices end up on the “correct” side more often than not. This mechanical accuracy boost does not translate into a directional edge (bot directional edge remains negative) but rather reflects the informativeness of prices at the time of trade.

Table IA.19: Edge by Trader Type: Election versus Non-Election Markets

This table regresses edge (dollars per token) on trader-type indicators for wallets active in both election and non-election markets. Column (1) pools all trades. Column (2) adds wallet-level controls. Column (3) adds election-market interactions to test whether type-level edge differences change in high-attention markets. Bot is the omitted category. WLS with trade-count weights; t -statistics in parentheses based on wallet-clustered standard errors. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

	(1) Pooled	(2) + Controls	(3) + Interaction
Sophisticated	−0.008*** (−21.10)	−0.009*** (−21.68)	−0.008*** (−21.05)
Active Retail	−0.008*** (−41.10)	−0.009*** (−28.33)	−0.008*** (−28.30)
Casual	−0.033*** (−82.75)	−0.033*** (−83.50)	−0.038*** (−95.00)
One-Shot	−0.060*** (−59.90)	−0.060*** (−60.40)	−0.089*** (−73.75)
Election			0.002** (1.99)
Soph. × Elec.			−0.001 (−1.03)
Retail × Elec.			0.000 (0.11)
Casual × Elec.			0.022*** (18.00)
One-Shot × Elec.			0.087*** (39.41)
Constant	0.002*** (8.75)	0.002*** (8.00)	0.002*** (8.05)
Controls	No	Yes	Yes
Observations	2.79M	2.79M	2.79M
R-squared	0.011	0.011	0.012

IA.24 Options Validation: Sample and Construction

This subsection and the three that follow report the full options analysis supporting Section 7. The unit of observation throughout is a type-by-underlying cell: for each trader type and underlying stock, we aggregate all contracts the type opened on that underlying, requiring at least 100 contracts per cell (20 in the short-dated at-the-money subsample). Forecasting accuracy θ is the volume-weighted fraction of a cell's contracts that expire in the money. Execution ε is the volume-weighted difference between the contract's lifecycle volume-weighted average price and the entry price, so positive values indicate entry below the contract's average traded price. Total edge decomposes as in Equation 3: payoff minus lifecycle VWAP (directional) plus lifecycle VWAP minus entry price (execution). Trader types follow the exchange origin code classification described in Section 7.1; market makers are excluded. Table IA.20 describes the sample.

Table IA.20: Options Sample Description by Exchange Origin Code

This table describes the CBOE equity options sample by exchange origin code classification. Contracts is the number of option contracts opened. Volume is aggregate premium in dollars. Entry price is the average premium paid per contract. Payoff is the average realized payoff (intrinsic value at expiration or exercise). DTE is the average days to expiration at entry. % Calls and % ITM report the fraction of contracts that are calls and in-the-money at expiry, respectively. Market makers are excluded from entry/payoff statistics because their positions are hedged and the data do not link hedges to individual contracts. Sample period: January 2011 to July 2022.

Type	Contracts	Volume (\$)	Entry (\$)	Payoff (\$)	DTE	% Calls	% ITM
Firm (Prop.)	1.8M	\$441M	1.77	1.78	61.3	48.4%	32.8%
Broker-Dealer	115,153	\$19M	2.21	2.95	60.8	40.8%	31.2%
Market Maker	14.7M	\$7.5B	—	—	—	—	—
Small Retail	9.7M	\$1.6B	2.74	2.92	43.3	62.2%	33.2%
Medium Retail	738,641	\$249M	1.64	1.77	43.9	59.2%	30.9%
Large Retail	373,752	\$610M	1.80	1.86	53.4	48.7%	29.4%
Prof. (<100)	1.3M	\$44M	3.33	3.60	60.4	52.1%	39.1%
Prof. (100–199)	28,356	\$6M	1.40	1.47	56.3	48.6%	33.9%
Prof. (>199)	15,487	\$15M	1.62	1.67	66.5	50.2%	29.5%

IA.25 Options Skill Persistence

The cross-skill persistence test in Section 7.3 splits each type-by-underlying cell at January 1, 2017, near the midpoint of the 2011–2022 sample by expiration date, and requires at least 50 contracts in each half. Table IA.21 reports the contemporaneous correlation between θ and ε and the four split-half correlations, with 95% confidence intervals from the Fisher transformation. Cross-skill persistence is small for every type: the largest estimate is 0.055 (Prof. <100, execution to subsequent accuracy), which explains less than half a percent of variance. Own-skill persistence is modest and heterogeneous, including negative own- θ persistence for Prof. <100 (−0.076), which is why Section 7.3 treats the near-zero cross-skill coefficients as corroborative rather than decisive. The contemporaneous correlations range from −0.161 to +0.155, matching the within-type range reported in Section 7.3.

Table IA.21: Options Skill Persistence and Cross-Skill Prediction by Trader Type

This table reports contemporaneous and split-sample correlations between forecasting accuracy θ and execution quality ε in equity options. The unit of observation is a type-by-underlying cell with at least 100 contracts. $\rho(\theta, \varepsilon)$ is the contemporaneous correlation across cells within each type, computed on n cells. The split-sample columns divide each cell at January 1, 2017, near the midpoint of the sample by expiration date, and require at least 50 contracts in each half (n_{split} cells); $\theta \rightarrow \theta$ and $\varepsilon \rightarrow \varepsilon$ correlate first-half with second-half values of the same skill, and $\theta \rightarrow \varepsilon$ and $\varepsilon \rightarrow \theta$ correlate first-half values of one skill with second-half values of the other. Brackets report 95% confidence intervals from the Fisher transformation. Trader types follow the exchange origin code classification; market makers are excluded. Sample period: January 2011 to July 2022.

Type	n	$\rho(\theta, \varepsilon)$	n_{split}	$\theta \rightarrow \theta$	$\varepsilon \rightarrow \varepsilon$	$\theta \rightarrow \varepsilon$	$\varepsilon \rightarrow \theta$
Firm (Prop.)	5,880	−0.015 [−0.040, 0.011]	3,122	0.118 [0.083, 0.153]	0.039 [0.004, 0.074]	−0.013 [−0.048, 0.022]	−0.011 [−0.046, 0.024]
Broker-Dealer	2,208	0.078 [0.036, 0.119]	781	0.024 [−0.047, 0.094]	0.106 [0.036, 0.175]	0.002 [−0.068, 0.072]	0.009 [−0.061, 0.079]
Small Retail	7,069	0.155 [0.132, 0.178]	3,595	0.111 [0.079, 0.143]	0.100 [0.067, 0.132]	−0.017 [−0.050, 0.015]	0.019 [−0.014, 0.051]
Medium Retail	6,024	0.074 [0.049, 0.099]	2,567	0.019 [−0.020, 0.058]	0.032 [−0.007, 0.070]	0.003 [−0.036, 0.042]	−0.003 [−0.041, 0.036]
Large Retail	5,108	0.128 [0.101, 0.155]	2,059	0.038 [−0.005, 0.081]	−0.058 [−0.101, −0.015]	0.036 [−0.007, 0.079]	−0.006 [−0.049, 0.037]
Prof. (<100)	4,406	−0.161 [−0.189, −0.132]	1,827	−0.076 [−0.121, −0.030]	0.010 [−0.036, 0.056]	0.029 [−0.017, 0.074]	0.055 [0.009, 0.101]
Prof. (100–199)	2,119	0.020 [−0.023, 0.063]	579	0.002 [−0.080, 0.083]	−0.142 [−0.221, −0.062]	−0.001 [−0.083, 0.080]	−0.023 [−0.104, 0.059]
Prof. (>199)	1,834	0.077 [0.032, 0.123]	422	−0.031 [−0.126, 0.064]	−0.272 [−0.358, −0.181]	0.002 [−0.093, 0.097]	0.034 [−0.062, 0.129]

IA.26 Options Entry Price Selection by Moneyness

Table IA.22 reports regressions of entry price on accuracy within each trader type, separately for ITM and OTM contracts. Under the table's convention, a negative coefficient indicates that a type pays less for the contracts it predicts correctly. Prof. <100 pays lower prices for correctly predicted contracts on both the ITM side ($\beta = -2.15$) and the OTM side ($\beta = -1.43$), consistent with informed contract selection across the moneyness range. The sign and magnitude vary across the other types because options traders operate across multiple choice margins; the gradient is theoretically signed only for types whose informational advantage operates directly on the entry-price margin.

Table IA.22: Options: Entry Price Selection by Moneyness

This table reports regressions of entry price on accuracy within each trader type, separately for in-the-money (ITM) and out-of-the-money (OTM) contracts. The dependent variable is the average entry price (dollars) per underlying; the independent variable is accuracy (fraction ITM at expiry) per underlying. A negative β indicates that a type pays less for contracts it predicts correctly. N is the number of underlyings with sufficient observations. The sample is CBOE equity options, January 2011 through July 2022. t -statistics in parentheses, based on classical OLS standard errors. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

Type	ITM Contracts		OTM Contracts		N
	β	t -stat	β	t -stat	
Firm (Prop.)	0.230	(1.10)	-0.270***	(-3.00)	5,658
Broker-Dealer	2.460***	(3.90)	0.450	(1.60)	1,774
Small Retail	1.170***	(5.10)	0.690***	(5.50)	6,970
Medium Retail	0.400	(1.30)	0.180	(1.50)	4,526
Large Retail	-0.330	(-1.20)	-0.240	(-1.20)	3,553
Prof. (<100)	-2.150***	(-7.70)	-1.430***	(-13.00)	4,072
Prof. (100-199)	-1.190**	(-2.00)	-0.450**	(-2.30)	1,062
Prof. (>199)	1.640**	(2.00)	0.090	(0.20)	765

IA.27 Options Accuracy Quintiles

Section 7 reports that the accuracy-profitability inversion does not replicate in options: sorting cells into quintiles of directional accuracy, more accurate cells are more profitable. Table IA.23 reports the full sort. In the pooled sample, volume-weighted total edge rises

from -59.4 cents per share in the bottom accuracy quintile to $+81.4$ cents in the top, a Q5–Q1 spread of $+140.8$ cents, and the regression of total edge on accuracy yields a slope of 269.2 cents. The gradient is positive in the short-dated at-the-money, above-chance, and type-demeaned samples as well. The Prof. <100 sample is the one exception on the volume-weighted spread (-71.8 cents); the equal-weighted spread ($+94.6$ cents) and the regression slope ($+112.8$ cents) are both positive, so the negative volume-weighted spread reflects a small number of high-volume, high-edge underlyings in the bottom accuracy quintile rather than a negative gradient.

Table IA.23: Options Accuracy Quintiles and the Accuracy–Profitability Gradient

This table tests whether the accuracy-profitability inversion replicates in equity options. The unit of observation is a type-by-underlying cell with at least 100 contracts (20 in the short-dated at-the-money sample). Cells are sorted into quintiles of forecasting accuracy θ , the volume-weighted fraction of contracts expiring in the money. Panel A reports volume-weighted quintile means for the full pooled sample: total edge (payoff minus entry price), its directional (payoff minus lifecycle VWAP) and execution (lifecycle VWAP minus entry price) components in cents per share, the equal-weighted total edge, and ROI (total edge over average entry price). Panel B reports OLS regressions of total edge on θ for five samples; the slope is in cents per share, t -statistics in parentheses based on classical OLS standard errors, ρ is the cell-level Spearman correlation between θ and total edge, and Q5–Q1 is the spread in volume-weighted total edge. The short-dated at-the-money sample restricts to moneyness between 0.95 and 1.05 and average days to expiration of 30 or fewer; the above-chance sample restricts to cells with $\theta > 0.343$; the type-demeaned sample subtracts volume-weighted type means from θ and edge. In the Prof. <100 sample the volume-weighted Q5–Q1 spread is negative while the equal-weighted spread ($+94.59$ cents) and the regression slope are positive: the negative volume-weighted spread reflects a small number of high-volume, high-edge underlyings in the bottom accuracy quintile, not a negative gradient. *, **, and *** denote significance at the 10%, 5%, and 1% levels. Sample period: January 2011 to July 2022.

<i>Panel A: Quintile means, full pooled sample (volume-weighted)</i>							
Quintile	N	Accuracy (%)	Total (¢)	Directional (¢)	Execution (¢)	EW total (¢)	ROI (%)
Q1 (Low)	6,928	8.0	-59.43	-58.13	-1.30	-88.08	-65.73
Q2	6,927	25.7	-8.07	-20.09	12.02	-43.67	-2.78
Q3	6,928	34.0	16.09	10.85	5.24	0.06	7.67
Q4	6,927	43.3	73.31	71.58	1.73	34.81	47.85
Q5 (High)	6,928	68.6	81.41	76.26	5.15	105.72	32.34
Q5–Q1			+140.84	+134.39	+6.45	+193.80	
<i>Panel B: Regressions of total edge on accuracy</i>							
Sample	N		β (¢)		R^2	Spearman ρ	Q5–Q1 (¢)
Full pooled	34,638		269.23*** (43.39)		0.052	0.49	+140.84
Short-dated ATM	24,298		159.56*** (51.80)		0.099	0.51	+118.65
Above chance ($\theta > 0.343$)	17,642		247.78*** (17.03)		0.016	0.20	+112.68
Prof. (<100) only	4,406		112.85*** (4.01)		0.004	0.36	-71.83
Type-demeaned	34,638		277.75*** (44.05)		0.053	0.47	+142.34

IA.28 Execution Ranking Stability Across Regimes

Section 7 reports that type-level execution rankings are stable across market regimes. Table IA.24 reports the underlying correlations. For each of five sample splits, we compute each type’s execution edge separately in the two halves and correlate the cross-type rankings (eight types per correlation). Premium-normalized execution, the lifecycle VWAP minus the entry price scaled by the entry price, yields Spearman rank correlations between 0.90 and 0.95 across all five splits. Execution measured in cents per share yields lower rank correlations (0.00 to 0.62) because cents-denominated edge scales with the premium level, which differs mechanically across regimes; we report both measures.

Table IA.24: Stability of Cross-Type Execution Rankings Across Market Regimes

This table tests whether the cross-type ranking of execution quality is stable across market regimes. For each of five sample splits, we compute each trader type’s execution edge separately in the two halves and correlate the cross-type rankings (eight types per correlation). Premium-normalized execution is the lifecycle VWAP minus the entry price, scaled by the entry price; cents-based execution is the same difference in cents per share. Cents-based rank correlations are lower because cents-denominated edge scales with the premium level, which differs mechanically across regimes. Sample period: January 2011 to July 2022.

Regime split	Premium-normalized		Cents per share	
	Spearman	Pearson	Spearman	Pearson
Calls vs. puts	0.905	0.934	0.619	0.946
Early (2011–2016) vs. late (2017–2022)	0.952	0.776	0.500	0.769
High vs. low premium	0.929	0.913	0.000	0.433
Large vs. small capitalization	0.929	0.935	0.452	0.888
At-the-money vs. other moneyness	0.929	0.898	0.524	0.870